Course Glossary

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This glossary defines key terms used in the course slides. The web version links each term to its first substantive appearance in the slides.

2

2008 financial crisis. The global financial disruption that weakened confidence in banks, financial governance, and centralized risk management. Aliases: 2008 Global Financial Crisis, financial crisis, global financial crisis.

Source: 03-IN - Framing.

21 million cap. Bitcoin's approximate maximum total supply limit of 21 million bitcoin. Aliases: 21 million cap (Bitcoin), 21M cap, hard cap, supply cap.

Source: 01-BF - Monetary Policy.

4

4byte.directory. A free signature database that maps function selectors back to their human-readable signatures. Aliases: 4byte.

Source: 11-FO - Looking up a selector: a name, not proof.

5

50 percent rule. OFAC's rule that a firm owned 50 percent or more by blocked persons is blocked too. Aliases: OFAC 50 percent rule, fifty percent rule.

Source: 05-FO - OFAC designates; every U.S. person blocks.

51% attack. An attack where a participant or coalition gains enough consensus power to manipulate accepted history or disrupt normal confirmation. Aliases: 51 percent attack, 51% attack (Bitcoin), 51% attack (blockchain consensus), 51% attack (proof of work), majority attack, majority hash attack, majority hash power attack.

Source: 03-IN - Why This History Matters.

A

Aave. A DeFi protocol that offers flash loans from its pools.

Source: 10-FO - A flash loan lasts one transaction.

ABI encoding. The Ethereum convention for packing function arguments, return values, and event data into byte sequences. Aliases: ABI encoding (Ethereum), ABI-encoded, encoded arguments.

Source: 04-EF - ABI, Encoding, and Function Selectors.

Access control. Rules that determine who may read data, propose transactions, endorse updates, order entries, or administer the ledger. Aliases: Access control (distributed ledgers), Access control (smart contracts), access-control policy, authorization logic, authorization policy.

Source: 04-DS - Transparency vs. Privacy.

Access policy. Rules that determine which participants may read, write, submit, approve, or administer specific resources. Aliases: access control policy.

Source: 01-PC - What each reader can see.

Account abstraction. ERC-4337's design in which a user signs a UserOperation and a bundler packs operations into one transaction to the EntryPoint contract's handleOps, so the transaction's sender is the bundler, not the actor. Aliases: ERC-4337, EntryPoint, UserOperation, bundler.

Source: 03-EF - Paymasters and relayers pay gas, not the bill.

Account model. A contract model where accounts hold state and programs operate by reading or writing those accounts during execution. Aliases: Account-centric model (smart contracts), account model, account-based model.

Source: 04-EF - Recognize It: Rust on Solana.

Account model (Ethereum). An accounting and execution model where balances, code, and storage are attached to accounts and updated directly. Aliases: Account model (Ethereum accounting), Account model (Ethereum transaction model), Ethereum Account Model, Ethereum accounts, accounts.

Source: 01-EF - Ethereum keeps accounts, not coins.

Accuracy. The degree to which reputation scores correlate with actual future peer behavior. Aliases: Accuracy (reputation systems).

Source: 03-DS - Properties of Reputation Systems.

Acknowledgment. A message confirming that another message was received. Aliases: ACK, Acknowledgment (network protocol), acknowledgment, acknowledgments.

Source: 05-IN - Omission Faults.

Across. An intents bridge: the user deposits on one chain, a relayer pays on the other from its own tokens, and the protocol repays relayers later through Ethereum. Aliases: Across Protocol.

Source: 08-EF - Liquidity bridges: someone pays you there.

Activation queue. The waiting process a staked node passes through before becoming an active validator. Aliases: Activation queue (PoS validator lifecycle), activation epoch.

Source: 03-CM - Proof of Stake: The Full Lifecycle.

AddedBlackList. The event USDT's contract emits when an address is blacklisted; a later RemovedBlackList undoes it. Aliases: RemovedBlackList, addBlackList, removeBlackList.

Source: 01-PC - A freeze is true only as of the read.

addr message. A Bitcoin network message that advertises known peer addresses to other nodes. Aliases: addr message (Bitcoin protocol), address message, addrv2 message.

Source: 03-BF - Gossip and Peer Connections.

Address (Bitcoin). A user-facing encoding of spending conditions, typically derived from public-key material, where value can be locked. Aliases: Bitcoin address.

Source: 02-BF - Script type says what kind of wallet paid.

Address (Ethereum). A 20-byte account identifier used to refer to EOAs, contract accounts, owners, recipients, and callers. Aliases: address.

Source: 04-EF - Types: Value Types.

Address Clustering. The process of grouping addresses or outputs that likely belong to the same actor based on transaction evidence and heuristics. Aliases: Address Clustering (blockchain forensics), Address clustering (forensics), address cluster analysis, clustering, wallet clustering.

Source: 02-FO - Co-spending links inputs, unless it's a CoinJoin.

Address format. The visible encoding of a Bitcoin address, whose prefix shows its script type: 1 for P2PKH, 3 for P2SH, bc1q for native SegWit, and bc1p for Taproot. Aliases: Address format (Bitcoin), Base58, Bech32, SegWit address, legacy address.

Source: 02-BF - Script type says what kind of wallet paid.

Address freeze. An issuer-controlled function that puts an address on a token contract's blacklist so it cannot send that token. Aliases: Address freeze (stablecoin), USDT freeze, blacklist, blacklisting, freeze, freeze capability, stablecoin freeze.

Source: 01-PC - USDT on Tron: same freeze, same owner rule.

Address gossip. The exchange of known peer addresses among connected Bitcoin nodes. Aliases: Address gossip (Bitcoin network), address relay, peer address gossip.

Source: 03-BF - Joining the Bitcoin Network.

Address index. The last derivation-path level: which address on the receiving or change chain, starting at 0. Aliases: address_index.

Source: 05-BF - The path names coin, account, chain and index.

Address payable. A Solidity address type that is allowed to receive ETH through value-transfer operations. Aliases: Address payable (Solidity), address payable, payable address.

Source: 04-EF - Types: Value Types.

Address poisoning. A theft in which the attacker plants an address that looks like one the victim has paid in the victim's history, hoping the victim copies it next time. Aliases: poisoning.

Source: 09-FO - Four ways out of a wallet look different.

Address reuse. Using the same Bitcoin address again, such as sending a transaction's change back to the address that paid. Aliases: Address reuse (Bitcoin).

Source: 02-BF - Reading Alice's payment, field by field.

Address zero. A special all-zero Ethereum address commonly used in token events to signal minting or burning rather than transfer from a normal owner. Aliases: 0x0, Address zero (Ethereum), zero address.

Source: 06-EF - What Is Minting?.

Admission control. The process for deciding who may join or act in the network before they receive credentials or roles. Aliases: Admission control (permissioned ledger), access admission, membership admission.

Source: 01-PC - Find the gatekeeper: who can stop or see it.

Adoption friction. The combined technical, institutional, economic, and user-level barriers that prevent blockchain from spreading broadly. Aliases: Adoption friction (blockchain), accumulated friction, barriers to adoption.

Source: 05-AS - The Real Question.

Adversary. A modeled opponent who may try to read, alter, forge, replay, or otherwise break a cryptographic system. Aliases: attacker.

Source: 01-CP - What Is Cryptography?.

Adversary assumption. The protocol assumption about what faulty or malicious participants are able and willing to do. Aliases: Adversary assumption (consensus), adversary model.

Source: 01-CM - Open chains buy openness with slower finality.

Adverse Selection. A market outcome where hidden information causes lower-quality or riskier participants to dominate.

Source: 04-IN - The "Market for Lemons": When Trust Breaks Down.

AES. A widely used symmetric block cipher standard for encrypting fixed-size blocks with a secret key. Aliases: AES (symmetric cipher), Advanced Encryption Standard.

Source: 03-CP - What Is Symmetric Cryptography?.

Agreement. The consensus property that honest nodes converge on the same result. Aliases: Agreement (consensus property).

Source: 04-AS - What Consensus Is Actually Trying to Guarantee.

AI error at scale. A failure mode where brittle, hallucinated, or overconfident AI output causes widespread damage when trusted in automated systems. Aliases: AI error at scale (blockchain automation), AI errors at scale.

Source: 05-AS - Risks.

AI-driven exploit. An attack where AI helps discover, prioritize, or execute weaknesses in contracts, wallets, protocols, or public code. Aliases: AI-driven exploit (blockchain security), AI-driven exploits.

Source: 05-AS - Risks.

Air-gapped computer. A computer kept off the internet so the keys it holds stay offline; one form of cold wallet. Aliases: air gap, air-gapped, air-gapped wallet.

Source: 05-BF - Hot or cold says where the key lives.

Algorithm deprecation. The process of retiring an algorithm from trusted use because its security margin or standard status is no longer adequate. Aliases: Algorithm deprecation (cryptography), cryptographic deprecation, deprecated algorithm.

Source: 05-CP - Cryptographic Assumptions Decay Over Time.

Algorithmic design failure. A collapse caused by automated economic or protocol mechanisms that fail under real market pressure. Aliases: Algorithmic design failure (crypto markets), algorithmic failure.

Source: 05-AS - When Blockchain Systems Broke.

Algorithmic stablecoin. A stablecoin with no backing asset, in which an algorithm trades tokens to hold the price; TerraUSD collapsed in May 2022. Aliases: algorithmic stablecoins.

Source: 05-EF - Three ways a stablecoin holds its peg.

Ali Baba cave protocol. A teaching analogy where repeated random challenges convince a verifier that a prover knows a secret without revealing it. Aliases: Ali Baba cave, cave protocol.

Source: 05-CP - Ali Baba Cave: The Protocol.

Allowance. An ERC-20 permission, set with approve, that lets another address, often a contract, spend up to a set amount of the holder's tokens through transferFrom. Aliases: Allowance (fungible token), allowances, spending allowance, spending allowances.

Source: 05-EF - ERC-20: six functions and two events.

Allowlist. A minting control that restricts mint access to pre-approved addresses. Aliases: Allowlist (minting), allow list, pre-approved list, whitelist.

Source: 06-EF - Minting on Ethereum.

Alternative hypotheses. Other plausible explanations for what the sources show, which ICD 203 asks analysts to identify and weigh. Aliases: alternative hypothesis.

Source: 14-FO - When sources disagree, say so.

AML program. A written, risk-based program an MSB must run under the Bank Secrecy Act, with a compliance officer, training and independent review. Aliases: anti-money-laundering program.

Source: 05-FO - KYC ties an account to a person.

AML typology. A named pattern of what the funds did, such as a peel chain, CoinJoin, chain-hop, OTC aggregation, scam cash-out or drain. It describes the movement and its on-chain cue; on its own it says nothing about who did it. Aliases: AML typologies, laundering typology.

Source: 12-FO - Six typologies you have already met.

Announce-request-transmit pattern. A relay sequence where a node advertises data, a peer requests it, and the node then sends the full item. Aliases: Announce-request-transmit pattern (Bitcoin relay), announce-request-transmit, inv-getdata relay.

Source: 03-BF - Transaction Relay.

Anomaly detection. A monitoring technique that flags sudden behavior changes that may indicate reputation exploitation. Aliases: Anomaly detection (trust systems).

Source: 03-DS - Threats: On–Off (Reputation Milking).

Anonymity. The condition of acting without linking an action or transaction to an identifiable person. Aliases: anonymous.

Source: 03-IN - The Cypherpunk Movement.

Anonymity layer. A network routing tool or service that obscures a user's real network address from the peers they contact. Aliases: Anonymity layer (network privacy), anonymity network, privacy routing layer.

Source: 04-BF - Mitigating Network Deanonymization.

Anonymity-enhanced cryptocurrency. FinCEN's term for privacy coins such as Monero; ransoms are most often in Bitcoin, but criminals increasingly ask for AECs. Aliases: AEC, AECs, anonymity-enhanced cryptocurrencies.

Source: 03-FO - A note leads to a portal, a price, an address.

ApeCoin. A separate ERC-20 governance and utility token used for ApeCoin DAO voting rather than direct BAYC NFT ownership. Aliases: APE, ApeCoin (BAYC ecosystem), ApeCoin DAO.

Source: 06-EF - Case Study: Bored Ape Yacht Club.

Append-only. A record design where new entries are added to the end and past entries are not silently edited in place. Aliases: append-only history, append-only record.

Source: 04-DS - From Logs to Ledgers.

Append-only ledger. A record structure where new entries can be added but past entries are not silently edited or deleted. Aliases: Append-only ledger (Bitcoin), append-only database, global append-only ledger.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Append-only state machine. A model of Ethereum where valid transactions advance a shared global state and accepted history is extended rather than rewritten. Aliases: Append-only state machine (Ethereum), global state machine, shared state machine.

Source: 02-EF - The Ethereum Operational Framework.

Application Binary Interface. The byte-level interface that specifies how callers encode function calls, return values, and event data for a contract. Aliases: ABI, Application Binary Interface (Ethereum).

Source: 04-EF - ABI, Encoding, and Function Selectors.

Approval. A permission a holder grants, with ERC-20's approve, letting another address or contract move up to an allowance of their tokens. Aliases: Approval (token permission), approvals, operator approval, over-broad approvals.

Source: 05-EF - ERC-20: six functions and two events.

Approval drain. A theft that reads in four steps: the victim's approval, the spender's take with transferFrom, the drainer's cut, and a swap out of tokens that can be frozen.

Source: 09-FO - Reading an approval drain, step by step.

Approval event. The log an ERC-20 token emits when an allowance is set, by approve in the owner's own transaction or by permit in the spender's.

Source: 09-FO - An approval lets someone else spend your tokens.

Approval phishing. Tricking a user into signing an approval so that a spender, not the owner, can move their tokens. It is how a drainer takes funds.

Source: 12-FO - AADAPT names most of the typologies.

APT38. A North Korea-linked group that the FBI named, with Lazarus Group, as responsible for the $620M Ronin theft on 14 April 2022.

Source: 09-FO - Ronin: the FBI named Lazarus, not the chain.

Arbitrage. An MEV strategy that profits from price differences for the same asset across venues such as decentralized exchanges. Aliases: Arbitrage (MEV).

Source: 03-EF - The mempool is public, so pending trades show.

Arbitrum. An optimistic rollup on Ethereum, with a Security Council that can act in an emergency.

Source: 08-EF - One sequencer orders each rollup's trades.

ARIP. Nigeria's program for onboarding virtual asset service providers into a more formal regulatory process. Aliases: Accelerated Regulatory Incubation Program.

Source: 03-AS - Sub-Saharan Africa: Impact Signals.

Array. An ordered collection of values that may have fixed or dynamic length. Aliases: Array (Solidity), arrays, dynamic array, fixed-size array.

Source: 04-EF - Types: Reference Types.

Artificial Intelligence. Automated systems that analyze data, find patterns, and make or support decisions at scale. Aliases: AI, Artificial Intelligence (blockchain future).

Source: 05-AS - Two Powerful Technologies Collide.

Arweave. A storage protocol used for content-addressed token metadata or media with paid permanence assumptions. Aliases: Arweave (token metadata storage), Arweave TX ID.

Source: 06-EF - Metadata, IPFS, and the Pointer Problem.

ASIC. Specialized hardware built to perform mining computations far more efficiently than general-purpose devices. Aliases: ASIC (mining hardware), Application-Specific Integrated Circuit, purpose-built ASIC.

Source: 02-CM - Hardware Evolution and Centralization Pressures.

ASIC concentration. The concentration of mining power among operators with specialized application-specific mining hardware. Aliases: ASIC centralization, ASIC concentration (mining), specialized hardware concentration.

Source: 04-BF - Bitcoin Gold 51% Attack Discovery.

ASIC resistance. A mining algorithm's design to keep chips built for one algorithm from dominating, so ordinary processors can still compete. Aliases: ASIC-resistant, resistant to ASIC-based mining.

Source: 08-FO - Most cryptojacking mines Monero, on CPUs.

Asset swap. A token trade, often executed through smart contracts rather than a traditional intermediary. Aliases: Asset swap (DeFi), swap, swap assets, swapping.

Source: 01-AS - What Users Actually Do.

Asset-backed token. A token whose value depends on something held outside its contract, such as an issuer's reserves or a custodian's holdings. Aliases: asset-backed tokens, reference token, reference tokens.

Source: 05-EF - Same Token Standard, Different Promise.

Associated data. Metadata or headers that are authenticated by an AEAD scheme but intentionally left unencrypted. Aliases: AAD, additional authenticated data.

Source: 03-CP - From Confidentiality to Authenticated Encryption (AEAD).

Associated token account. The usual token account, whose address is derived from the owner's address and the mint's address. Aliases: ATA, associated token accounts.

Source: 08-EF - Solana keeps each token in its own account.

Asymmetric cryptography. Cryptography that uses a matched pair of keys: a secret private key, and a public key computed from it that anyone can use to check the private key's work. Aliases: public key cryptography, public-key cryptography.

Source: 04-CP - A private key is a number that controls coins.

Async PayJoin. A PayJoin variant, BIP 77, that replaces the receiver's own server with an untrusted directory server, so a receiver need not run one. Aliases: BIP 77, directory server.

Source: 06-FO - PayJoin hides a co-spend inside a payment.

Atomic transaction. A transaction that all succeeds or none of it does; a reentrancy drain or a flash loan's borrow, use and repayment happen as one. Aliases: atomic, atomicity.

Source: 10-FO - Reading a reentrancy: one call nested in itself.

Attack surface. The set of code paths, inputs, calls, and assumptions through which a contract can be misused or exploited. Aliases: Attack surface (smart contracts).

Source: 04-EF - From Bitcoin Script to Solidity.

Attacker contract. A contract the attacker deploys and calls the victim pool through; its code runs when the pool pays it. Aliases: attacker contracts.

Source: 10-FO - Reentrancy: the contract pays before it updates.

Attestation (Ethereum). A validator vote that supports blocks or checkpoints in Ethereum's Proof-of-Stake consensus process. Aliases: Attestation (Ethereum PoS vote), attestations, validator attestation, validator vote.

Source: 01-EF - Pending, then included, then final.

Attestation (Proof of Stake). A stake-weighted validator vote about a block, chain head, or checkpoint relationship. Aliases: Attestation (PoS vote), Attestation (Proof of Stake vote), attest, attestations, validator vote.

Source: 03-CM - The Committee: Attestation and Validation.

Attestation weight. The total validator voting power represented by attestations supporting a block or branch. Aliases: Attestation weight (Ethereum), attested stake weight, stake-weighted votes.

Source: 02-EF - The Ethereum Operational Framework.

Attester. A validator assigned to vote on block validity, chain head, or checkpoint relationships. Aliases: Attester (Ethereum PoS role), attesters, committee attester.

Source: 02-EF - The Ethereum Operational Framework.

Attribution. Linking on-chain activity to the service or actor behind it. Aliases: Attribution (blockchain forensics), actor attribution, entity attribution, identity attribution, real-world attribution.

Source: 01-FO - Step 4 ended Low: now name the service.

Attribution uncertainty. The remaining doubt when a transaction trace points to an address or cluster but not yet to a confirmed real-world actor. Aliases: Attribution uncertainty (blockchain forensics), attribution risk.

Source: 01-FO - Challenges in Forensic Analysis.

Auditability. The ability to inspect evidence that records or transactions followed expected rules and were not silently altered. Aliases: Auditability (blockchain use cases), Auditability (distributed ledgers), audit trail, auditable history, auditable record, public auditability.

Source: 04-DS - Case Study B — Public Sector: EBSI & Estonia (KSI){.smaller}.

Audited cryptographic library. A cryptographic implementation reviewed and maintained so applications can rely on safe defaults instead of custom security code. Aliases: OpenSSL, audited libraries, libsodium.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Austrian Economics. An economic framework that favors hard money, limited intervention, and the liquidation of malinvestment after credit booms.

Source: 04-IN - The Ideological Divide: Austrian vs. Keynesian Economics.

Authenticated encryption with associated data. An encryption design that provides confidentiality, integrity, and authentication in one API while allowing some data to be authenticated but not encrypted. Aliases: AEAD, authenticated encryption.

Source: 03-CP - From Confidentiality to Authenticated Encryption (AEAD).

Authentication path. The ordered list of sibling hashes needed to recompute a Merkle root from a target item. Aliases: Authentication path (Merkle proof), path.

Source: 02-CP - Merkle Proofs (Membership).

Authenticity. A security property that supports checking whether data or a message came from the claimed source. Aliases: authentication.

Source: 01-CP - What Is Cryptography?.

Authorization. The application's decision to grant a role, resource, or session after successful ownership or identity verification. Aliases: Authorization (token gating), access authorization.

Source: 06-EF - Token Gating: Access Flow.

Automated large-scale attack. An attack that uses automation to probe many contracts, protocols, wallets, or users at once. Aliases: Automated large-scale attack (blockchain security), automated attack, large-scale automated attack.

Source: 05-AS - Risks.

Automated market maker. A DEX mechanism where smart contracts price swaps from token reserves instead of matching buyers and sellers through an order book. Aliases: AMM, AMMs.

Source: 01-FO - Decentralized Exchanges (DEXs).

Automated provenance extraction. The automated parsing of on-chain events to recover asset type, amount, source chain, destination chain, and related movement history. Aliases: Automated provenance extraction (blockchain forensics), provenance extraction.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Autonomous economic agent. A software system that evaluates conditions and executes economic actions, such as on-chain transactions, without continuous human direction. Aliases: autonomous economic agents, software agent.

Source: 05-AS - Opportunities.

Autonomous system. A routing domain on the Internet, usually controlled by one organization, that announces paths for reaching networks. Aliases: AS, ASN, Autonomous system (Internet routing), autonomous system number.

Source: 04-BF - Partitioning Attacks.

Autonomy. A design priority favoring user control, censorship resistance, self-custody, permissionless access, and auditability. Aliases: Autonomy (monetary system design).

Source: 04-IN - Design Trade-Offs: Efficiency vs. Autonomy.

Availability. The fraction of time a service is operational during the period users need it. Aliases: Availability (distributed systems), uptime.

Source: 01-DS - Availability, MTTF, and MTTR.

Avalanche effect. The property that a tiny input change causes a large and unpredictable change in the hash output. Aliases: avalanche, avalanche behavior.

Source: 02-CP - Hash Function Properties.

avoid_reuse. A Bitcoin Core wallet flag, added in version 0.19 and off by default, that defends against dust attacks. Aliases: avoid_reuse flag.

Source: 06-FO - Dust is bait: a dust input is not a link.

B

b-money. Wei Dai's proposed decentralized money design that anticipated distributed accounting but did not become an operational consensus system. Aliases: b money.

Source: 01-BF - Lessons from Precursors.

Balance attack. A network-layer attack that partitions or delays validators so competing branches gain roughly balanced attestation weight. Aliases: Balance attack (Ethereum PoS), balancing attack.

Source: 02-EF - Balance Attack.

Balance-transfer invariant. The accounting rule that a valid transfer subtracts from one balance and adds to another without creating or losing units unexpectedly. Aliases: Balance-transfer invariant (token contract), conservation of balances.

Source: 04-EF - Example: Coin (Code, Part 2).

Ballot stuffing. A manipulation tactic where fake or coordinated accounts give each other high ratings to inflate reputation scores. Aliases: Ballot stuffing (reputation attack), rating inflation, vote flooding.

Source: 03-DS - PeerTrust: Handling Manipulation.

Bandwidth. The finite data-carrying capacity of network links, shared unevenly across participants. Aliases: Bandwidth (distributed systems), throughput.

Source: 01-DS - Fallacy #3: Bandwidth Is Infinite.

Bank for International Settlements. An international institution described as the central bank for central banks that promotes cooperation among monetary authorities. Aliases: BIS.

Source: 04-IN - The Modern Financial Architect: Central Banks.

Base fee. The protocol-determined part of an Ethereum transaction fee that is burned under EIP-1559. Aliases: Base fee (EIP-1559), Base fee (Ethereum fee market), Base fee (Ethereum gas), base-fee burn, base_fee, burned base fee.

Source: 01-EF - Gas prices the work; a failed call still pays.

Base58Check. An encoding with a checksum that turns address bytes into text; Tron applies it to the 0x41 prefix plus Ethereum's 20 bytes, giving a 34-character address starting with T. Aliases: Base58.

Source: 04-CP - A Tron address is an Ethereum address plus 41.

Basis vector. A vector used with others to generate the points of a lattice. Aliases: Basis vector (lattice cryptography), basis vectors.

Source: 05-CP - Lattices: Intuition Without the Math.

Beacon block. The outer consensus-layer block container that validators propose and attest to in post-Merge Ethereum. Aliases: Beacon block (Ethereum consensus layer), beacon blocks.

Source: 02-EF - Beacon Blocks and Execution Payloads.

Best-effort propagation. A message-spreading model where nodes try to relay information but delivery, ordering, and timing are not guaranteed. Aliases: Best-effort propagation (blockchain), best-effort delivery, best-effort transport.

Source: 01-DS - Fallacy #1: The Network Is Reliable.

BGP hijack. An Internet routing attack where traffic is redirected by falsely announcing network paths. Aliases: Border Gateway Protocol hijack, route hijack.

Source: 04-BF - Partitioning Attacks.

Bias. A systematic tendency that shapes what a source notices, values, highlights, or omits. Aliases: Bias (information evaluation), biases, source bias.

Source: 01-AS - Reading Critically.

Binding. The property that it should be infeasible to open one commitment as two different underlying messages. Aliases: Binding (commitment), binding commitment.

Source: 02-CP - Hashing as Commitment.

BIP 67. The standard that sorts a multi-sig's public keys so the same set of keys always gives the same redeem script and address. Aliases: BIP67, key sorting, sorted keys.

Source: 06-BF - Key order in the script carries no meaning.

BIP39. The standard that turns 128 to 256 random bits plus a checksum into a 12- to 24-word seed phrase from a 2048-word list, then runs the words through 2048 rounds of HMAC-SHA512 to make a 512-bit seed. Aliases: BIP 39, BIP39 mnemonic, mnemonic, mnemonic code.

Source: 05-BF - A seed phrase is the whole wallet as words.

BIP84. The derivation scheme for native SegWit (P2WPKH) addresses, whose paths start m/84'; it also publishes a test phrase and its first addresses. Aliases: BIP 84, purpose 84'.

Source: 05-BF - The path names coin, account, chain and index.

Birthday paradox. The principle that collisions among random-looking hash outputs become likely after about 2^(n/2) trials for an n-bit digest. Aliases: Birthday paradox (hashing), birthday attack, birthday bound.

Source: 02-CP - Math Spotlight: Birthday Paradox and Work Factors.

Bisq. A P2P market that, as of v1.2 (2019), locks both traders' coins in a 2-of-2 multisig escrow. A dispute goes to a mediator, then to arbitration once a trader publishes a time-locked transaction.

Source: 04-FO - P2P markets: two people trade, an escrow holds.

Bit gold. Nick Szabo's proposed work-based digital money concept that helped frame scarcity through costly computation. Aliases: bit gold.

Source: 01-BF - Lessons from Precursors.

Bit string. A sequence of 0s and 1s used as the standard representation of digital inputs, outputs, keys, and messages. Aliases: bit strings.

Source: 01-CP - Sets and Functions.

Bitcoin. The first widely adopted blockchain system and a reference point for understanding how blockchain designs became visible to the public. Aliases: Bitcoin (blockchain platform), Bitcoin (historical synthesis), Bitcoin (monetary alternative), Bitcoin (permissionless digital cash system), Bitcoin (technical system), Bitcoin network, bitcoin.

Source: 01-IN - By day 10 you can do nine things.

Bitcoin Core. The open-source reference implementation commonly used to operate a full Bitcoin node.

Source: 03-BF - Setting Up a Full Node.

Bitcoin Fog. A custodial mixer on Bitcoin, whose operator was sentenced to more than 12 years.

Source: 06-FO - A mixer pools coins and pays out others.

Bitcoin fork. A chain derived from Bitcoin's code or history that follows different rules or community choices after a split. Aliases: Bitcoin-derived chain, forked chain.

Source: 04-BF - Bitcoin Gold 51% Attack Discovery.

Bitcoin Gold. A 2017 fork of Bitcoin that changed its mining algorithm but kept far less hash rate and economic scale than Bitcoin. Aliases: BTG.

Source: 04-BF - Bitcoin Gold 51% Attack Discovery.

Bitcoin ledger. Bitcoin's public permissionless record of payments, authorized by digital signatures and validated by nodes following common rules. Aliases: Bitcoin's ledger.

Source: 04-DS - Case Study C — Bitcoin Ledger (Permissionless).

Bitcoin overflow bug. A 2010 Bitcoin implementation flaw that allowed a transaction in block 74638 to create about 184 billion BTC. Aliases: 2010 overflow bug, block 74638 overflow bug.

Source: 04-BF - Bitcoin Overflow Bug.

Bitcoin peer-to-peer network. The decentralized network of Bitcoin nodes that connect directly to relay transactions, blocks, and peer information without a central server. Aliases: Bitcoin P2P network.

Source: 03-BF - Bitcoin's Peer-to-Peer Network.

Bitcoin Pizza Day. The May 22, 2010 purchase in which Laszlo Hanyecz paid 10,000 BTC for two pizzas, creating an early real-world price signal. Aliases: 10,000 BTC pizza, pizza purchase.

Source: 01-BF - The 10,000 BTC Pizza.

Bitcoin protocol message. A structured message exchanged between Bitcoin nodes over long-lived TCP connections. Aliases: P2P message.

Source: 03-BF - Communication Protocol Basics.

Bitcoin Script. Bitcoin's stack-based scripting language for defining and validating the conditions required to spend transaction outputs. Aliases: Script.

Source: 03-BF - Introduction to Bitcoin Script.

Bitcoin whitepaper. The 2008 proposal by Satoshi Nakamoto for peer-to-peer electronic cash that does not depend on trusted financial intermediaries. Aliases: Bitcoin whitepaper (Bitcoin), Bitcoin: A Peer-to-Peer Electronic Cash System, whitepaper.

Source: 03-IN - Framing.

Bitget hack. The 24 September 2026 attack in which attackers reached the exchange Bitget's wallet servers and forged withdrawal commands, taking about $388 million across 11 blockchains by Bitget's count. Aliases: Bitget, Bitget theft.

Source: 08-EF - Bitget: the hack was at the exchange.

Blacklister role. In USDC's contract, a separate role, not the owner, that keeps the blacklist; each listing emits Blacklisted, and transfers from or to a listed address are refused. Aliases: Blacklisted event, blacklister.

Source: 05-EF - What a USDT freeze leaves on chain.

Blind signature. A cryptographic signing technique that lets a signer authorize a message without learning the message contents. Aliases: blind signatures.

Source: 03-IN - Framing.

Blinded block header. A block header and bid value forwarded without revealing the full transaction body before validator selection. Aliases: Blinded block header (Ethereum MEV), blinded bid, blinded header.

Source: 03-EF - Searchers, builders and relays set the order.

Blob data. Temporary Ethereum data space introduced to make rollup batch publication cheaper without expanding permanent state. Aliases: Blob data (Ethereum scaling), EIP-4844 blob data, blob, blobs.

Source: 04-AS - Layer 2s, Rollups, and Blobs.

Block. A batch of transactions with a header; the header holds the hash of the previous block, which links the block into the chain. Aliases: Block (blockchain data structure), blocks.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Block (Bitcoin). A package of transactions that can be appended to the Bitcoin chain after satisfying proof-of-work and validation rules. Aliases: Bitcoin block (data structure), block, blocks.

Source: 01-BF - Primitive 2: The Public Ledger via P2P Networking.

Block assembly. The process where a miner selects waiting transactions and packs them into a block, whose first transaction, the coinbase, pays the miner. Aliases: candidate block assembly.

Source: 02-IN - A miner packs waiting payments into a block.

Block body. The part of a block that contains the transactions or state updates being recorded in the ledger. Aliases: block transactions, body.

Source: 02-CM - Block Body Anatomy.

Block builder. A specialist that combines ordinary transactions and searchers' bundles into full blocks, choosing their order. Aliases: Block builder (Ethereum block building), Block builder (Ethereum), builder, builders.

Source: 03-EF - A DEX trade is visible before it lands.

Block cipher. A symmetric cipher that transforms fixed-size blocks of data under a secret key. Aliases: block ciphers.

Source: 03-CP - Building Blocks of Block Ciphers.

Block header. The compact metadata portion of a blockchain block that can include a Merkle root committing to the block's transactions. Aliases: Block header (Bitcoin), Block header (blockchain), block headers, header.

Source: 02-CP - Where You'll See Them (Non-Exhaustive).

Block header hash. The hash of a block's header that miners compare to the difficulty target when proving work. Aliases: Block header hash (Bitcoin mining), block hash, header hash.

Source: 01-BF - Primitive 3: Consensus via Proof-of-Work.

Block interval. The intended average time between new blocks in a chain. Aliases: block time, target interval.

Source: 02-CM - Difficulty Adjustment: Control-Loop Intuition.

Block linking. The use of a previous block hash inside a later block so changing earlier data breaks later references. Aliases: Block linking (hashing), hash linking, hash-linked blocks, hash-linked records.

Source: 04-AS - Hashing in Blockchain Practice.

Block log. The durable ordered ledger history that records accepted blocks, including transactions later marked valid or invalid. Aliases: Block log (Hyperledger Fabric), block history, ledger history.

Source: 02-PC - From Permissioned Concepts to Fabric Terms.

Block propagation. The process by which newly found blocks spread across the peer-to-peer network. Aliases: Block propagation (Bitcoin network), block relay.

Source: 02-CM - Block Propagation and Stale/Orphan Blocks.

Block proposer (Ethereum). The validator selected for a slot to build and broadcast a candidate block. Aliases: Block proposer (Ethereum PoS role), Block proposer (Ethereum PoS), proposer, slot proposer.

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Block proposer (Proof of Stake). The validator selected for a slot to choose transactions, build a candidate block, and propose it to the network. Aliases: Block proposer (PoS role), leader, proposer.

Source: 03-CM - The Block Proposer: Selection and Building.

Block relay. The process by which newly found blocks spread among Bitcoin nodes and miners. Aliases: Block relay (Bitcoin network), block broadcast, block propagation.

Source: 04-BF - Partitioning Attacks.

Block reward. The miner compensation for finding a valid block, made up of newly issued bitcoin and transaction fees. Aliases: Block reward (Bitcoin), mining reward.

Source: 01-BF - Incentives: Making Honesty Profitable.

Block size. The amount of transaction data a block can carry under the protocol's limits. Aliases: block capacity.

Source: 02-CM - Parameter Choices and Tradeoffs.

Block space. The limited capacity inside blocks that users compete for when they want transactions included. Aliases: Block space (Bitcoin), blockspace.

Source: 01-BF - Monetary Policy.

Block subsidy. Newly issued coins paid to the miner of a valid block as part of the block reward. Aliases: Block subsidy (Bitcoin monetary policy), Block subsidy (Bitcoin), block reward, block reward subsidy, mining subsidy, subsidy.

Source: 02-CM - Security Model and Economics.

Block weight. Bitcoin's consensus measure for block size limits, capped at 4 million weight units. Aliases: Block weight (Bitcoin), weight units.

Source: 02-BF - Fees are priced per virtual byte, not per BTC.

Block withholding. A pool attack where a miner submits valid shares but hides full valid blocks, reducing the pool's rewards. Aliases: Block withholding (mining pool), pool block withholding.

Source: 02-CM - Block Withholding in Pools.

Blockchain. A shared technical system for recording and verifying data without relying only on a central trusted authority. Aliases: blockchains.

Source: 01-IN - Start by saying what cases you work.

Blockchain abstraction layer. Software or services that hide raw blockchain operations behind wallets, apps, exchanges, or other user-facing tools. Aliases: abstraction layer, interface layer.

Source: 01-AS - What Does a User Actually Interact With?.

Blockchain address. A public identifier built from a public key, usually by hashing it; each chain hashes and encodes it differently. Aliases: account address, address, addresses, wallet address.

Source: 04-CP - A Bitcoin address is a hash of the public key.

Blockchain forensics. Analysis of blockchain transactions, addresses, and patterns to connect on-chain activity with real-world actors or behavior. Aliases: Blockchain forensics (forensics), Blockchain forensics (investigative analysis), blockchain analytics, forensics, on-chain forensics.

Source: 03-IN - Promises vs. Reality.

Blockchain network. A distributed network of machines that propagate and validate blockchain state without a single central operator. Aliases: blockchain P2P network.

Source: 01-DS - Distributed Computing and Blockchain.

Blockchain stack. The layered combination of cryptography, networking, consensus, ledger structures, and applications that makes a blockchain system work. Aliases: blockchain layers, technical stack.

Source: 04-AS - The Blockchain Stack at a Glance.

Blockchain Trilemma. The design trade-off among decentralization, security, and scalability in blockchain systems. Aliases: decentralization-security-scalability trilemma.

Source: 04-IN - The Engineer's Dilemma: The Blockchain Trilemma.

Blockchain vulnerability. A weakness in blockchain software, protocols, networks, or assumptions that can create security or reliability risk. Aliases: vulnerabilities.

Source: 01-IN - By day 10 you can do nine things.

BLS signature. A signature scheme Ethereum uses because many validator signatures can be aggregated into one compact proof. Aliases: BLS, BLS signature (Ethereum), Boneh-Lynn-Shacham signature.

Source: 02-EF - Slots and Epochs.

BOINC. A generalized volunteer-computing platform that coordinates work distribution, result upload, validation, and credit. Aliases: Berkeley Open Infrastructure for Network Computing.

Source: 01-DS - Volunteer Computing Pipeline: BOINC.

Bootstrap. The process by which new participants enter a trust system and begin earning reputation from low or neutral starting trust. Aliases: Bootstrap (reputation systems), bootstrapping, newcomer handling.

Source: 03-DS - Properties of Reputation Systems.

Bootstrapping. An FHE technique that refreshes noisy ciphertexts so more encrypted operations can be performed. Aliases: Bootstrapping (FHE), noise refresh, noise reset.

Source: 05-CP - Fully Homomorphic Encryption (FHE).

Bored Ape Yacht Club. An Ethereum NFT collection that became a membership club, event network, artwork license system, and entertainment brand. Aliases: BAYC, Bored Ape Yacht Club (membership NFT project).

Source: 06-EF - Case Study: Bored Ape Yacht Club.

Branching factor. The number of child hashes combined at each internal node of a Merkle tree. Aliases: Branching factor (Merkle tree), binary tree, k-ary tree.

Source: 02-CP - Design Choices & Variants.

Bretton Woods. The post-World War II monetary system in which the U.S. dollar was convertible to gold and other major currencies were pegged to the dollar. Aliases: Bretton Woods system.

Source: 04-IN - The Turning Point: Ending Bretton Woods.

Bribery. An attack where an adversary pays validators to vote for, build, or support a chosen block or chain. Aliases: Bribery (PoS attack), validator bribery.

Source: 03-CM - Censorship, Bribery, and Eclipse.

Bridge. Infrastructure that represents or moves assets or messages across chains using contracts, custodians, relayers, or other trust assumptions. Aliases: Bridge (cross-chain infrastructure), Bridge (cross-chain movement), Bridge (cross-chain), blockchain bridge, bridges, cross-chain bridge.

Source: 06-EF - Centralization Chokepoints.

Broadcast. The act of sending a signed transaction or block outward to peers so the wider network can learn about it. Aliases: Broadcast (blockchain network), broadcasting, transmission.

Source: 02-IN - Every payment takes the same five steps.

Broadcast storm. A network condition where repeated or duplicate broadcasts consume so much capacity that useful traffic is crowded out. Aliases: broadcast storms.

Source: 01-DS - Case Study: Gnutella & "Flooding".

Browser mining. Cryptojacking by a mining script that runs in a website visitor's browser without their consent; it stops when the visitor leaves the site. Aliases: in-browser mining.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

BTC-e. An internet-based exchange, fined $110 million by FinCEN in 2017, whose accounts needed only a username, password and email address. Aliases: BTC-e (case study), Bitcoin Exchange.

Source: 01-FO - BTC-e: Weak KYC Made the Cluster Actionable.

Builder. A participant that assembles candidate blocks from pending transactions for possible proposal by validators. Aliases: Builder (Ethereum block building), block builder, builders.

Source: 02-EF - Censorship.

Burn event. The log a Uniswap v2 pool writes when liquidity is withdrawn and LP tokens are burned. Aliases: Burn log.

Source: 07-EF - A swap leaves Swap and Sync logs.

Bybit hack. North Korea's theft of about $1.5 billion in crypto from the exchange Bybit on or about 21 February 2025, which Elliptic calls the largest theft in history. Aliases: Bybit, Bybit theft.

Source: 07-EF - Bybit: stolen tokens became ETH first.

Byte-exact hashing. Hashing that depends on the exact serialized bytes, field order, and encoding specified by the protocol. Aliases: bit-for-bit deterministic hashing.

Source: 02-CM - Block Header Anatomy: What Gets Hashed.

Bytecode. The low-level instructions produced from smart contract languages and executed by the EVM. Aliases: Bytecode (EVM), Bytecode (Ethereum), EVM bytecode, compiled bytecode, contract bytecode, unique bytecode.

Source: 01-EF - Every node runs the same program: the EVM.

Byzantine agreement. A protocol problem where honest participants must settle on one value even while traitors send conflicting values.

Source: 05-IN - Mini-Activity: Byzantine Agreement.

Byzantine fault. A failure where a node behaves arbitrarily, deceptively, or maliciously, including sending conflicting messages to different peers. Aliases: Byzantine faults, arbitrary fault, malicious fault.

Source: 05-IN - Byzantine Faults.

Byzantine fault threshold. The maximum number of Byzantine participants a protocol's quorum math is designed to tolerate. Aliases: Byzantine fault threshold (BFT quorum), fault threshold, t.

Source: 01-CM - Open chains buy openness with slower finality.

Byzantine Fault Tolerance. The ability of a protocol to keep reaching correct agreement even when some participants behave Byzantine. Aliases: BFT, BFT-based systems, Byzantine Fault Tolerance (consensus), Byzantine Fault Tolerant, Byzantine Fault Tolerant consensus protocol.

Source: 05-IN - Byzantine Faults.

Byzantine Generals Problem. The problem of getting loyal distributed actors to agree on one plan even when some actors or channels are faulty or malicious. Aliases: BGP, Byzantine Generals Problem (Bitcoin), Byzantine agreement problem.

Source: 05-IN - Byzantine Generals Problem.

Byzantine-fault tolerance. A consensus family designed to remain safe when some known participants may lie, equivocate, or act maliciously. Aliases: BFT, BFT orderer, Byzantine fault tolerance, Byzantine-fault tolerance (Hyperledger Fabric ordering).

Source: 01-CM - Open chains buy openness with slower finality.

bZx attack. The 18 February 2020 oracle-manipulation attack in which one transaction borrowed 7,500 ETH, raised sUSD's price on two thin exchange pools, borrowed against that price on bZx and repaid, keeping 2,381.41 ETH. Aliases: bZx.

Source: 10-FO - bZx, Feb 2020: borrow, move a price, borrow.

C

C2PA. A content provenance standard that helps creators and platforms preserve media origin and edit history. Aliases: C2PA (content provenance), C2PA standard, Coalition for Content Provenance and Authenticity.

Source: 05-AS - Opportunity: Authenticity and Provenance.

Calldata. The read-only input bytes supplied by the caller, including encoded function arguments. Aliases: Calldata (Solidity data location), call data, transaction input payload.

Source: 04-EF - Types: Reference Types.

Candidate block. A proposed block assembled by a miner from pending transactions before a valid proof-of-work has been found. Aliases: Candidate block (Bitcoin mining), candidate blocks.

Source: 01-BF - Primitive 3: Consensus via Proof-of-Work.

Canonical block. A block treated as the accepted branch by the protocol after receiving enough stake-weighted support. Aliases: Canonical block (PoS), canonical acceptance.

Source: 03-CM - The Committee: Attestation and Validation.

Canonical chain. The ledger branch that honest nodes treat as the accepted history under the consensus rule. Aliases: Canonical chain (blockchain), accepted chain, canonical branch, canonical history, main chain.

Source: 01-CM - Consensus decides which history we all accept.

Canonical history. The transaction history currently accepted by nodes under Bitcoin's chain-selection and validation rules. Aliases: Canonical history (Bitcoin), Canonical history (Ethereum), accepted history, authoritative history, canonical chain, canonical head, shared history, valid history.

Source: 01-BF - Bitcoin, Simplified.

Capability. In the Diamond Model, the tools and techniques used in an event. An AADAPT technique fits here. Aliases: capability vertex.

Source: 12-FO - The Diamond: four features of one event.

Capital at risk. Validator funds locked under protocol rules that can be lost or delayed if the validator misbehaves or fails duties. Aliases: Capital at risk (PoS), bonded capital, locked collateral, staked capital.

Source: 03-CM - Orientation: Proof of Work vs Proof of Stake.

Capital Controls. Legal restrictions on moving money into, out of, or within a financial system during stress or by policy choice.

Source: 04-IN - The Problem of Control: Chokepoints & Censorship.

Capital flow control. State efforts to restrict, monitor, or steer money moving across borders or out of the domestic financial system. Aliases: capital controls.

Source: 02-AS - Why Governments Care.

Cash-out. Turning cryptocurrency into money or goods the criminal can use, at a service such as an exchange, an OTC broker or a darknet market. Aliases: cash out, cashing out.

Source: 03-FO - A ransom cashes out at a service.

Casper-FFG. Ethereum's finality gadget that justifies and finalizes checkpoints when enough total stake votes consistently across epochs. Aliases: Casper Friendly Finality Gadget, Casper-FFG (Ethereum finality), FFG, Friendly Finality Gadget.

Source: 02-EF - Gasper: Fork Choice + Finality.

CBDC pilot. A limited operational test of a CBDC design before full-scale production deployment. Aliases: pilot.

Source: 02-AS - Central Bank Digital Currencies (CBDCs).

CCTP. Circle's own bridge for USDC, an issuer's bridge. Aliases: Circle's CCTP.

Source: 08-EF - Bitget: bridges moved it faster than freezes.

Censorship. The intentional filtering, dropping, or blocking of messages or transactions by a participant or adversary. Aliases: Censorship (Ethereum), Censorship (PoS attack), Censorship (blockchain), Censorship (distributed systems), soft censorship, transaction censorship.

Source: 05-IN - Why Failures Become Security Risks.

Censorship resistance. A system property that makes it difficult for a party to block, freeze, or prevent valid transactions. Aliases: Censorship resistance (Bitcoin network), Censorship resistance (blockchain network), Censorship resistance (decentralization), censorship-resistant.

Source: 03-IN - Promises vs. Reality.

Central authority. A single organization or service that can vouch for participants, enforce rules, or resolve disputes for the network. Aliases: Central authority (P2P trust), central referee.

Source: 03-DS - What Is "Trust" in P2P?.

Central Bank. An institution that manages a fiat monetary system through policy, emergency liquidity, and payment-system oversight. Aliases: central banks.

Source: 04-IN - The Modern Financial Architect: Central Banks.

Central bank digital currency. A digital form of central bank money issued or governed by a monetary authority. Aliases: CBDC, CBDCs, central bank digital currencies, central bank digital currency.

Source: 03-IN - Why This History Matters.

Central server. A server that provides a key coordination function for a system, such as indexing files or routing users to resources. Aliases: central index.

Source: 03-IN - P2P Normalizes Decentralization.

Centralization. The concentration of control, speed, or privacy decisions in fewer hands or a single owner. Aliases: Centralization (ledger design), centralized design.

Source: 04-DS - Trade-offs.

Centralization chokepoint. A dependency above or around a decentralized base layer where control, access, or failure risk concentrates in a small number of operators. Aliases: centralization chokepoints, centralized chokepoint.

Source: 06-EF - Centralization Chokepoints.

Centralized architecture. A system design where one authority coordinates work, controls state, and becomes the main trust dependency.

Source: 05-IN - Centralized vs. Distributed Architectures.

Centralized database. A database controlled by a trusted owner or administrator who can define, update, and manage the authoritative record. Aliases: admin-trusted database, central database.

Source: 04-DS - Distributed Ledger vs. Centralized Database (Compare/Contrast).

Centralized exchange. An exchange that needs an account and a listing for the asset, and can freeze deposits; after a deposit to its address the record is off-chain. Aliases: CEX, centralized exchanges.

Source: 07-EF - Where funds can leave a chain.

Centralized index. A central service that stores searchable metadata or routing information for many participants. Aliases: central index, centralized discovery.

Source: 01-DS - Client–Server: Centralized Index and Control.

Centralized issuer. A single organization that creates, backs, or administers a digital money system. Aliases: central issuer, centralized company.

Source: 01-BF - Lessons from Precursors.

Centralized system. A system organized around a clear control point that simplifies coordination but creates a point of trust or failure. Aliases: Centralized system (architecture), centralized architecture.

Source: 04-AS - Centralized vs Distributed Architectures.

Centralized trust. A trust model where security depends heavily on one authority or a small set of authorities behaving correctly.

Source: 05-IN - What is "trust" in digital systems?.

Centralized trust model. A model where institutions verify identity, keep records, enforce rules, and resolve disputes on behalf of users. Aliases: centralized trust, digital trust.

Source: 03-IN - Digital Trust: The Centralized Model.

Certificate Authority. The identity service that enrolls users, peers, admins, and organizations by issuing certificates and keys. Aliases: CA, Certificate Authority (Hyperledger Fabric), Fabric CA.

Source: 02-PC - From Permissioned Concepts to Fabric Terms.

ChaCha20. A modern stream cipher that generates a fast keystream for symmetric encryption. Aliases: ChaCha20 (symmetric cipher).

Source: 03-CP - What Is Symmetric Cryptography?.

Chain head. The block a node currently treats as the tip of the best chain to extend. Aliases: Chain head (PoS fork choice), current head, head.

Source: 03-CM - Fork Choice Rules.

Chain hopping. Moving assets across different blockchains to force investigators into a new ledger, asset format, or tracing environment. Aliases: Chain hopping (blockchain forensics), Chain hopping (obfuscation), asset hopping, cross-chain hopping.

Source: 01-FO - Decentralized Exchanges (DEXs).

Chain of ownership. The sequence of signed transfers that links a coin's prior transaction to its current authorized owner. Aliases: Chain of ownership (Bitcoin), chain of signatures, signature chain.

Source: 01-BF - Primitive 1: Ownership via Digital Signatures.

Chain reorganization. A change in which a competing branch replaces recent blocks that were previously treated as part of the accepted chain. Aliases: Chain reorganization (Bitcoin), block reorganization, ledger rewrite, reorg.

Source: 04-BF - What Can Consensus Attacks Achieve?.

Chain-reaction analysis. Striking an output known to be spent out of every other ring where it appears as a decoy, which can leave only the real input. Aliases: chain reaction.

Source: 07-FO - Old Monero leaked: zero decoys, newest is real.

Chaincode. Fabric's term for application code that implements business rules and proposes ledger updates during transaction simulation. Aliases: Chaincode (Hyperledger Fabric), Fabric smart contract, smart contract.

Source: 02-PC - Fabric-Specific Concepts To Watch For.

Chaincode definition. The channel-level agreement on a chaincode's name, version, sequence number, endorsement policy, and related settings. Aliases: Chaincode definition (Hyperledger Fabric), chaincode approval definition, contract definition.

Source: 02-PC - Chaincode Lifecycle.

Chaincode event. A notification emitted by chaincode so external observers can react to a business action after the transaction is processed. Aliases: Chaincode event (Hyperledger Fabric), Fabric event, event.

Source: 02-PC - Chaincode Example: CreateAsset.

Chaincode lifecycle. The governance process for installing a chaincode package, approving a channel definition, and committing that definition before invocation. Aliases: Chaincode lifecycle (Hyperledger Fabric), Fabric chaincode lifecycle, lifecycle.

Source: 02-PC - Chaincode Lifecycle.

Chaincode package. The deployable artifact containing smart-contract code and metadata needed to run chaincode on endorsing peers. Aliases: Chaincode package (Hyperledger Fabric), deployable chaincode artifact, package.

Source: 02-PC - Chaincode Lifecycle.

Chaining state. The intermediate hash state that is passed from one compression round to the next. Aliases: Chaining state (hashing), final internal state, internal state.

Source: 02-CP - Merkle–Damgård: How SHA-256 is Built.

Challenge period. The roughly seven days an optimistic rollup waits before a withdrawal back to Ethereum completes, during which anyone may challenge a batch with a fraud proof. Aliases: challenge window, withdrawal period.

Source: 08-EF - Optimistic waits a week; ZK waits for a proof.

Change address. The sender-controlled address that receives a transaction's change output, often a new one. Aliases: Change address (Bitcoin).

Source: 02-BF - Reading Alice's payment, field by field.

Change Address Detection. A heuristic process for identifying which output in a UTXO transaction likely returns leftover value to the sender. Aliases: Change Address Detection (Bitcoin forensics), change detection, change output detection.

Source: 02-FO - Every hop in your trace names its heuristic.

Change output. A new transaction output that returns leftover value to the sender because selected UTXOs must be spent in full. Aliases: Change output (Bitcoin transaction), Change output (Bitcoin), change.

Source: 02-BF - A trace is read one transaction at a time.

Change undetermined. The call an analyst makes when no clue shows which output of a transaction is change. Both outputs are followed, and that is a correct answer. Aliases: no call on change.

Source: 03-FO - Each hop gets a heuristic and a confidence.

Channel. A ledger or communication partition that limits which participants can see or interact with a subset of data. Aliases: Channel (Hyperledger Fabric), Channel (permissioned ledger privacy), channels, partition, partitions.

Source: 01-PC - The trace stops at the operator's edge.

Channel configuration. The governance data that defines channel membership, policies, administrative rules, and related configuration for a Fabric channel. Aliases: Channel configuration (Hyperledger Fabric), channel config, channel policy configuration.

Source: 02-PC - From Permissioned Concepts to Fabric Terms.

Channel MSP definition. The channel-level MSP information that tells a channel which organizations exist and which signatures satisfy their trust rules. Aliases: Channel MSP definition (Hyperledger Fabric), channel MSP, channel MSP definitions.

Source: 02-PC - What is an MSP?.

Checkpoint (Ethereum). An epoch-boundary block used as a target for finality votes. Aliases: Checkpoint (Ethereum finality), checkpoint link, epoch checkpoint.

Source: 02-EF - Gasper: Fork Choice + Finality.

Checkpoint (Proof of Stake). A block at an epoch boundary used as an anchor for finality votes. Aliases: Checkpoint (PoS finality), Checkpoint (Proof of Stake finality), checkpoints, epoch checkpoint, finality anchor.

Source: 03-CM - Epochs and Checkpoints: Organizing Slots.

Child DAO. The new DAO a split moved a holder's ether into; its 27-day creation window kept the ether from moving.

Source: 10-FO - The DAO: one split called many times over.

Chinese-language laundering networks. Laundering networks that Chainalysis reported, as of January 2026, consistently launder over 10% of pig-butchering proceeds while centralized-exchange use declines. Aliases: Chinese-language money laundering networks.

Source: 12-FO - Scam cash-out: away from what can freeze.

Chinese-language money laundering network. Laundering networks that run OTC services, collecting many small deposits into wholesale amounts that can re-enter legitimate finance, as of January 2026. Aliases: Chinese-language laundering network.

Source: 04-FO - Some OTC desks launder: small in, large out.

CHIPS. A private US dollar clearing and settlement system used for large-value payments among banks. Aliases: CHIPS (payment settlement), Clearing House Interbank Payments System.

Source: 02-AS - Global Payment Infrastructure: Overview.

Chokepoint. A central place in a financial network where an authority or intermediary can block, freeze, or redirect activity. Aliases: Chokepoint (financial system), central chokepoint, chokepoints, single point of control.

Source: 04-IN - The Problem of Control: Chokepoints & Censorship.

Chord. A ring-based distributed hash table where hashed peers and keys are arranged in identifier order and lookups follow shortcut links. Aliases: Chord (DHT).

Source: 02-DS - DHT Example: Chord.

Chosen collision. A collision attack where the attacker deliberately constructs different inputs that hash to the same digest. Aliases: Chosen collision (hashing), chosen-prefix collision, crafted collision.

Source: 05-CP - Hash Fragility: Collision Resistance Isn't Forever.

Churn. The continual joining, leaving, rebooting, moving, and reconnecting of nodes in a distributed network. Aliases: Churn (P2P membership), Churn (P2P networking), Churn (network topology), membership churn, peer churn.

Source: 01-DS - Fallacy #5: Topology Doesn't Change.

Cipher Block Chaining. A block-cipher mode that XORs each plaintext block with the previous ciphertext block before encryption. Aliases: CBC.

Source: 03-CP - Modes of Operation Overview.

Cipher Feedback. A legacy mode that turns a block cipher into a self-synchronizing stream-like construction. Aliases: CFB.

Source: 03-CP - Modes of Operation Overview.

Ciphertext. The encrypted output that should hide the plaintext from anyone without the correct key or keystream. Aliases: Ciphertext (FHE), ciphertexts, encrypted data.

Source: 03-CP - Infinite-length One Time Pad.

CIPS. China's cross-border yuan payment system that combines messaging and settlement for renminbi transactions. Aliases: CIPS (cross-border payment system), Cross-Border Interbank Payment System.

Source: 02-AS - Global Payment Infrastructure: Overview.

Classical consensus. A consensus family that usually assumes known participants, message exchange, quorum rules, and explicit fault models. Aliases: Classical consensus (consensus family), permissioned consensus.

Source: 04-AS - Classical vs Economic Consensus.

Classified holdings. Classified reporting that may bear on an assessment. In an unclassified class it is taught as a process only: it is judged with the same factors, counted as one line, and checked for echoes like any other source. Aliases: classified reporting.

Source: 14-FO - Classified holdings: the process only.

Client diversity. The presence of multiple independent software clients participating in the same blockchain network. Aliases: Client diversity (Ethereum network health), Client diversity (Ethereum operations), Client diversity (blockchain), consensus client diversity, diverse clients, execution client diversity, multi-client diversity.

Source: 01-DS - Fallacy #8: The Network Is Homogeneous.

Client-server architecture. A centralized design where clients request data or service from a server or server cluster that holds authority. Aliases: client-server, client-server model.

Source: 01-DS - Client–Server: Centralized Index and Control.

Clipboard hijacker. Malware on the victim's computer that swaps an address the victim copies for the thief's, so the victim signs a real payment to the wrong recipient. Aliases: clipper.

Source: 09-FO - Four ways out of a wallet look different.

Cloud cryptojacking. Cryptojacking in a stolen or fraudulently opened cloud account, where the victim pays the cloud bill and that bill far exceeds the coins the attacker earns.

Source: 08-FO - In the cloud the victim pays the bill.

CNAME. A DNS alias: an attacker's subdomain that points to a known mining pool, so a blocklist of pool names misses it. Aliases: DNS alias, domain alias, domain aliases.

Source: 08-FO - The wallet in the sample is the pivot.

Code as speech. The legal argument that source code can be expressive activity protected by the First Amendment.

Source: 03-IN - The Crypto Wars: U.S. Gov Position (1990s).

Code-based cryptography. A post-quantum cryptography family based on hard decoding problems from error-correcting codes.

Source: 05-CP - Post-Quantum Cryptography: The New Toolbox.

Coercive Extraction. Gaining value through theft, extortion, or forced illicit transfer rather than normal market participation.

Source: 03-AS - North Korea: Overview.

Coercive Revenue Generation. A state strategy that turns theft, extortion, or illicit transfer into funding for sanctioned activity. Aliases: coercive crypto revenue.

Source: 03-AS - North Korea: Overview.

Coin type. The derivation-path level that names the coin: 0' for Bitcoin and 1' for testnet. Aliases: coin_type.

Source: 05-BF - The path names coin, account, chain and index.

Coinbase transaction. The first transaction in a Bitcoin block, which creates new bitcoin and pays the miner according to protocol rules. Aliases: Coinbase transaction (Bitcoin mining), Coinbase transaction (Bitcoin), coinbase, coinbase tx.

Source: 01-BF - Incentives: Making Honesty Profitable.

Coincidence of Wants. The barter problem where each trader must want what the other trader is offering at the same time.

Source: 04-IN - What is Money? A Social Technology.

CoinJoin. A collaborative transaction technique that combines multiple users' inputs and outputs to make ownership links harder to trace. Aliases: CoinJoin (Bitcoin privacy technique), CoinJoin (Bitcoin privacy).

Source: 04-BF - Mitigating Network Deanonymization.

CoinJoin coordinator. The server that gathers users into each CoinJoin round. Enforcement in 2024 ended the main coordinators, not the CoinJoin technique. Aliases: coordinator, coordinator server.

Source: 06-FO - Equal outputs mark a CoinJoin on sight.

Cold wallet. A wallet whose key is not connected to the internet, such as an air-gapped computer or a hardware wallet, built to hold high-value funds securely at rest; a person must act, such as pressing a button, to sign. Aliases: cold storage, cold wallets.

Source: 05-BF - Hot or cold says where the key lives.

Collateral. Assets posted to back trading positions and payouts in a market. Aliases: Collateral (prediction markets), market collateral.

Source: 01-AS - Prediction Markets Case: Polymarket.

Collateralized Debt Obligation. A complex financial product built from pools of debt instruments, including mortgage-related securities. Aliases: CDO, collateralized debt obligations.

Source: 04-IN - Case Study: The 2008 Global Financial Crisis.

Collector buy-back. A real sale followed by a later purchase back at a new price. Time between the sales and outside funding separate it from a wash trade. Aliases: buy-back, buy-backs.

Source: 11-FO - Look-alikes that are not wash trades.

Collision resistance. The property that it should be infeasible to find any two distinct inputs with the same digest.

Source: 02-CP - Hash Function Properties.

Collusion. A coordinated attack where peers rate each other positively, or rate competitors negatively, to distort reputation scores. Aliases: Collusion (P2P reputation threat), collusion ring, coordinated praise.

Source: 03-DS - Properties of Reputation Systems.

Collusion resistance. A trust system's ability to limit the effect of peers that coordinate ratings to boost themselves or harm others.

Source: 03-DS - EigenTrust: Anchors and Collusion Resistance.

Commission fault. A failure where a node remains active but returns an incorrect or corrupted result. Aliases: commission faults.

Source: 05-IN - Commission Faults.

Commit. The step where a valid ordered transaction is written to the ledger and its effects update world state. Aliases: Commit (Hyperledger Fabric), committing, valid commit.

Source: 02-PC - One Invoke: Proposal to Commit.

Commit phase. The phase where enough matching votes allow the protocol to accept a value as decided for that position. Aliases: Commit phase (consensus), commit, committed.

Source: 01-CM - Open chains buy openness with slower finality.

Commitment. A compact cryptographic reference that can later prove whether a specific piece of data matches what was recorded. Aliases: Commitment (ledger verification), commitments, hash commitment.

Source: 01-PC - The trace stops at the operator's edge.

Committee (Ethereum). A randomly assigned group of validators that attests to block validity, chain head, and checkpoints. Aliases: Committee (Ethereum PoS), committees, validator committee.

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Committee (Proof of Stake). A randomly assigned subset of validators that checks a proposed block and votes on it for a slot or epoch. Aliases: Committee (PoS validator group), Committee (Proof of Stake group), committees, validator committee.

Source: 03-CM - The Committee: Attestation and Validation.

Committing peer. A peer acting in the commit role by receiving ordered blocks, rechecking policy and state-version rules, and updating ledger state. Aliases: Committing peer (Hyperledger Fabric), committer, committers, committing peers.

Source: 02-PC - The Big Picture: End-to-End Fabric Path.

Commodity Money. Money whose value is tied to the useful or scarce physical good itself, such as salt, cattle, or shells.

Source: 04-IN - A Brief History of Money's Form.

Common-input ownership heuristic. A heuristic that treats multiple inputs to one transaction as likely controlled by the same actor. Aliases: CIOH, Common-input ownership heuristic (Bitcoin forensics), common-input ownership.

Source: 04-AS - Graph Thinking and Heuristics.

Common-input-ownership heuristic. A tracing rule of thumb that assumes inputs spent together in one transaction are likely controlled by the same entity. Aliases: CIOH, Common-Input-Ownership Heuristic (Bitcoin forensics), Common-input-ownership heuristic (blockchain forensics), common input ownership, common-input heuristic, multi-input heuristic.

Source: 01-FO - Challenges in Forensic Analysis.

Compact block relay. A block propagation protocol that sends compact block information and only transmits full transactions the peer is missing. Aliases: Compact block relay (Bitcoin), cmpctblock, compact blocks.

Source: 03-BF - Block Propagation.

Completeness. The property that an honest prover can convince the verifier when the statement being proved is true. Aliases: Completeness (zero-knowledge proofs).

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

Composability. The ability for smart contracts to call and combine with other contracts in a single coordinated workflow. Aliases: Composability (smart contracts), atomic workflows, contract composability.

Source: 01-EF - A contract is code plus storage at an address.

Compression function. A fixed-size function that mixes one message block with the current hash state to produce the next state. Aliases: Compression function (hashing), compress.

Source: 02-CP - Merkle–Damgård: How SHA-256 is Built.

Compromised node. A node that has been taken over or corrupted so it may produce bad data or act against the protocol. Aliases: compromise, hijacked node.

Source: 05-IN - Why Failures Become Security Risks.

Computational hardness. The assumption that some problems are so difficult that no efficient known algorithm can solve them at relevant sizes. Aliases: hardness, intractability.

Source: 01-CP - Computational Hardness.

Computational irreversibility. The condition where reversing a transaction becomes infeasible because an attacker would need to redo and surpass accumulated proof-of-work. Aliases: Computational irreversibility (Bitcoin), computationally impractical to reverse, practical irreversibility.

Source: 01-BF - Bitcoin: The Mission Statement.

Computationally infeasible. A security claim that an attack is possible in theory but too expensive for realistic attackers using known methods. Aliases: infeasible.

Source: 02-CP - Pre-Image Resistance.

Compute Hijacking. The ATT&CK sub-technique, T1496.001, for using a victim's computing power, one common purpose being to validate cryptocurrency transactions and earn coins. Aliases: ATT&CK T1496.001, T1496.001.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

Concatenation. The operation of joining byte strings in order before hashing them together. Aliases: Concatenation (hash input), parallel operator, ||.

Source: 02-CP - What Is a Merkle Tree?.

Confidence word. The word on the course's scale that an assessment states to say how sure it is; an inference from chain behaviour alone, such as calling a wallet cold, is low confidence. Aliases: confidence scale, confidence words, course confidence scale.

Source: 05-BF - What each wallet type leaves the analyst.

Confidential smart contract. A smart contract design that keeps user data or account state private while still allowing contract logic to execute. Aliases: confidential smart contracts.

Source: 05-CP - Fully Homomorphic Encryption (FHE).

Confidentiality. A security property that keeps information hidden from parties who are not authorized to read it.

Source: 01-CP - What Is Cryptography?.

Confidentiality boundary. A Fabric design boundary that limits which members can see ledger data, policies, or private payloads. Aliases: Confidentiality boundary (Hyperledger Fabric), data-separation boundary, visibility boundary.

Source: 02-PC - Channels: Confidentiality Boundary.

Confirmation. A count of a transaction's own block and every block built on top of it; each one makes reversing the transaction harder. Aliases: Confirmation (Bitcoin), Confirmation (blockchain), Confirmation (proof of work), block confirmations, confirmation depth, confirmations.

Source: 01-CM - A stale block is routine, not an attack.

Confirmation requirement. The number of blocks a service waits after a transaction before treating it as sufficiently settled. Aliases: Confirmation requirement (Bitcoin services), confirmation depth, required confirmations.

Source: 04-BF - Bitcoin Gold 51% Attack Discovery.

Confirmed block. On Solana, a block that more than two-thirds of the network's active stake has voted on; finalized is the stronger level. Aliases: confirmed (Solana).

Source: 08-EF - Solana: programs own the accounts.

Confusion. A cipher design goal that obscures the relationship between the secret key and the resulting ciphertext. Aliases: Confusion (Shannon principle).

Source: 03-CP - Shannon's Principles: Confusion and Diffusion.

Consensus. The rule-based process nodes use to agree which transactions and blocks are valid and should become official history. Aliases: Consensus (blockchain layer), Consensus (blockchain mechanism), Consensus (distributed ledger ordering), Consensus (distributed systems), Consensus (distributed-systems goal), agreement, agreement process, consensus mechanism, consensus mechanisms, consensus process, consensus protocol.

Source: 02-IN - A miner packs waiting payments into a block.

Consensus attack. An attack that tries to manipulate which recent ledger history honest nodes accept as the valid chain. Aliases: Consensus attack (Bitcoin), consensus-layer attack.

Source: 04-BF - What Can Consensus Attacks Achieve?.

Consensus client. Software that runs consensus-layer logic such as duty assignment, block verification, committees, and finality. Aliases: CL client, Consensus client (Ethereum client), beacon node.

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Consensus family. A broad category of consensus protocols grouped by identity assumptions, adversary model, and finality style. Aliases: consensus families.

Source: 01-CM - Open chains buy openness with slower finality.

Consensus layer. The Ethereum layer responsible for validators, attestations, fork choice, and finality. Aliases: CL, Consensus layer (Ethereum architecture), Consensus layer (Ethereum).

Source: 02-EF - The Merge: Beacon Chain to PoS.

Consensus mechanism. A rule system that lets many independent participants agree on one accepted result or history. Aliases: Consensus mechanism (blockchain), consensus mechanisms.

Source: 03-IN - Why This History Matters.

Consensus protocol. A protocol that lets distributed nodes agree on shared ledger history or state without relying on one central referee. Aliases: Consensus protocol (blockchain mechanism), consensus, consensus protocols.

Source: 01-CP - How Blockchain Uses Cryptography.

Consensus rules. The protocol rules that determine whether Bitcoin transactions and blocks are valid. Aliases: Consensus rules (Bitcoin), Consensus rules (blockchain), consensus rule, network rules, validity rules.

Source: 03-BF - What Is a Bitcoin Node?.

Consistency. The property that participants converge on the same correct version of the record despite simultaneous writes or failures. Aliases: Consistency (distributed ledgers).

Source: 04-DS - From Logs to Ledgers.

Consolidation. The gathering of funds from many addresses or outputs into a smaller number of controlled destinations. Aliases: Consolidation (blockchain forensics), UTXO consolidation, fund aggregation.

Source: 02-FO - A peel chain: a small peel, the rest rolls on.

Constant product formula. An AMM pricing rule that keeps the product of two token reserves constant as trades change each reserve. Aliases: Constant product formula (AMM), constant product market maker, x*y=k.

Source: 01-FO - Decentralized Exchanges (DEXs).

ConstitutionDAO. A DAO formed to pool funds for a rare U.S. Constitution printing, whose token later persisted as a social symbol after the bid failed. Aliases: ConstitutionDAO (single-purpose DAO), PEOPLE.

Source: 06-EF - Case Study: ConstitutionDAO.

Constructor. A deployment-time function that runs once to initialize contract state before the runtime bytecode is stored on-chain. Aliases: Constructor (Solidity), constructor logic.

Source: 04-EF - Contract Deployment.

Content addressing. A storage addressing model where the address is derived from the content itself, so changing the file changes the address. Aliases: Content addressing (storage), content-addressed storage.

Source: 06-EF - Metadata, IPFS, and the Pointer Problem.

Content authenticity and provenance. Evidence about where digital content came from, how it changed, and who or what issued it. Aliases: authenticity and provenance, content provenance.

Source: 05-AS - Opportunity: Authenticity and Provenance.

Content Credentials. Metadata attached to media to preserve information about origin, edits, ownership, or other provenance claims. Aliases: content credential.

Source: 05-AS - Opportunity: Authenticity and Provenance.

Content identifier. A hash-derived identifier that points to specific content in IPFS, changing if the content changes. Aliases: CID, CIDs, Content identifier (IPFS).

Source: 06-EF - Metadata, IPFS, and the Pointer Problem.

Content-Addressable Network. A DHT design that maps peers and keys into a virtual coordinate space divided into node-owned zones. Aliases: CAN.

Source: 02-DS - DHT Example: CAN.

Content-addressed data system. A system that identifies stored content by hashes of the content rather than only by location or filename. Aliases: content-addressed data systems, content-addressed storage.

Source: 02-CP - Where You'll See Them (Non-Exhaustive).

Context factor. A weighting factor that accounts for the type, stakes, and norms of the transaction being rated. Aliases: context factors, context weight, transaction context.

Source: 03-DS - PeerTrust: How Trust Is Calculated.

Continuous security. The view that auditing, monitoring, upgrades, and migration are ongoing requirements rather than one-time setup tasks. Aliases: Continuous security (blockchain systems), adaptive security, security is continuous.

Source: 05-AS - Key Lessons.

Contract account. An Ethereum account that holds code and storage and runs only when activated by a transaction or contract call. Aliases: Contract account (Ethereum account type), Contract account (Ethereum), contract, contract accounts, contract address, smart contract account.

Source: 01-EF - A key or code controls each account.

Contract address. The Ethereum address where a deployed contract account's code and storage can be reached. Aliases: Contract address (Ethereum), Contract address (NFT verification), contract addresses.

Source: 01-EF - A key or code controls each account.

Contract declaration. The Solidity syntax that names the deployable unit of code and state. Aliases: Contract declaration (Solidity), contract Name, contract block.

Source: 04-EF - Solidity: File Structure, From Top to Bottom.

Contract deployment. The process of creating a contract account by sending creation code that runs once and returns runtime bytecode. Aliases: Contract deployment (Ethereum), deployment.

Source: 04-EF - Contract Deployment.

Contract event. A record emitted by smart contract execution that applications and indexers can use to track meaningful activity. Aliases: Contract event (blockchain indexing), contract events, event log, logs.

Source: 01-AS - Indexing: Making Blockchain Usable.

Control loop. A mechanism that observes actual output and changes an input to steer the system toward a target behavior. Aliases: Control loop (difficulty adjustment), feedback loop.

Source: 02-CM - Difficulty Adjustment: Control-Loop Intuition.

Control surface. The part of a token ecosystem through which a holder can actually exercise power or access a privilege. Aliases: Control surface (token governance), control surfaces.

Source: 06-EF - Case Study: Bored Ape Yacht Club.

Convertibility. The right or ability to exchange a monetary claim for a specified amount of an underlying asset.

Source: 04-IN - The Turning Point: Ending Bretton Woods.

Coordination cost. The added effort required for distributed participants to discover one another, exchange information, and stay coherent. Aliases: coordination complexity.

Source: 01-DS - Centralized vs. Decentralized: Trade-offs.

Correlated slashing. A penalty design where validators lose more stake when many validators misbehave in the same period or pattern. Aliases: Correlated slashing (PoS penalty), correlation penalty.

Source: 03-CM - Slashing — Making Misbehavior Expensive.

Correlation. Linking events across systems using patterns such as timing, amount, destination, or infrastructure rather than a direct shared identifier. Aliases: Correlation (forensics), cross-system correlation, event correlation.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Correlation penalty. A slashing penalty that increases when many validators are slashed in the same period. Aliases: Correlation penalty (Ethereum slashing), correlated slashing penalty.

Source: 02-EF - Nothing-at-Stake.

Correspondent banking. A banking arrangement where banks hold accounts with one another to route international payments. Aliases: correspondent banks.

Source: 02-AS - Global Payment Infrastructure: Overview.

Cosigner. One of the parties in a multi-sig wallet; in Electrum, cosigners exchange master public keys so each wallet can build the same addresses. Aliases: co-signer, cosigners.

Source: 06-BF - Wallet software builds, signs and broadcasts.

Counter. A value that increments across generator or encryption rounds so each keyed function input is distinct. Aliases: Counter (cryptography), ctr.

Source: 03-CP - Counter-Mode Pseudorandom Generator (PRG).

Counter mode. A block-cipher mode that encrypts counter values to generate a keystream. Aliases: CTR.

Source: 03-CP - Modes of Operation Overview.

Counter-Heuristic. A wallet, protocol, or user behavior designed to break the assumptions behind common forensic heuristics. Aliases: Counter-Heuristic (blockchain forensics), anti-clustering technique, counter heuristic.

Source: 02-FO - Co-spending links inputs, unless it's a CoinJoin.

Counter-mode PRG. A generator that evaluates a keyed function on successive counter values to produce a keystream. Aliases: CTR-style PRG, counter-mode pseudorandom generator.

Source: 03-CP - Counter-Mode Pseudorandom Generator (PRG).

Counterfeit token. A poisoning lure: a fake token the attacker made, sent to the look-alike for the same amount the victim really paid. Aliases: fake token.

Source: 09-FO - Address poisoning plants a look-alike in history.

Crash fault. A failure where a node stops responding completely and sends no further messages. Aliases: crash failure, crash faults.

Source: 05-IN - Crash Faults.

Crash-fault tolerance. A consensus family that tolerates nodes crashing, restarting, or going silent, but not lying or colluding. Aliases: CFT, Crash-fault tolerance (Hyperledger Fabric ordering), crash-fault-tolerant.

Source: 01-CM - Open chains buy openness with slower finality.

CREATE. The standard EVM deployment opcode that derives a new contract address from the creator address and nonce. Aliases: CREATE (EVM opcode).

Source: 04-EF - Contract Deployment.

CREATE2. An EVM deployment opcode that derives a contract address from the creator address, a salt, and the initcode hash. Aliases: CREATE2 (EVM opcode).

Source: 04-EF - Contract Deployment.

Credential. Cryptographic proof that ties an action to a recognized participant, organization, and role. Aliases: Credential (permissioned ledger), credentials, operational certificate.

Source: 01-PC - Permissioned: the operator decides who writes.

Credibility weighting. A defense that scales ratings by the rater's own credibility so untrusted raters have little influence.

Source: 03-DS - PeerTrust: Handling Manipulation.

Credit Rating Agency. An institution that evaluates the risk of financial products and assigns ratings such as AAA. Aliases: credit rating agencies.

Source: 04-IN - Case Study: The 2008 Global Financial Crisis.

Critical evaluation. The practice of judging whether a blockchain project actually solves a trust problem better than simpler alternatives.

Source: 03-IN - Why This History Matters.

Critical information gap. Something the analysis needed but did not have. ICD 203 asks analysts to identify and address these gaps. Aliases: critical gaps, information gap.

Source: 14-FO - Classified holdings: the process only.

Cross-chain bridge. Infrastructure that locks, verifies, or represents value from one blockchain so related value can move or appear on another. Aliases: Cross-chain bridge (blockchain forensics), bridge, cross-chain bridge contract.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Cross-chain swap. Swapping a coin on one chain for a coin on another, as THORChain does, without a relayer and without minting a copy. Aliases: cross-chain swaps.

Source: 08-EF - Liquidity bridges: someone pays you there.

Cross-chain tracing. Following asset movement across multiple blockchains by matching bridge events, swaps, timings, amounts, and destinations. Aliases: Cross-chain tracing (blockchain forensics), chain-by-chain reconstruction, cross-chain forensics.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Cross-checking. Comparing claims across independent sources to test whether evidence and interpretations hold up. Aliases: Cross-checking (information evaluation), cross-check, multiple sources.

Source: 01-AS - Reading Critically.

Crypto intermediary. A company or platform that stands between users and blockchain systems by providing custody, trading, lending, or other services. Aliases: crypto intermediaries.

Source: 05-AS - When Blockchain Systems Broke.

Crypto Wars. The 1990s political and legal conflict over whether strong cryptography would remain restricted or become broadly available to the public.

Source: 03-IN - The Crypto Wars: U.S. Gov Position (1990s).

Crypto-collateralized stablecoin. A stablecoin backed by other crypto worth more than the coins issued, as with DAI and USDS. Aliases: crypto-backed, crypto-backed stablecoin, crypto-collateralized.

Source: 05-EF - Three ways a stablecoin holds its peg.

Crypto-native environment. A system whose users, assets, and incentives mostly remain on-chain rather than depending on traditional institutions. Aliases: crypto-native ecosystem, crypto-native system.

Source: 05-AS - Future 2: Specialized Crypto Ecosystem.

CryptoAPI. A Windows cryptographic interface whose default key storage behavior mattered in the CryptoDefense failure. Aliases: CryptoAPI (Windows), Windows CryptoAPI.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Cryptocurrency. A digital asset ecosystem built around blockchain-based tokens, markets, and networks. Aliases: Cryptocurrency (digital money), crypto, crypto markets, cryptocurrencies, traditional cryptocurrency.

Source: 01-IN - Start by saying what cases you work.

CryptoDefense. A ransomware case where private keys were generated with strong RSA but left accessible in the victim's own key store.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Cryptoeconomic penalty. A punishment that combines cryptographic proof of a rule violation with an economic loss imposed by the protocol. Aliases: Cryptoeconomic penalty (PoS), economic penalty, protocol penalty.

Source: 03-CM - Slashing — Making Misbehavior Expensive.

Cryptographic agility. The ability to replace or upgrade cryptographic algorithms without redesigning the whole protocol or application. Aliases: algorithm agility, hash agility, signature agility.

Source: 05-CP - Cryptographic Assumptions Decay Over Time.

Cryptographic assumption. A belief that a cryptographic problem remains too expensive for realistic attackers to solve with known methods and hardware. Aliases: cryptographic assumptions, hardness assumption.

Source: 05-CP - Cryptographic Assumptions Decay Over Time.

Cryptographic commitment. A compact value, often a hash or Merkle root, that binds a system to underlying data without carrying all of that data directly. Aliases: Cryptographic commitment (blockchain), commitment, compact commitment.

Source: 04-AS - Hashing: What Problem Does It Solve?.

Cryptographic hash. A function that turns data into a compact fingerprint that changes completely if the input changes. Aliases: digital fingerprint, hash, hashes.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Cryptographic hash function. A deterministic algorithm that maps data of any length to a fixed-length digest used for integrity, commitments, and verification. Aliases: Cryptographic hash function (PRG building block), cryptographic hash, hash, hash function.

Source: 02-CP - What Is a Cryptographic Hash Function?.

Cryptographic link. A reference from one block to another using a cryptographic hash so later blocks depend on earlier block contents. Aliases: hash link, mathematical link.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Cryptographic Proof. Mathematical evidence produced by cryptographic mechanisms that can verify claims without relying solely on institutional trust. Aliases: Cryptographic proof (Bitcoin), cryptographic proof instead of trust, cryptographic proofs.

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

Cryptographic trust. Trust grounded in cryptographic checks that verify identity, authenticity, and message integrity. Aliases: crypto trust layer.

Source: 03-DS - Cryptographic vs. Social Trust.

Cryptographic verification. The use of hashes, signatures, and related checks to confirm data integrity and sender identity.

Source: 03-DS - Trust but Verify: The Complete Trust Loop.

Cryptographically relevant quantum computer. A quantum computer powerful enough to break important public-key cryptographic systems used by today's blockchains. Aliases: CRQC, cryptographically relevant quantum threat.

Source: 05-AS - Cryptographically Relevant Quantum Threat.

Cryptographically secure pseudorandom number generator. A random generator designed so attackers cannot feasibly predict outputs even after seeing other outputs. Aliases: CSPRNG, cryptographically secure PRNG.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Cryptography. The mathematical toolkit blockchains use to prove authorization, protect integrity, and make tampering evident. Aliases: Cryptography (blockchain layer), Cryptography (blockchain pillar), Cryptography (mathematical security), crypto.

Source: 02-IN - Four beliefs to correct when you brief.

Cryptography export controls. Government restrictions that limited the export or public distribution of strong cryptographic tools. Aliases: crypto export controls, export controls.

Source: 03-IN - Privacy as a Prerequisite for Open Society.

Cryptography Mailing List. The specialist mailing list where the Bitcoin whitepaper was first circulated to a community already debating privacy, trust, and digital cash.

Source: 01-BF - Satoshi Nakamoto's Whitepaper.

Cryptojacking. Using someone else's electricity, hardware or computing power to mine cryptocurrency without their consent. The attacker takes no coins from the victim; the reward goes to the attacker's own wallet. Aliases: cryptojacker, cryptojackers, malicious cryptomining.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

Cumulative work. The total proof-of-work backing a chain of blocks, used to decide which history nodes treat as canonical. Aliases: Cumulative work (Bitcoin), highest-work chain, most cumulative work, most work.

Source: 01-BF - Bitcoin, Simplified.

Curated participation. A participation model where membership is granted only to approved entities instead of anyone who connects. Aliases: controlled participation, curated admission.

Source: 01-PC - Find the gatekeeper: who can stop or see it.

Currency layer. The layer concerned with what unit of value people use, such as dollars, yuan, bitcoin, or a CBDC. Aliases: Currency layer (state control).

Source: 02-AS - The Permission Layer: Three Challenges to State Control.

Currency Transaction Report. A report the Bank Secrecy Act requires of a money services business, where it applies, alongside Suspicious Activity Reports. Aliases: Currency Transaction Reports.

Source: 05-FO - The BSA makes the business keep records.

Custodial failure. A breakdown where a platform holding user assets loses, mismanages, or blocks access to those assets. Aliases: Custodial failure (crypto intermediaries), custody failure.

Source: 05-AS - When Blockchain Systems Broke.

Custodial model. An arrangement where a platform holds the user's keys and provides account access, services, and possible recovery. Aliases: Custodial model (key custody), custodial custody, custodial wallet, platform custody.

Source: 01-AS - Custody Models.

Custody. The safekeeping and control of an underlying asset or keys by a party trusted to support a tokenized claim. Aliases: Custody (asset-backed token), custodian, custodians.

Source: 05-EF - Same Token Standard, Different Promise.

Custom error. A named Solidity failure type that can return structured information when a call reverts. Aliases: Custom error (Solidity), error.

Source: 04-EF - Example: Coin (Code, Part 2).

Cypherpunk Manifesto. Eric Hughes's 1993 public statement arguing that privacy in the electronic age must be built with code rather than granted by institutions. Aliases: A Cypherpunk's Manifesto, Cypherpunk Manifesto.

Source: 03-IN - Framing.

Cypherpunk movement. A community of activists, computer scientists, and cryptographers who argued that strong cryptography could protect privacy and individual autonomy online. Aliases: cypherpunk ethos, cypherpunks.

Source: 03-IN - The Cypherpunk Movement.

Cypherpunk vision. The political and technical belief that cryptography can support privacy, autonomy, decentralization, and resistance to centralized control. Aliases: cypherpunk thinking, original vision.

Source: 05-AS - The Original Vision.

D

DAI. A crypto-collateralized stablecoin governed through MakerDAO rather than a single traditional issuer. Aliases: DAI (stablecoin), MakerDAO DAI.

Source: 02-AS - Types of Digital Money.

Damping. A control on iterative trust propagation that limits drift and keeps global scores tied to trusted reference points. Aliases: Damping (trust propagation).

Source: 03-DS - PeerTrust vs. EigenTrust: Summary.

Dandelion-style relay. A transaction relay approach that first sends a transaction along a short random path before broadcasting it widely. Aliases: Dandelion relay, Dandelion-style relay (Bitcoin privacy), stem-and-fluff relay.

Source: 04-BF - Mitigating Network Deanonymization.

DAO split. The DAO's function that let a holder leave with their ether into a child DAO; the attacker called it inside itself before their balance was updated. Aliases: split function.

Source: 10-FO - The DAO: one split called many times over.

DAO Token. The token The DAO sold, about 1.15 billion of them for about 12 million ETH in 2016; holders voted on the projects the fund backed. Aliases: DAO Tokens.

Source: 10-FO - The DAO: one split called many times over.

Darknet market. An online illicit marketplace accessed through privacy-focused infrastructure such as Tor. Aliases: dark web market, illicit marketplace.

Source: 01-FO - Challenges in Forensic Analysis.

DarkSide. The ransomware-as-a-service group behind the May 2021 Colonial Pipeline attack, whose developers took a share of each ransom from the affiliates who ran the attacks. Aliases: DarkSide ransomware.

Source: 03-FO - Colonial: shut down at dawn, paid the next day.

Data location. A label that specifies where reference-type data lives during execution, such as storage, memory, or calldata. Aliases: Data location (Solidity), data locations.

Source: 04-EF - Types: Reference Types.

dDAI. Euler's debt token for DAI.

Source: 10-FO - Euler, March 2023: a loan, a gift, a liquidation.

Deanonymization. Linking pseudonymous blockchain activity to a real-world identity, operator, organization, or accountable entity. Aliases: Deanonymization (blockchain forensics), de-anonymization, identity linkage.

Source: 01-FO - The chain shows a service; records show who.

Decentralization. The distribution of participation, control, and validation across multiple parties rather than one central authority. Aliases: Decentralization (Bitcoin), Decentralization (blockchain design), Decentralization (ledger design), Decentralization (system control), decentralized, decentralized design.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Decentralized application. An application that combines a user interface, wallet interaction, and one or more on-chain smart contracts. Aliases: DApp, DApps, Dapp, dApp, dApps, dapp, decentralized app.

Source: 01-EF - Most users reach the chain through a wallet.

Decentralized Autonomous Organization. A token-governed organization that coordinates voting, treasury control, protocol changes, or shared work through on-chain and off-chain processes. Aliases: DAO, DAOs, decentralized autonomous organization.

Source: 06-EF - Three Patterns: DAOs, Membership NFTs, and Social Tokens.

Decentralized coordination. A system's ability to let many independent participants organize shared activity without a central controller.

Source: 03-IN - P2P Normalizes Decentralization.

Decentralized exchange. An on-chain marketplace where users trade tokens through smart contracts rather than a centralized exchange operator. Aliases: DEX, DEXes, DEXs, Decentralized exchange (DeFi).

Source: 03-EF - A DEX trade is visible before it lands.

Decentralized finance. Financial applications such as trading, lending, payments, or settlement built around blockchain assets and smart contracts. Aliases: DeFi.

Source: 01-AS - Finance (High-Level Overview).

Decentralized infrastructure network. A protocol-based network that coordinates independent providers of services such as data feeds or storage. Aliases: infrastructure network, protocol service network.

Source: 01-AS - Infrastructure as a Use Case.

Decentralized physical infrastructure network. A network that coordinates real-world hardware contributions from many independent participants. Aliases: DePIN, decentralized physical infrastructure networks.

Source: 05-IN - Everyday Examples of Distributed Systems.

Decentralized system. A system that spreads authority, operation, or communication across many participants instead of one central controller. Aliases: decentralized design, decentralized network.

Source: 01-DS - Centralized vs. Decentralized: Trade-offs.

decimals(). The number of decimal places a token's raw integer amounts use, such as 18 for WETH and 6 for USDC. Aliases: decimals, token decimals.

Source: 07-EF - A swap leaves Swap and Sync logs.

Decompiled code. Code-like output that a tool rebuilds from a contract's bytecode. It is a reconstruction of the evidence, not the evidence itself, and it carries the tool's guesses. Aliases: decompiled output.

Source: 11-FO - The chain stores bytecode, not source.

Decompiler. A tool that turns bytecode back into something like code: functions by selector, storage reads and writes, calls and conditions. It cannot recover names, comments or the author's intent, and it can get types, internal functions and large contracts wrong. Aliases: decompilation, decompilers, decompiling.

Source: 11-FO - A decompiler rebuilds structure, not intent.

Decoupling. A situation where related trends move independently rather than rising and falling together. Aliases: Decoupling (token adoption).

Source: 06-EF - Hidden Infrastructure Growth.

Decoy output. An earlier output from the Monero chain, belonging to someone else, that the spender adds to a ring so the real input is hidden among it and the other decoys. Aliases: decoy, decoys, mixin, ring member, ring members.

Source: 07-FO - Monero hides sender, receiver and amount.

Decryption. The process of recovering plaintext from ciphertext using the correct cryptographic key. Aliases: decrypt.

Source: 03-CP - The Shared Key.

Deep pool. A pool with large reserves, where a sale moves the price only a little and arbitrage pulls it back to the market. Aliases: deep market, deep pools.

Source: 07-EF - Arbitrage holds a deep pool near the market.

Deep reorganization. A chain reorganization that replaces many recent blocks rather than only one or two competing tips. Aliases: Deep reorganization (proof of work), deep reorg, multi-block reorganization.

Source: 04-BF - Bitcoin Gold 51% Attack Discovery.

Defense Evasion. The ATT&CK and AADAPT tactic under which almost every laundering typology falls, because laundering is the adversary hiding where the value went.

Source: 12-FO - ATT&CK: tactics are why, techniques how.

DeFi pool. A contract in which a DeFi protocol keeps its users' deposits; anyone may call it, including a contract an attacker writes.

Source: 10-FO - A DeFi pool is a contract anyone can call.

Delegate contract. The contract an EIP-7702 authorization names, whose code then runs as the delegating account.

Source: 09-FO - EIP-7702 lets a plain account run code.

Delegatecall. An EVM operation that executes another contract's code while using the caller's storage and execution context. Aliases: Delegatecall (EVM), delegatecall.

Source: 04-EF - Upgradeability and Proxy Patterns.

Delegated minter role. A contract permission that lets a trusted operator or bridge address mint tokens on behalf of the system. Aliases: MINTER_ROLE.

Source: 06-EF - Minting on Ethereum.

Delegation. Assigning voting power to another address without transferring token ownership. Aliases: Delegation (token governance), delegate, delegated voting.

Source: 06-EF - Case Study: ENS DAO.

Delegation indicator. The code EIP-7702 writes to a delegated account: 0xef0100 followed by the delegate contract's address.

Source: 09-FO - EIP-7702 lets a plain account run code.

Delisting. Treasury lifting sanctions it had placed on a person or entity, removing it from the list. Aliases: delisted, sanctions delisting.

Source: 05-FO - Tornado Cash: sanctioned 2022, delisted 2025.

Denial and deception. The ICD 203 factor that asks whether the subject could have planted a source, such as a fake identity or a borrowed alias. Aliases: possible denial and deception.

Source: 14-FO - Judge each source before you count it.

Denial-of-service attack. An attack that prevents a node or service from responding normally. Aliases: DoS, DoS attack.

Source: 05-IN - Crash Faults.

Deployment bytecode. The bytecode sent in a deployment transaction that the EVM executes to initialize a contract and produce runtime bytecode. Aliases: Deployment bytecode (Ethereum), creation bytecode, creation code.

Source: 04-EF - Contract Deployment.

Deposit address. An address an exchange assigns to one customer for deposits, while the exchange itself holds its key. Aliases: deposit addresses, exchange deposit address.

Source: 05-BF - Whoever holds the key controls the coins.

Deposit contract. The execution-layer contract where a participant deposits 32 ETH to join the validator set. Aliases: Deposit contract (Ethereum staking), staking deposit contract, validator deposit contract.

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Deposit-withdrawal linking. Pairing a mixer deposit with a later withdrawal from user habits: the same address on both sides, a direct transfer between the two addresses, matching amounts across pools, or a withdrawal soon after a deposit in a quiet pool. Aliases: deposit-withdrawal pairing, mixer user-habit heuristics.

Source: 06-FO - User habits link deposit to withdrawal.

Derivation path. A key's address in an HD wallet's tree, written m / purpose' / coin_type' / account' / change / address_index, for example m/84'/0'/0'/0/0. Aliases: BIP 44, BIP44, BIP44 path, derivation paths.

Source: 05-BF - The path names coin, account, chain and index.

DES. An older block cipher that is now treated as legacy because its security margin is no longer adequate. Aliases: DES (legacy symmetric cipher), Data Encryption Standard.

Source: 03-CP - What Is Symmetric Cryptography?.

Deshielding. A Zcash transaction out of the shielded pool to a transparent address; the ledger shows the receiving address and the amount coming out. Aliases: deshielding transaction, z to t transaction.

Source: 07-FO - Zcash has two pools: one public, one shielded.

Destination tag. On the XRP Ledger, a number sent with a payment to one exchange address that says which customer to credit. Aliases: XRP destination tag.

Source: 04-FO - Shared deposit addresses link their senders.

destroyBlackFunds. USDT's owner-only function that zeroes a blacklisted address's balance and cuts total supply, emitting DestroyedBlackFunds and no Transfer log. Aliases: DestroyedBlackFunds.

Source: 05-EF - What a USDT freeze leaves on chain.

Determinism. The property that the same input to a hash function always produces the same digest. Aliases: Determinism (chaincode), Determinism (hashing), deterministic, deterministic execution.

Source: 02-CP - What Is a Cryptographic Hash Function?.

Deterministic execution. Execution where the same input and prior state always produce the same result on every node. Aliases: Deterministic execution (Ethereum), Deterministic execution (smart contracts), determinism, deterministic, deterministic behavior.

Source: 01-EF - Every node runs the same program: the EVM.

Deterministic finality. A finality model where a protocol step or ordering service declares an entry final once required conditions are met. Aliases: declared finality, deterministic commit, final commit, immediate finality.

Source: 04-DS - Permissioned vs. Permissionless: Comparison.

Developer's cut. The share of each ransom that a ransomware-as-a-service group's developers take from the affiliate. DarkSide reportedly took 25% of ransoms under $500,000, falling to 10% above $5 million. Aliases: developer share, developer's share.

Source: 03-FO - Developer and affiliate split it on-chain.

DEX aggregator. A service that routes one trade through many pools for the best price, such as 1inch; like a router, it is only in the path. Aliases: aggregator, aggregators.

Source: 03-EF - Routers and aggregators are only in the path.

DEX router. A contract traders go through to reach a pool, which can show up in a pool's logs in place of the person trading. Aliases: router, routers.

Source: 07-EF - A swap leaves Swap and Sync logs.

DFIR firm. A digital forensics and incident response firm, which may negotiate with the criminals and pay the ransom on the victim's behalf. Aliases: DFIR, digital forensics and incident response firm.

Source: 03-FO - A note leads to a portal, a price, an address.

Diamond Model. An intrusion-analysis model by Caltagirone, Pendergast and Betz (2013) that frames one event by four core features: adversary, capability, infrastructure and victim. Each feature has its own source and confidence, and an event can be recorded with features unknown. Aliases: Diamond Model of Intrusion Analysis, the Diamond.

Source: 12-FO - The Diamond: four features of one event.

Difficulty. The protocol setting that controls how hard it is to find a valid proof-of-work block. Aliases: Difficulty (Proof of Work), difficulty adjustment, difficulty target, target.

Source: 04-AS - Difficulty, Work, and Probabilistic Finality.

Difficulty adjustment. A protocol rule that adjusts the target based on recent block timing so average block intervals stay near the design goal. Aliases: Difficulty adjustment (proof of work), difficulty retargeting, retargeting.

Source: 02-CM - Difficulty Adjustment: Control-Loop Intuition.

Difficulty target. The threshold a block hash must fall below for the block to satisfy the PoW validity rule. Aliases: Difficulty target (Bitcoin mining), Difficulty target (proof of work), difficulty threshold, target, target threshold.

Source: 02-CM - The PoW Rule at a Glance.

Diffusion. A cipher design goal that spreads each plaintext bit's influence across many ciphertext bits. Aliases: Diffusion (Shannon principle).

Source: 03-CP - Shannon's Principles: Confusion and Diffusion.

Digest. The fixed-length output of a hash function that acts as a compact fingerprint of the input data. Aliases: Digest (hashing), fingerprint, hash, hash digest, hash output, hash value.

Source: 02-CP - What Is a Cryptographic Hash Function?.

DigiCash. David Chaum's early digital cash company that used blind signatures for private payments but still depended on a central company and banks. Aliases: Chaumian digital cash.

Source: 03-IN - Early Attempts at Digital Money.

Digital asset. A blockchain-recorded item of value or ownership that can be transferred through valid transactions. Aliases: Digital asset (blockchain), digital assets.

Source: 02-IN - A blockchain is a record, not a currency.

Digital cash. Money represented digitally so it can be transferred online, with Bitcoin aiming to make that transfer work without a trusted bank. Aliases: Digital cash (Bitcoin precursor), decentralized digital cash, digital money, electronic cash, internet-native cash.

Source: 01-BF - The Emergence of Bitcoin.

Digital Euro. The European Central Bank's proposed public digital-money project for euro-area payments. Aliases: digital euro project.

Source: 03-AS - EU: Overview.

Digital money. Money or payment value represented and transferred through digital systems rather than physical cash. Aliases: digital cash, digital currencies, digital currency.

Source: 03-IN - Early Attempts at Digital Money.

Digital Product Passport. A product-linked digital record used to track provenance, compliance, lifecycle, or supply-chain information. Aliases: DPP, Digital Product Passports.

Source: 06-EF - Hidden Infrastructure Growth.

Digital Ruble. Russia's state-run central-bank digital currency project for domestic payment control and possible settlement experiments. Aliases: Russia's digital ruble.

Source: 03-AS - Russia: Overview.

Digital Rupee. India's central-bank-led digital-money project for safer payments, settlement efficiency, and state oversight. Aliases: India's digital rupee, e-rupee.

Source: 03-AS - India: Overview.

Digital Scarcity. Verifiable scarcity in a digital system, such as Bitcoin's protocol-limited supply.

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

Digital signature. A cryptographic proof that a message or transaction was authorized by the holder of a corresponding private key. Aliases: Digital signature (Bitcoin ownership), Digital signature (Bitcoin), Digital signature (Ethereum transaction), digital signatures, signature, signatures, signed payload, signing, unforgeable signature.

Source: 01-IN - Week 1 teaches how it works, week 2 traces.

Digital signature workflow. Hashing a transaction, signing the hash with the private key and a fresh random nonce, and letting anyone verify the result with the public key. Aliases: sign-and-verify workflow.

Source: 04-CP - A signature binds the key to these exact bytes.

Direct trust. Trust based on a participant's own prior interactions with a peer. Aliases: direct experience, first-hand trust.

Source: 03-DS - Direct vs. Indirect Trust.

Discrete logarithm. The hard problem of recovering a hidden exponent from a public group relation. Aliases: Discrete logarithm (public-key cryptography), discrete log, discrete logarithms.

Source: 05-CP - Shor's Algorithm (1994): Breaking RSA and ECC.

Distributed architecture. A system design where many peer nodes share responsibility for coordination and state without a central hub.

Source: 05-IN - Centralized vs. Distributed Architectures.

Distributed Consensus. Agreement among many independent participants about valid system state or transaction history. Aliases: Distributed Consensus (blockchain).

Source: 04-IN - Design Trade-Offs: Efficiency vs. Autonomy.

Distributed hash table. A structured P2P system that maps keys to responsible peers so data can be found through efficient routing. Aliases: DHT, DHTs, Distributed Hash Table (P2P networking), structured overlay routing.

Source: 02-DS - Structured Overlay Networks.

Distributed Ledger. A record-keeping system replicated across multiple participants rather than maintained only by one central database. Aliases: DLT, distributed ledger technology, distributed ledgers.

Source: 04-IN - The Institutional Counterpoint: The BIS Critique.

Distributed system. A system made of multiple independent computers that coordinate over a network to accomplish shared behavior. Aliases: Distributed system (architecture), distributed architecture, distributed systems.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Distributed trust. A trust model where dependence is spread across participants, cryptography, consensus rules, and incentives instead of one central authority.

Source: 05-IN - What is "trust" in digital systems?.

DNS seed. A publicly maintained domain name that returns IP addresses for reachable Bitcoin peers during node startup. Aliases: DNS seed (Bitcoin peer discovery), DNS seed (Bitcoin), DNS seeds.

Source: 03-BF - Joining the Bitcoin Network.

Dollar-like Digital Instrument. A digital asset or payment instrument that gives users exposure to U.S. dollar value without necessarily using a traditional bank account. Aliases: digital dollar instrument, dollar-like instrument.

Source: 03-AS - What We're Analyzing Today.

Domain separation. The practice of marking different kinds of hash inputs with distinct constants so they cannot be confused. Aliases: LeafTag, NodeTag, tags.

Source: 02-CP - Building a Merkle Tree.

Donate function. Euler's function that let users donate to the reserves; added in July 2022 and audited, it lacked a health check. Aliases: donate.

Source: 10-FO - Euler, March 2023: a loan, a gift, a liquidation.

Donation wallet. A wallet built into stock mining tools that mines a share, often 2–5%, for the tool's developer. Aliases: developer donation wallet.

Source: 08-FO - Characterize: for, against, and alternatives.

Doodles. An Ethereum NFT project that evolved toward a media, entertainment, and collectibles brand with holder benefits and limited participation. Aliases: Doodles (membership NFT project).

Source: 06-EF - Case Study: Doodles.

Double attestation. A slashable offense where a validator signs two conflicting attestations for the same target epoch. Aliases: Double attestation (Ethereum slashing), attester equivocation.

Source: 02-EF - Nothing-at-Stake.

Double proposal. A slashable offense where a validator signs two different blocks for the same slot. Aliases: Double proposal (Ethereum slashing), proposer equivocation.

Source: 02-EF - Nothing-at-Stake.

Double spend. An attack where an adversary tries to replace an accepted payment with a conflicting history that excludes or redirects it. Aliases: double-spend attack.

Source: 02-CM - Attacks: Double Spend.

Double spending. An attack goal where a payment is accepted and then a conflicting history is published to erase or reverse it. Aliases: Double spending (Ethereum threat model), double-spend, double-spend attack.

Source: 02-EF - Double Spending (Threat Model).

Double-spend. An attack attempt that uses the same bitcoin value in conflicting payments by reversing, delaying, or replacing a recent transaction. Aliases: Double-spend (Bitcoin), double spend, double-spending.

Source: 04-BF - Why Attack Bitcoin?.

Double-spend attack. An attempt to spend funds and then replace history so the original payment is excluded. Aliases: Double-spend attack (blockchain payments), double spending, double-spend attempt.

Source: 04-AS - Common PoW Attack Themes.

Double-spend check. The validation rule that an output can be consumed at most once. Aliases: Double-spend check (Bitcoin), double-spend prevention.

Source: 02-BF - Replace-by-fee: unconfirmed can still change.

Double-spend problem. The digital money problem of preventing the same unit of value from being spent more than once. Aliases: double spending.

Source: 01-CP - Example: The Double-Spend Problem.

Double-spending. The attempt to spend the same digital asset more than once. Aliases: Double-spending (Bitcoin), Double-spending (blockchain), double spend, double spending, double-spend, double-spend attempt, double-spend problem, double-spending problem.

Source: 03-IN - The Synthesis: Bitcoin.

Drainer. The phishing site and the code behind it that get a victim to sign an approval or a transfer, which the thief then uses to empty the wallet. Aliases: drainer flow, drainer flows, drainer kit, drainer kits, drainers, wallet drainer.

Source: 09-FO - An approval lets someone else spend your tokens.

Drainer admin address. The address that receives a drainer developer's cut of each theft. Aliases: admin account, admin address.

Source: 09-FO - A drainer kit is phishing sold as a service.

Drainer profit split. The fixed share of each theft a drainer pays its developers, for Inferno Drainer 20% to an admin account and 80% to the customer who ran the site. Aliases: 20/80 split, drainer cut, profit sharing.

Source: 09-FO - A drainer kit is phishing sold as a service.

Drainer-as-a-service. The model in which a drainer's developers rent their kit to customers who run the phishing sites, and take a cut of every theft. Aliases: scam-as-a-service.

Source: 09-FO - A drainer kit is phishing sold as a service.

Drex. Brazil's central-bank pilot for tokenized digital financial infrastructure and programmable settlement. Aliases: Brazil Drex, Brazil's Drex pilot.

Source: 03-AS - Latin America: Impact Signals.

Drift Security Council. Drift's five signers who approve admin actions. Two of them pre-signed transactions that hid admin authority, which handed control to the attacker.

Source: 13-FO - Drift: a week of preparation, twelve minutes of theft.

Drift theft. The 1 April 2026 theft of about $285 million from Drift, a Solana exchange protocol, with no bug in its code. Two of five Security Council signers pre-signed transactions that handed over admin control, and 31 withdrawals followed in about 12 minutes. Aliases: Drift hack.

Source: 13-FO - A state actor steals to fund a state.

Dummy element. The empty first item in a multi-sig spend, there because an old bug in OP_CHECKMULTISIG pops one item too many; since SegWit (BIP 147) it must be empty. Aliases: OP_0 dummy, dummy item, multi-sig dummy.

Source: 06-BF - The rule is written in the script.

Durable nonce account. An account used so that signers' approvals can be sent later rather than at once. At Drift, a Solana protocol, the attacker created them between 23 and 30 March 2026, about a week before the theft. Aliases: durable nonce, durable nonce accounts.

Source: 13-FO - A state actor steals to fund a state.

Dust attack. Sending a tiny amount to an address that has already been paid, in the hope that the wallet spends it together with the owner's other coins and so reveals them. Aliases: dust, dust input, dusting attack.

Source: 06-FO - Dust is bait: a dust input is not a link.

DVN. A verifier in a LayerZero message bridge that vouches a message is genuine. Aliases: DVNs.

Source: 08-EF - Message bridges trust a message: Kelp DAO.

dYdX. A DeFi protocol that, like Aave, offers flash loans from its pools; Qin et al. study both.

Source: 10-FO - A flash loan lasts one transaction.

E

e-CNY. China's retail central-bank digital currency pilot built inside a state-directed payment architecture. Aliases: China's retail CBDC, digital renminbi, digital yuan.

Source: 03-AS - China: Overview.

e-gold. A centralized digital money service that let users transfer ownership of gold-denominated balances. Aliases: E-gold, e-gold.

Source: 03-IN - Early Attempts at Digital Money.

ECDSA. A digital signature scheme used by major blockchains to authorize transactions and prove control of keys. Aliases: ECDSA (blockchain signatures), Elliptic Curve Digital Signature Algorithm.

Source: 05-AS - Cryptographically Relevant Quantum Threat.

Echo. A source that agrees with another only because it copied it, so the two are one line. Count the origins, not the labels. Aliases: echo rule, echoes.

Source: 14-FO - An echo is not a second line.

Eclipse attack. An attack where a node's peer connections are manipulated so it receives a distorted view of the network. Aliases: Eclipse attack (Bitcoin network), Eclipse attack (Bitcoin), Eclipse attack (P2P threat), Eclipse attack (network attack), Eclipse attack (network security), eclipse, eclipse attacks, eclipse attempt, eclipse-style isolation, eclipsed node, network isolation, network-level eclipse, node isolation attack, peer isolation attack.

Source: 01-DS - Fallacy #4: The Network Is Secure.

Economic consensus. A consensus family that uses scarce resources, incentives, or penalties to shape behavior in open systems. Aliases: Economic consensus (consensus family), blockchain economic consensus.

Source: 04-AS - Classical vs Economic Consensus.

Economic Instrument of Statecraft. The use of financial, trade, monetary, and market tools to project power or impose costs. Aliases: economic instrument of national power, economic statecraft.

Source: 03-AS - Financial Rails Are Power.

Economic Shelter. Use of an asset or instrument to protect savings from inflation, devaluation, or weak local currency conditions. Aliases: savings shelter, stablecoin shelter.

Source: 03-AS - Latin America: Impact Signals.

Ecosystem component. A service around the chain that improves usability, interoperability, performance, or access to outside information. Aliases: Ecosystem component (blockchain systems), supporting component, supporting service.

Source: 01-AS - The Full Blockchain System.

eDAI. Euler's deposit token for DAI.

Source: 10-FO - Euler, March 2023: a loan, a gift, a liquidation.

Effective balance. The capped validator balance used by the consensus layer to weight selection probability and voting influence. Aliases: Effective balance (Ethereum staking), effective validator balance.

Source: 02-EF - Validator Selection & Randomness.

Effective gas price. The actual per-gas price paid after applying the sender's fee ceiling, the base fee, and the priority fee. Aliases: Effective gas price (Ethereum fee market), actual per-unit fee.

Source: 03-EF - Fees set the queue: base fee burned, tip paid.

Efficiency. A design priority favoring high speed, low cost, high throughput, and reversibility in payments. Aliases: Efficiency (monetary system design).

Source: 04-IN - Design Trade-Offs: Efficiency vs. Autonomy.

EigenTrust. A global reputation algorithm that normalizes local trust values and iteratively propagates them through the network to produce shared trust scores.

Source: 03-DS - EigenTrust: From Local Opinions to Global Scores.

Eight fallacies of distributed computing. A checklist of common false assumptions engineers make about networks, such as reliability, zero latency, and homogeneous nodes. Aliases: eight fallacies, fallacies of distributed computing.

Source: 01-DS - The Eight Fallacies.

EIP-1559. An Ethereum upgrade that changed transaction fees by introducing a burned base fee plus a tip paid to the block's fee recipient. Aliases: EIP-1559 (Ethereum fee market).

Source: 01-EF - Gas prices the work; a failed call still pays.

EIP-1559 fee mechanism. Ethereum's fee model that combines an automatically adjusted base fee with a user-selected priority tip and fee ceiling. Aliases: EIP-1559, EIP-1559 fee mechanism (Ethereum fee market), Ethereum Improvement Proposal 1559.

Source: 03-EF - Fees set the queue: base fee burned, tip paid.

EIP-7702. The Ethereum change, live since the Pectra upgrade, that lets an ordinary account sign one authorization making it run a delegate contract's code. Aliases: 7702.

Source: 09-FO - Four ways out of a wallet look different.

EIP-7702 authorization. The message an account's owner signs under EIP-7702, naming a chain, a delegate contract and the account's nonce; it is carried in a type 0x04 transaction. Aliases: 7702 authorization, signed authorization.

Source: 09-FO - Four ways out of a wallet look different.

EIP-7702 delegation. The state of an account whose code, set by an EIP-7702 authorization, points to a delegate contract, so calls to the account run the delegate's code. Aliases: 7702 delegation.

Source: 09-FO - Four ways out of a wallet look different.

Elastic Currency. Money whose supply can be expanded or contracted to meet changing economic needs. Aliases: elastic money.

Source: 04-IN - The Ideological Divide: Austrian vs. Keynesian Economics.

Electronic Codebook. A block-cipher mode that encrypts each block independently under the same key. Aliases: ECB.

Source: 03-CP - Modes of Operation Overview.

Electronic coin. In Nakamoto's framing, a chain of digital signatures that transfers value from one owner to the next. Aliases: Electronic coin (Bitcoin), coin.

Source: 01-BF - Primitive 1: Ownership via Digital Signatures.

Electrum. Open-source self-custody wallet software that creates a user's keys and addresses. Aliases: Electrum wallet.

Source: 05-BF - Most people hold bitcoin one of two ways.

Electrum server. A server that lightweight wallets such as Electrum and Sparrow ask about their addresses instead of running their own node; anyone can run one. Aliases: Electrum servers, private Electrum server.

Source: 06-BF - Wallet software builds, signs and broadcasts.

Elliptic curve cryptography. Public-key cryptography built on arithmetic with points on an elliptic curve. Aliases: ECC, Elliptic Curve Cryptography (quantum vulnerability), elliptic-curve cryptography.

Source: 04-CP - A private key is a number that controls coins.

Elliptic-curve discrete logarithm problem. The problem of working back from a public key, a point on an elliptic curve, to the private key, the number that multiplied the curve's fixed point G. Aliases: ECDLP, elliptic curve DL, elliptic-curve discrete log.

Source: 04-CP - A private key is a number that controls coins.

Encrypt-then-MAC. A composition pattern that encrypts data first and then authenticates the ciphertext with a message authentication code.

Source: 03-CP - From Confidentiality to Authenticated Encryption (AEAD).

Encrypted mempool. A proposed transaction pool design that hides transaction contents until ordering or inclusion decisions are harder to censor or exploit. Aliases: Encrypted mempool (Ethereum censorship mitigation), encrypted mempools.

Source: 02-EF - Censorship.

Encryption. The process of transforming plaintext into ciphertext using a cryptographic key. Aliases: encrypt.

Source: 03-CP - The Shared Key.

Endianness. The convention for ordering bytes when serializing numbers or fields before hashing or validation. Aliases: Endianness (block serialization), byte order.

Source: 02-CM - Block Header Anatomy: What Gets Hashed.

Endorsement. An endorser's signed approval of a simulated chaincode result and its read-write set. Aliases: Endorsement (Hyperledger Fabric), endorsements, signed endorsement.

Source: 02-PC - One Invoke: Proposal to Commit.

Endorsement mismatch. A failure where endorsers return different results for the same proposal, preventing the client from satisfying endorsement policy. Aliases: Endorsement mismatch (Hyperledger Fabric), mismatched endorsements.

Source: 02-PC - Determinism and Safety.

Endorsement policy. A rule specifying which peers or organizations must approve a transaction before it can be committed. Aliases: Endorsement policy (Hyperledger Fabric), approval policy, endorsement policies.

Source: 04-DS - Case Study A — Hyperledger Fabric (Permissioned).

Endorsing peer. A peer acting in the approval role by simulating chaincode, producing a read-write set, and signing the proposed result. Aliases: Endorsing peer (Hyperledger Fabric), endorser, endorsers, endorsing peers.

Source: 02-PC - The Big Picture: End-to-End Fabric Path.

Endpoint. A system around the chain, such as a wallet or exchange, where users interact with a blockchain and where thefts happen. Aliases: Endpoint (blockchain security), endpoints.

Source: 02-IN - Four beliefs to correct when you brief.

Energy. The Tron resource that pays for a contract call's computation, such as a USDT transfer; it comes from staked TRX, Energy another account delegates, or burned TRX. Aliases: Tron Energy, delegated Energy.

Source: 01-PC - Energy pays for a USDT move: who staked it?.

Energy consumption. The electricity and physical resources consumed by blockchain operation, especially proof-of-work mining. Aliases: Energy consumption (proof-of-work blockchains), electricity consumption, resource use.

Source: 05-AS - External Constraints.

Energy externality. A cost or impact of mining energy use that falls outside the protocol's direct accounting, such as grid or emissions effects. Aliases: Energy externality (proof of work), environmental externality, externality.

Source: 02-CM - Energy, Externalities, and Design Responses.

ENS DAO. The DAO that governs Ethereum Name Service protocol infrastructure, token voting processes, working groups, and treasury decisions. Aliases: ENS, ENS DAO (protocol governance), Ethereum Name Service DAO.

Source: 06-EF - Case Study: ENS DAO.

Entanglement. A quantum relationship where the state of one qubit is linked with another in a way classical bits cannot reproduce. Aliases: Entanglement (quantum computing).

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Entity Cluster. A set of addresses the analyst believes one entity controls. Aliases: Entity Cluster (blockchain forensics), address cluster, cluster, entity.

Source: 02-FO - Co-spending links inputs, unless it's a CoinJoin.

Entropy. A measure of how much unpredictability a source or process provides. Aliases: Entropy (key generation), Entropy (randomness), bad entropy, randomness, randomness quality.

Source: 01-CP - Probability Refresher: Independence & Random Variables.

Entropy quality. The practical unpredictability of the source feeding key, seed, or nonce generation. Aliases: randomness quality.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Enum. A value type that restricts a variable to a fixed set of named states. Aliases: Enum (Solidity), enumeration.

Source: 04-EF - Types: Value Types.

Epoch (Ethereum). A group of 32 slots used to reshuffle committees and evaluate finality checkpoints. Aliases: Epoch (Ethereum PoS time unit), epochs.

Source: 02-EF - Slots and Epochs.

Epoch (Proof of Stake). A group of consecutive slots used to organize validator selection, checkpoints, and finality updates. Aliases: Epoch (PoS time unit), epochs.

Source: 03-CM - Proof of Stake: The Full Lifecycle.

Equivocation. Signing conflicting blocks or votes for the same slot, height, or consensus decision. Aliases: Equivocation (Ethereum PoS), Equivocation (PoS offense), conflicting signatures, conflicting votes, double proposal, double vote, equivocating.

Source: 03-CM - Slashing — Making Misbehavior Expensive.

ERC-1155. A multi-token Ethereum standard that tracks balances by both address and token ID, allowing fungible and unique-like assets in one contract. Aliases: ERC-1155 (Ethereum token standard), ERC-1155 (token standard), ERC1155.

Source: 06-EF - Fungible vs. Non-Fungible.

ERC-165. An Ethereum standard that lets contracts advertise which interfaces they implement. Aliases: ERC-165 (Ethereum interface standard).

Source: 04-AS - Tokens and Standards.

ERC-20. The Ethereum token standard: six functions (totalSupply, balanceOf, transfer, approve, allowance, transferFrom) and two events (Transfer, Approval) that a token offers. Aliases: EIP-20, ERC-20 (Ethereum token standard), ERC-20 (token standard), ERC20.

Source: 05-EF - ERC-20: six functions and two events.

ERC-721. The Ethereum non-fungible token standard, in which ownerOf returns the one owner of each token ID and every move emits a Transfer event naming that token ID. Aliases: ERC-721 (Ethereum token standard), ERC-721 (token standard), ERC721.

Source: 05-EF - An NFT is a token ID with one owner.

Escrow wallet. A wallet that holds a trader's crypto while a P2P trade is pending. A platform may hold its key, as Paxful did, or the traders may lock coins in a multisig. Aliases: escrow, platform escrow.

Source: 04-FO - P2P markets: two people trade, an escrow holds.

Ether. Ethereum's native asset, used to pay gas and represent value on the network. Aliases: ETH, Ether (Ethereum native asset).

Source: 01-EF - A key or code controls each account.

Ethereum. A blockchain platform emphasized in the course for smart contracts, tokens, and application-oriented design. Aliases: Ethereum (blockchain platform), Ethereum (programmable blockchain platform), Ethereum network.

Source: 01-IN - By day 10 you can do nine things.

Ethereum Discovery v5. Ethereum's peer discovery protocol, based on Kademlia-like ideas for finding nodes in the P2P layer. Aliases: Discovery v5, Ethereum node discovery.

Source: 02-DS - The Blockchain Link: Efficient Routing.

Ethereum Improvement Proposal. A public specification for a proposed Ethereum protocol change, process change, or ecosystem standard. Aliases: EIP, EIPs, Ethereum Improvement Proposal (Ethereum governance), Ethereum Improvement Proposals.

Source: 01-EF - A key or code controls each account.

Ethereum Request for Comment. A category of Ethereum standard that defines shared application interfaces such as token behavior and metadata conventions. Aliases: ERC, ERCs, Ethereum Request for Comment (Ethereum standard), Ethereum Request for Comments.

Source: 06-EF - Fungible vs. Non-Fungible.

Ethereum Request for Comments. An Ethereum application-level standard that sets shared conventions contracts and tools can rely on. Aliases: ERC, ERC standard, ERC standards, ERCs, Ethereum Request for Comments (Ethereum standard).

Source: 04-EF - Upgradeability and Proxy Patterns.

Ethereum Virtual Machine. The stack-based virtual machine that every Ethereum node uses to execute contract bytecode and reproduce state transitions. Aliases: EVM, Ethereum Virtual Machine (Ethereum execution).

Source: 01-EF - Ethereum keeps accounts, not coins.

Etherscan. A block explorer whose decompiler tab runs Panoramix; the course names it for awareness.

Source: 11-FO - A decompiler rebuilds structure, not intent.

Euler Finance hack. The 13 March 2023 attack in which Euler lost about $197 million through a donate function that lacked a health check, funded by a flash loan; the funds were later returned. Aliases: Euler, Euler exploit.

Source: 10-FO - A DeFi pool is a contract anyone can call.

European Blockchain Services Infrastructure. An EU initiative using shared permissioned ledger infrastructure for trusted cross-border services such as credentials and notarization. Aliases: EBSI.

Source: 04-DS - Case Study B — Public Sector: EBSI & Estonia (KSI){.smaller}.

Event. A contract-declared log format that lets successful transactions emit structured data for off-chain consumers. Aliases: Event (Solidity), event emission, events.

Source: 04-EF - Solidity: File Structure, From Top to Bottom.

Event log. Structured output emitted during transaction execution and recorded in the transaction receipt. Aliases: Event log (Ethereum), emitted logs, event logs, events, logs.

Source: 03-EF - A DEX trade is visible before it lands.

Event signature hash. The keccak-256 hash of an event signature that appears as the first topic for ordinary Solidity events. Aliases: Event signature hash (Ethereum log), event selector, topic 0.

Source: 04-EF - Events and Logs.

Eventual synchrony. A timing assumption that the network may be unreliable for a while but eventually delivers messages within usable bounds. Aliases: Eventual synchrony (distributed-systems assumption), eventually synchronous network.

Source: 03-CM - Safety and Liveness — PoS Invariants.

Evil Contract. The AADAPT sub-technique, under Execution, about phishing users into signing a contract. AADAPT files approval phishing here; it has no entry named for drainer kits. Aliases: ADT3012.002.

Source: 12-FO - AADAPT names most of the typologies.

EVM jump. The instruction the EVM uses to implement every call, return, loop and conditional. Because all control flow is jumps, rebuilding functions and structure from bytecode is hard. Aliases: EVM jumps.

Source: 11-FO - A decompiler rebuilds structure, not intent.

EVM sandbox. The restricted execution environment that prevents contracts from directly accessing local files, networks, clocks, or operating-system randomness. Aliases: EVM sandbox (Ethereum execution), isolation, sandboxed EVM, sandboxed execution.

Source: 01-EF - Every node runs the same program: the EVM.

Exact lookup. A search for one specific item using its known key, hash, or identifier. Aliases: exact-match search, key-based lookup.

Source: 02-DS - Structured Overlay Networks.

Exact match. Sourcify's strongest verification result: the published source recompiles to the on-chain bytecode byte for byte, metadata hash included, so these are exactly the files that were compiled. A plain match agrees on everything but the metadata hash, so behavior is the same but names, comments or file paths may differ. Aliases: exact-match verification.

Source: 11-FO - Verified source: a match someone checked.

eXch. A service through which, with others, most of the ETH stolen from Bybit was converted to bitcoin by 5 March 2025.

Source: 07-EF - Bybit: stolen tokens became ETH first.

Exchange platform. A service that gives users account-based access to crypto trading, custody, or other blockchain-related products. Aliases: Exchange platform (crypto access), crypto exchange, exchange, platform.

Source: 01-AS - What Does a User Actually Interact With?.

Exchange record. An exchange's records of who opened and used an account. It is usually the strongest kind of line an analyst receives, but it records what the customer typed, so the name on it can be false. Aliases: account record, exchange records.

Source: 14-FO - The chain says what moved; others say who.

Exclusive OR. A bit operation that outputs 1 when exactly one input bit is 1 and outputs 0 otherwise. Aliases: XOR, $\hat{}$, .

Source: 03-CP - Exclusive OR (XOR) Logic.

Execute-Order-Validate. A Fabric architecture that separates transaction simulation, ordering, and validation into distinct stages. Aliases: EOV, Execute-Order-Validate (Hyperledger Fabric), Execute-order-validate (Hyperledger Fabric), execute order validate, execute-order-validate architecture.

Source: 04-DS - Case Study A — Hyperledger Fabric (Permissioned).

Execution client. Software that handles Ethereum transactions, state, and EVM execution. Aliases: EL client, Execution client (Ethereum client).

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Execution layer. The Ethereum layer responsible for transactions, accounts, smart contracts, and state updates. Aliases: EL, Execution layer (Ethereum architecture), Execution layer (Ethereum).

Source: 02-EF - The Merge: Beacon Chain to PoS.

Execution payload. The execution-layer block data inside a post-Merge beacon block, including the header, ordered transactions, and execution data. Aliases: Execution payload (Ethereum), execution block, payload.

Source: 02-EF - Execution Payload: Header Structure.

Execution payload body. The variable-size portion of an execution payload that carries ordered transactions and execution-layer data. Aliases: Execution payload body (Ethereum), body, payload body.

Source: 02-EF - Execution Payload: Header Structure.

Execution payload header. The compact portion of an execution payload containing commitments and metadata for verifying transactions, receipts, and resulting state. Aliases: Execution payload header (Ethereum), header, payload header.

Source: 02-EF - Execution Payload: Header Structure.

Exit. A place value can leave a chain or change form: a deep DEX pool, a centralized exchange, an OTC desk or a bridge; some balances have no exit at all. Aliases: exit point, exits.

Source: 07-EF - Where funds can leave a chain.

Expected work. The average number of hash attempts needed before finding a valid block at a given target. Aliases: Expected work (proof of work), expected hashes, expected trials.

Source: 02-CM - Difficulty, Hash Rate, and Expected Work.

Explorer label. A blockchain explorer's grouping or naming of addresses, for example listing many addresses as one wallet. The label can be wrong, so a hop resting on it alone is Low confidence. Aliases: block explorer label, explorer's label.

Source: 03-FO - Each hop gets a heuristic and a confidence.

Exposed public key. A public key that has been revealed in a way that could become dangerous if attackers can derive the corresponding private key. Aliases: Exposed public key (quantum threat), exposed public keys.

Source: 05-AS - Cryptographically Relevant Quantum Threat.

Extended public key. A key (xpub, or zpub for native SegWit) that derives every public key and address on a branch, including ones not used yet, but no private key. Aliases: extended public keys, master public key, xpub, zpub.

Source: 05-BF - An xpub reveals every address, past and future.

External chain. The branch of an HD wallet account that makes receiving addresses; change level 0 in the derivation path. Aliases: receiving chain.

Source: 05-BF - One seed grows a whole tree of keys.

External data. Information from outside the blockchain, such as prices, reserves, outcomes, or timestamps, that contracts may need to use. Aliases: External data (oracle input), off-chain data, real-world data.

Source: 01-AS - Oracles: Connecting to the Real World.

Externally Owned Account. An Ethereum account controlled by a private key that can initiate transactions and pay gas. Aliases: EOA, EOAs, Externally Owned Account (Ethereum account type), Externally Owned Account (Ethereum), externally owned accounts, wallet account.

Source: 01-EF - A key or code controls each account.

F

Factoring. The problem of finding the prime factors that multiply together to form a given composite number. Aliases: Factoring (hard problem), factoring large primes, integer factoring.

Source: 01-CP - Proofs and Reasoning.

Failure model. The set of failure behaviors a protocol assumes it must tolerate. Aliases: fault model.

Source: 01-CM - Open chains buy openness with slower finality.

Fallback function. A special Solidity function that can run when calldata does not match a known selector or when ETH is received in certain cases. Aliases: Fallback function (Solidity), fallback.

Source: 04-EF - ABI, Encoding, and Function Selectors.

Fallback peer. A known peer address included in node software as a backup way to connect when ordinary discovery is insufficient. Aliases: Fallback peer (Bitcoin), fallback node, hardcoded fallback peer.

Source: 03-BF - Joining the Bitcoin Network.

False Merge. An error where unrelated users or addresses are incorrectly combined into one inferred entity. Aliases: False Merge (blockchain forensics), false clustering, incorrect cluster merge.

Source: 02-FO - One false link merges two actors into one.

False positive. An incorrect match or attribution that wrongly links activity, funds, or identities. Aliases: False positive (forensic analysis), incorrect attribution, incorrect match.

Source: 01-FO - Challenges in Forensic Analysis.

Fan-In. A pattern where value from many sources is gathered into one destination. Aliases: Fan-In (blockchain forensics), fan in, many-to-one flow.

Source: 02-FO - A peel chain: a small peel, the rest rolls on.

Fan-in pattern. A transaction graph pattern where many sources converge into one service, wallet, or cluster. Aliases: Fan-in pattern (graph analysis), fan-in, funds pooling.

Source: 01-FO - BTC-e: Weak KYC Made the Cluster Actionable.

Fan-in/fan-out. A transaction graph pattern showing many inputs converging, many outputs dispersing, or both. Aliases: Fan-in/fan-out (forensics pattern), fan-in, fan-out.

Source: 04-AS - Graph Thinking and Heuristics.

Fan-Out. A pattern where one source distributes value to many destination addresses. Aliases: Fan-Out (blockchain forensics), fan out, one-to-many flow.

Source: 02-FO - Batched payouts look busy but aren't a peel.

Fault isolation. A design property where a failure in one part of a system is contained instead of spreading through the whole service. Aliases: failure isolation.

Source: 01-DS - Why Distribute? Motivations and Risks.

Fault tolerance. A system's ability to keep operating correctly despite component failures or misbehavior. Aliases: Fault tolerance (Bitcoin network), fault tolerance spectrum, network robustness.

Source: 05-IN - Fault Tolerance Spectrum.

Federal Reserve. The central bank of the United States, responsible in this lesson for pursuing price stability and maximum employment. Aliases: Fed, U.S. Federal Reserve.

Source: 04-IN - The Modern Financial Architect: Central Banks.

Fediverse. A collection of independently operated social-network servers that communicate through shared protocols. Aliases: Mastodon, federated social networks.

Source: 05-IN - Everyday Examples of Distributed Systems.

Fedwire. A US Federal Reserve settlement system where dollar payments are finally transferred between participating institutions. Aliases: Fedwire (payment settlement).

Source: 02-AS - Global Payment Infrastructure: Overview.

Fee-adjusted amount matching. A correlation technique that matches likely related transfers after accounting for fees, slippage, or service deductions. Aliases: Fee-adjusted amount matching (cross-chain tracing), amount matching, fee-adjusted matching.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Feedback scope. The amount of interaction history behind a rating or reputation score. Aliases: evidence depth, transaction scope.

Source: 03-DS - PeerTrust: How Trust Is Calculated.

Feedback score. A measure of how satisfied peers were with past interactions with a participant. Aliases: satisfaction, satisfaction score.

Source: 03-DS - PeerTrust: How Trust Is Calculated.

Fiat Money. Government-issued money not backed by a physical commodity and sustained by legal decree, policy, and broad acceptance. Aliases: fiat, fiat currency.

Source: 04-IN - A Brief History of Money's Form.

Fiat off-ramp. A service or path used to convert crypto assets into fiat currency or spendable equivalents. Aliases: Fiat off-ramp (crypto laundering), cash-out path, cash-out venue, off-ramp.

Source: 01-FO - Decentralized Exchanges (DEXs).

Fiat-backed stablecoin. A stablecoin whose issuer holds reserves, such as dollars, behind the tokens, as with USDT and USDC; the issuer can freeze it through the contract's code. Aliases: fiat-backed, fiat-backed stablecoins, fiat-collateralized stablecoin.

Source: 05-EF - Three ways a stablecoin holds its peg.

Finality. The point at which a ledger entry can reasonably be treated as permanent under the consensus model. Aliases: Finality (blockchain settlement), Finality (consensus), Finality (distributed ledgers), finalization, finalized, ledger finality, settlement, transaction finality.

Source: 04-DS - Immutability Source.

Finality (Ethereum). The point at which Ethereum history becomes economically and explicitly irreversible under the Proof-of-Stake consensus rules. Aliases: Finality (Ethereum PoS), finalization, finalized checkpoint.

Source: 01-EF - Pending, then included, then final.

Finality delay. A condition where Ethereum continues producing blocks but lacks enough validator participation to finalize checkpoints. Aliases: Finality delay (Ethereum operations), delayed finality, finality stall.

Source: 02-EF - Real-World Ethereum Security Examples.

Finalization. The protocol step that makes a checkpoint and its prior history economically irreversible under the stated assumptions. Aliases: Finalization (PoS finality), finality, finalized checkpoint.

Source: 03-CM - Finalization: Irreversible History.

Finalized checkpoint. A checkpoint treated as economically irreversible because reverting it would require large-scale slashable validator behavior. Aliases: Finalized checkpoint (Ethereum finality), finality, finalized, finalized history.

Source: 02-EF - Gasper: Fork Choice + Finality.

Financial Exclusion. A condition where an actor is denied practical access to banks, payment networks, settlement systems, or related financial services. Aliases: financial isolation, payment exclusion.

Source: 03-AS - Royal Funds Frozen!.

Financial experiment. A way of understanding blockchain as a live environment for new assets, markets, incentives, and failure modes. Aliases: Financial experiment (blockchain), financial substrate.

Source: 05-AS - What Is Blockchain Really?.

Financial inclusion. Expanding access to useful financial services for people or businesses underserved by existing banking systems.

Source: 02-AS - Why Governments Care.

Financial Rail. A payment or settlement pathway through which value moves between participants. Aliases: financial rails, payment rail, settlement rail.

Source: 03-AS - Financial Rails Are Power.

FinCEN. The U.S. body whose rules under the Bank Secrecy Act set what a money services business must do. Businesses register with it, it fines firms that break AML laws, and it takes Section 311 actions. Aliases: Financial Crimes Enforcement Network.

Source: 05-FO - A VASP moves crypto for others.

Finger table. A Chord node's list of shortcut peers used to jump across the ring toward a target identifier. Aliases: Finger table (Chord), finger tables, shortcuts.

Source: 02-DS - DHT Example: Chord.

First Bitcoin transaction. The January 12, 2009 transfer of 10 BTC from Satoshi Nakamoto to Hal Finney. Aliases: First Bitcoin transaction (Bitcoin), Satoshi-to-Hal transaction.

Source: 01-BF - The Network Comes Alive: The First Blocks.

First principles. A learning approach that starts from core concepts and assumptions before moving to complex systems or applications.

Source: 01-IN - Week 1 teaches how it works, week 2 traces.

First service. The first exchange, OTC broker, darknet market or other service that the traced money reaches. The trace ends there, or where the money stops moving. Aliases: first service reached.

Source: 03-FO - This trace stopped at a wallet, not a service.

First-spy heuristic. A deanonymization method that treats the first peer observed relaying a transaction as the best guess for its source. Aliases: First-spy heuristic (Bitcoin forensics), first spy, first-spy.

Source: 04-BF - Timing & Latency Attacks.

Fiscal Policy. Government choices about spending and taxation used to influence economic conditions.

Source: 04-IN - The Ideological Divide: Austrian vs. Keynesian Economics.

Fix fingers. A maintenance task where peers refresh shortcut entries that may have become stale after other peers leave or join. Aliases: Fix fingers (Chord maintenance), finger repair, fix_fingers().

Source: 02-DS - The "Self-Healing" Solutions.

Fixed-supply rule. Bitcoin's monetary rule that the protocol should not permit creation beyond its capped supply schedule. Aliases: Fixed-supply rule (Bitcoin), capped supply, supply cap.

Source: 04-BF - Bitcoin Overflow Bug.

Flash loan fee. The fee a flash loan borrower returns with the amount borrowed, 0.05% on Aave at deployment and changeable by governance vote.

Source: 10-FO - A flash loan lasts one transaction.

Flash loan pool. A DeFi pool whose deposits are lent out within one transaction; it decides only from whether the loan comes back.

Source: 10-FO - A DeFi pool is a contract anyone can call.

Flooding. A forwarding pattern where a peer sends a query to all neighbors, who repeat the process until a limit is reached. Aliases: Flooding (P2P routing), Flooding (P2P search), blind broadcast, flood, flooded query, flooding search, query flooding.

Source: 01-DS - Case Study: Gnutella & "Flooding".

Foreign exchange. The exchange of one currency for another, especially in cross-border payment and settlement workflows. Aliases: FX.

Source: 02-AS - mBridge So Far.

Foreign Reserves. State-held external assets used to support currency stability, international payments, and crisis response. Aliases: international reserves, reserve assets.

Source: 03-AS - Royal Funds Frozen!.

Forensic analysis. The investigation of blockchain activity, artifacts, and transaction patterns to support analysis or attribution. Aliases: Forensic analysis (blockchain), blockchain forensics, forensics.

Source: 01-IN - You will say what activity is, and how sure.

Forensic substrate. The persistent, signed, timestamped transaction record that supports later tracing and analysis. Aliases: Forensic substrate (blockchain), auditable financial record.

Source: 06-EF - Analyst Implications.

Fork. A divergence where separated parts of a blockchain network extend different versions of the chain. Aliases: Fork (Ethereum PoS), Fork (blockchain partition), blockchain fork, competing branch, short-term fork, temporary fork.

Source: 02-DS - The Blockchain Link.

Fork choice. The protocol logic nodes use to choose the current best chain head among competing valid branches. Aliases: Fork choice (Ethereum), fork-choice rule, head selection.

Source: 02-EF - Forks in Proof of Stake — Causes and Resolution.

Fork choice rule (Proof of Stake). The deterministic rule nodes use to choose the current chain head among competing viable branches. Aliases: Fork choice rule (PoS consensus), fork choice, fork-choice rule, fork-choice rules.

Source: 03-CM - Fork Choice Rules.

Fork race. A competition between competing valid branches where whichever branch spreads and gains work faster becomes the accepted history. Aliases: Fork race (proof of work), fork-race win, race between branches.

Source: 04-BF - Selfish Mining.

Formalization. Movement from informal or lightly governed activity into licensed, recognized, and supervised channels. Aliases: formal integration, regulatory formalization.

Source: 03-AS - Sub-Saharan Africa: Overview.

Fragmentation. The splitting of users, assets, standards, and infrastructure across many partially incompatible chains or platforms. Aliases: Fragmentation (crypto ecosystem), ecosystem fragmentation, market fragmentation.

Source: 01-AS - Complications and Risks.

Fragmented adoption. A future where blockchain use survives unevenly across isolated sectors, regions, or communities without one coherent ecosystem. Aliases: Fragmented adoption (blockchain futures), constrained adoption, fragmentation.

Source: 05-AS - A Spectrum of Futures.

Fragmented Oversight. A regulatory condition where several agencies or institutions share authority without one unified rulemaking center. Aliases: fragmented regulation, fragmented supervision.

Source: 03-AS - United States: Overview.

Fraud detection and monitoring. Using models and transaction analysis to identify suspicious flows, risky patterns, or possible abuse in blockchain activity. Aliases: Fraud detection and monitoring (blockchain analytics), fraud monitoring, transaction monitoring.

Source: 05-AS - Opportunities.

Fraud proof. Evidence submitted to challenge an incorrect Layer 2 result so the base chain can reject or correct it. Aliases: Fraud proof (Layer 2 verification), fraud proofs.

Source: 01-AS - Scaling and Interoperability.

Fraud tactic. The one tactic AADAPT adds to ATT&CK's list: the adversary trying to illicitly create, acquire or use value-form. It exists because much digital-asset crime takes value without an intrusion. Aliases: ADTA0001, Fraud (AADAPT tactic).

Source: 12-FO - AADAPT: ATT&CK's shape for digital assets.

Free identity. The condition in an open P2P network where creating new peer identities has little or no cost. Aliases: Free identity (P2P security), cheap identity, identity is free.

Source: 02-DS - The Great P2P Weakness: Sybil & Eclipse Attacks.

Freeze authority. The account a Solana mint names as authorized to freeze tokens in a token account. Aliases: freeze authorities.

Source: 08-EF - Solana keeps each token in its own account.

Freeze window. The time during which funds stay in an issuer's blacklistable token and can still be frozen; it closes once they are swapped out of that token.

Source: 05-EF - Freezable? Which issuer, and how fast?.

Fresh address. A new address a wallet hands out for each payment or customer, so one wallet ends up holding outputs at many addresses. Aliases: fresh addresses, new address per payment.

Source: 05-BF - A wallet manages many keys and addresses.

Friends With Benefits. A Web3-focused social DAO where application review and FWB token ownership gate membership spaces and participation. Aliases: FWB, Friends With Benefits (social DAO).

Source: 06-EF - Case Study: Friends With Benefits.

Frosties. An NFT project whose January 2022 sellout raised about $1.1 million before operators abandoned the project and drained proceeds. Aliases: Frosties (rug pull case study).

Source: 06-EF - The Frosties Rug Pull.

Full node. A blockchain node that downloads and validates every transaction and block, often serving data to weaker peers. Aliases: Full node (blockchain network), full nodes, validating node.

Source: 01-DS - The Blockchain Link.

Full node (Bitcoin). A Bitcoin node that independently enforces validity rules and rejects invalid transactions or blocks. Aliases: Full node (Bitcoin network), full nodes, full validating node, node.

Source: 01-BF - Security Model & Threats.

Full validation. Verification of the entire relevant transaction and block history under Bitcoin consensus rules. Aliases: Full validation (Bitcoin).

Source: 03-BF - Types of Nodes.

Full viewing key. A Zcash read-only key that shows all transactions to and from an address, without giving any power to spend. Aliases: Zcash full viewing key.

Source: 07-FO - A view key shows payments in, not spends.

Full-node wallet. A wallet run on the user's own full node, which scans the chain itself and tells no server which addresses it holds. Aliases: own-node wallet, wallet on its own node.

Source: 06-BF - Servers learn which addresses a wallet has.

Fully Homomorphic Encryption. Encryption that lets a system compute directly on encrypted data and later decrypt the result to the same answer as a plaintext computation. Aliases: FHE, fully-homomorphic encryption.

Source: 05-CP - Fully Homomorphic Encryption (FHE).

Function. A rule that maps each input from one set to one output in another set. Aliases: Function (mathematics), functions.

Source: 01-CP - Sets and Functions.

Function modifier. A Solidity construct used to add reusable checks or behavior around function execution. Aliases: Function modifier (Solidity), modifier, modifiers.

Source: 04-EF - Solidity: File Structure, From Top to Bottom.

Function mutability. A function annotation that communicates whether a function may read contract state, write contract state, or avoid state entirely. Aliases: Function mutability (Solidity), mutability, pure, view.

Source: 04-EF - Visibility, Mutability, and Payable.

Function selector. The first four bytes of the keccak-256 hash of a function signature, used to route calldata to the intended function. Aliases: 4-byte selector, Function selector (Ethereum ABI), selector.

Source: 04-EF - ABI, Encoding, and Function Selectors.

Function signature. The canonical function name and parameter-type string that is hashed to derive a function selector. Aliases: Function signature (Ethereum ABI), name(types).

Source: 04-EF - ABI, Encoding, and Function Selectors.

Function visibility. A function or variable annotation that controls whether code can be accessed externally, internally, by derived contracts, or only within the same contract. Aliases: Function visibility (Solidity), external, internal, private, public, visibility.

Source: 04-EF - Visibility, Mutability, and Payable.

Fungible token. A token type where each unit is interchangeable with any other unit of the same asset. Aliases: ERC-20 token, ERC-20 tokens, Fungible token (tokenization), fungible tokens.

Source: 05-EF - ERC-20: six functions and two events.

G

Galois/Counter Mode. A modern mode that combines CTR-style encryption with an authentication tag. Aliases: AES-GCM, GCM.

Source: 03-CP - Modes of Operation Overview.

Gap limit. The rule that a restoring wallet stops looking after 20 unused addresses in a row. Aliases: address gap limit.

Source: 05-BF - The path names coin, account, chain and index.

Garantex. An exchange that received nearly $6 million from the Conti ransomware gang, as of April 2022.

Source: 03-FO - A ransom cashes out at a service.

Gas (Ethereum). The unit Ethereum uses to measure and charge for computation, storage, and other EVM operations. Aliases: Gas (Ethereum execution), Gas (Ethereum resource unit), gas metering.

Source: 01-EF - A key or code controls each account.

Gas limit. A cap on how much gas a transaction or block may consume. Aliases: Gas limit (Ethereum), block gas limit, transaction gas limit.

Source: 01-EF - Gas prices the work; a failed call still pays.

Gas metering. The EVM rule that charges gas for operations so contract execution consumes bounded network resources. Aliases: Gas metering (EVM), Gas metering (Ethereum), gas-metered computation, metering computation.

Source: 01-EF - Every node runs the same program: the EVM.

Gasper. Ethereum's consensus protocol combining LMD-GHOST fork choice with Casper-FFG finality. Aliases: Ethereum full consensus protocol, Gasper (Ethereum consensus).

Source: 02-EF - Gasper: Fork Choice + Finality.

Genesis Block. The first Bitcoin block, mined by Satoshi Nakamoto on January 3, 2009. Aliases: Block 0, Genesis Block (Bitcoin), genesis block.

Source: 01-BF - The Network Comes Alive: The First Blocks.

GENIUS Act. The 2025 U.S. framework establishing a federal path for regulated payment stablecoins.

Source: 03-AS - United States: Impact Signals.

getdata message. A Bitcoin protocol message used to request full data for an announced transaction or block. Aliases: getdata, getdata message (Bitcoin protocol).

Source: 03-BF - Communication Protocol Basics.

getReserves(). The Uniswap v2 pool function that returns its two reserves at the moment it is called. Aliases: getReserves.

Source: 07-EF - A swap leaves Swap and Sync logs.

Getter. A read function that returns contract state, sometimes generated automatically for public state variables. Aliases: Getter (Solidity), auto-generated getter, getter function.

Source: 04-EF - Example: SimpleStorage (Code).

Gitcoin. A Web3 funding ecosystem that uses grants, matching pools, and token governance to support open-source and public-goods work. Aliases: GTC, Gitcoin (public-goods DAO).

Source: 06-EF - Case Study: Gitcoin.

Global settlement layer. A blockchain-based or blockchain-inspired layer for settling payments, tokenized claims, and financial obligations across institutions. Aliases: shared settlement system.

Source: 05-AS - Future 1: Global Settlement Layer.

Global state. The shared Ethereum record of account balances, nonces, code, and storage after accepted transactions are applied. Aliases: Ethereum state, Global state (Ethereum), shared global state.

Source: 01-EF - Ethereum keeps accounts, not coins.

Global trust vector. The final EigenTrust score set in which each entry represents a peer's global reputation. Aliases: global reputation vector, trust vector.

Source: 03-DS - EigenTrust: From Local Opinions to Global Scores.

Gnutella. An early decentralized P2P file-sharing network that searched by forwarding queries through peers instead of using a central index. Aliases: Gnutella (case study).

Source: 01-DS - Case Study: Gnutella & "Flooding".

Gold-Exchange Standard. A monetary arrangement where currencies are linked to a reserve currency that is itself convertible into gold.

Source: 04-IN - The Turning Point: Ending Bretton Woods.

Gossip. A propagation method where nodes forward new transactions or blocks to a small set of peers, who validate and relay them onward. Aliases: Gossip (P2P propagation), Gossip (blockchain network), controlled flood, gossip protocol, gossiping.

Source: 01-DS - Gossip in a Blockchain Network.

Gossip membership. A maintenance technique where peers exchange small samples of known peers to keep neighbor lists fresh. Aliases: membership gossip, peer-list gossip.

Source: 02-DS - The P2P Playbook: Surviving Churn.

Gossip protocol. A propagation method where each node forwards valid messages to neighboring peers until information spreads through the network. Aliases: Gossip protocol (Bitcoin network), gossip, peer relay.

Source: 02-IN - Nodes check a payment before passing it on.

Gossip relay. A dissemination pattern where peers forward new transactions or blocks broadly through neighbor-to-neighbor communication. Aliases: Gossip relay (P2P networking), gossip, gossip propagation, unstructured relay.

Source: 04-AS - Gossip vs DHTs.

Governance. The rules and institutions that decide membership, upgrades, dispute handling, and who can change the system. Aliases: Governance (distributed ledgers), decision rights, ledger governance.

Source: 04-DS - Distributed Ledger vs. Centralized Database (Compare/Contrast).

Governance coordination. The social process of reaching agreement on protocol changes, upgrades, or responses to system problems. Aliases: Governance coordination (blockchain protocols), coordination and agreement, protocol coordination.

Source: 05-AS - Human & Economic Constraints.

Governance manipulation. Attempts to distort protocol decisions through bribery, messaging, voting strategies, or other pressure on governance processes. Aliases: Governance manipulation (token-governed systems), governance attack, voting manipulation.

Source: 05-AS - Risks.

Governed membership. A participation model where admission, authorization, and membership changes are handled by governance outside the voting protocol. Aliases: controlled membership, known participants.

Source: 01-CM - Open chains buy openness with slower finality.

Government attribution. A government's public statement naming who carried out a theft, such as the FBI naming North Korea for Bybit. The analyst quotes it with the agency and date and does not attribute a state from the chain alone. Aliases: attribution to a state, state attribution.

Source: 13-FO - You characterize; the government attributes.

Government release. An official statement that names an actor or service, often with addresses, such as OFAC's Hydra addresses or the FBI's Bybit notice. It rarely states its basis, and lists change, so cite its date. Aliases: government designation release, government listing.

Source: 14-FO - The chain says what moved; others say who.

Graceful degradation. A failure-handling approach where a system keeps providing reduced service instead of collapsing completely.

Source: 01-DS - Why Distribute? Motivations and Risks.

graffiti. An optional 32-byte metadata field in a beacon block. Aliases: graffiti (beacon block field), graffiti field.

Source: 02-EF - Beacon Blocks and Execution Payloads.

Great Inflation. The high and volatile inflation period from about 1965 to 1982 highlighted as a post-Bretton Woods policy stress.

Source: 04-IN - The Problem of Value: The Specter of Inflation.

Greedy routing. A forwarding rule that sends a query to the neighboring zone that appears closest to the target coordinate. Aliases: Greedy routing (CAN), greedy walk.

Source: 02-DS - DHT Example: CAN.

Grover's Algorithm. A quantum search algorithm that gives a quadratic speedup for unstructured search problems. Aliases: Grover's algorithm.

Source: 05-CP - Grover's Algorithm (1996).

Guess-newest heuristic. Guessing that the newest member of a Monero ring is the real input, which was right about 80% of the time in early Monero because wallets drew decoys from older outputs. Aliases: guess-newest, newest is real.

Source: 07-FO - Old Monero leaked: zero decoys, newest is real.

Gwei. A common subunit of Ether, where one ETH equals one billion gwei. Aliases: Gwei (Ether unit).

Source: 01-EF - Gas prices the work; a failed call still pays.

H

Halving. The scheduled event that cuts Bitcoin's block subsidy in half every 210,000 blocks, roughly every four years. Aliases: Halving (Bitcoin monetary policy), Halving (Bitcoin), halvings, reward halving, subsidy halving.

Source: 01-BF - Monetary Policy.

Hard fork. An incompatible protocol change that can split a network into separate chains if different groups keep following different rules. Aliases: Hard fork (Ethereum governance), Hard fork (blockchain governance), Hard fork (blockchain protocol), chain split, fork, hard-fork split, network split.

Source: 04-DS - Governance Models.

Hard Money. Money with a difficult-to-expand and predictable supply, often contrasted with discretionary fiat currency.

Source: 04-IN - The Problem of Value: The Specter of Inflation.

Hardened derivation. A derivation-path level marked ' (purpose, coin type and account) that no public key can derive across. Aliases: hardened key, hardened level, hardened levels.

Source: 05-BF - An xpub reveals every address, past and future.

Hardware security module. A dedicated device or service designed to protect private keys and perform cryptographic operations without exposing raw key material. Aliases: HSM, HSMs.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Hardware wallet. A device that stores cryptocurrency private keys separately from general-purpose computers and signs transactions without exposing the key. Aliases: hardware wallets.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Harmony Horizon bridge hack. A 2022 theft from Harmony's Horizon bridge that the FBI said on 23 January 2023 the Lazarus Group carried out. Over $96M of its proceeds went through Tornado Cash. Aliases: Harmony's Horizon bridge, Horizon bridge hack.

Source: 13-FO - The names come from the government.

Harvest Now, Decrypt Later. An attack strategy where encrypted data is collected today so it can be decrypted later if future cryptanalytic or quantum advances break its protection. Aliases: HNDL, harvest-now-decrypt-later.

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Hash anchor. A hash recorded on the shared channel so peers can verify that private data has not been silently changed. Aliases: Hash anchor (private data), hash anchoring, integrity anchor.

Source: 02-PC - Private Data Collections.

Hash collision. A case where two distinct inputs produce the same hash digest. Aliases: collision, collisions.

Source: 02-CP - What Is a Cryptographic Hash Function?.

Hash commitment. A published digest that binds someone to underlying data while allowing the data to remain hidden until later disclosure. Aliases: commitment, hashing as commitment.

Source: 02-CP - Hashing as Commitment.

Hash function. A function that maps data to a compact value used to identify data and detect changes. Aliases: cryptographic hashing, hash functions, hashing.

Source: 01-CP - Sets and Functions.

Hash pointer. A hash reference in a block header that points to the previous block and links the chain together. Aliases: Hash pointer (Bitcoin), previous block hash, previous block header hash.

Source: 02-BF - The Merkle root commits to every transaction.

Hash power. The amount or share of proof-of-work computation controlled by a miner, pool, or network. Aliases: Hash power (proof of work), hash rate, hash share, mining power.

Source: 04-BF - Selfish Mining.

Hash puzzle. A PoW puzzle where miners search for a nonce so the hash of candidate data falls below a target. Aliases: hash-based puzzle.

Source: 02-CM - Proof-of-work Puzzles.

Hash rate. The number of hash attempts miners can perform per second, either individually or across the whole network. Aliases: H, network hash rate.

Source: 02-CM - Difficulty, Hash Rate, and Expected Work.

Hash-based cryptography. A post-quantum approach, especially for signatures, that builds security from hash-function assumptions. Aliases: Hash-based cryptography (post-quantum), hash-based signatures.

Source: 05-CP - Post-Quantum Cryptography: The New Toolbox.

Hash-chaining. A chain of hashes where each later entry depends on earlier data, causing past edits to break later links. Aliases: Hash-chaining (immutability source), hash chain, hash-chained history.

Source: 04-DS - Immutability Source.

Hash-linked chain. A sequence of blocks where each block includes the hash of the previous block, making earlier changes visible and costly to repair. Aliases: Hash-linked chain (Bitcoin), chain, cryptographically linked chain.

Source: 01-BF - How Double-Spending is Solved.

Hash-linking. A structure where each entry commits to the previous entry with a hash so later history depends on earlier contents. Aliases: Hash-linking (distributed ledgers), cryptographic linking, cryptographic links, hash linking.

Source: 04-DS - From Logs to Ledgers.

Hashcash. An anti-spam system that used computational work to make abuse costly and later influenced Bitcoin's Proof-of-Work design.

Source: 03-IN - Framing.

HD wallet. A hierarchical deterministic wallet: it derives an unlimited number of key pairs from one seed as a tree, inside the wallet software, with no registry or issuer. Aliases: BIP 32, BIP32, HD wallets, hierarchical deterministic wallet.

Source: 05-BF - One seed grows a whole tree of keys.

Health check. The check Euler's donate function lacked, so a user could make their own account insolvent and eligible for liquidation.

Source: 10-FO - Euler, March 2023: a loan, a gift, a liquidation.

Heaviest chain. The valid chain with the most accumulated proof-of-work, used as the surviving history in Nakamoto-style consensus. Aliases: Heaviest chain (Proof-of-Work), most-work chain.

Source: 01-CM - A stale block is routine, not an attack.

heimdall-rs. An open-source (MIT) decompiler for EVM bytecode, the one the course names for exercises. Aliases: heimdall.

Source: 11-FO - A decompiler rebuilds structure, not intent.

Helium. A decentralized wireless network built from individual hotspots that provide connectivity over unlicensed radio frequencies. Aliases: Helium (IoT network).

Source: 05-IN - Everyday Examples of Distributed Systems.

Heterogeneity. The normal condition that network participants differ in hardware, software, operating systems, and connection quality. Aliases: Heterogeneity (distributed systems), heterogeneous network, heterogeneous nodes.

Source: 01-DS - Fallacy #8: The Network Is Homogeneous.

Heuristic. An evidence-based rule of thumb used to infer likely relationships or behavior when direct proof is unavailable. Aliases: Heuristic (forensic analysis), Heuristic (forensics), forensic heuristic, heuristics, inference rule, rule of thumb.

Source: 01-FO - Challenges in Forensic Analysis.

Hidden infrastructure. The services users usually do not see, such as RPC providers, nodes, indexers, oracles, bridges, and scaling layers. Aliases: Hidden infrastructure (blockchain applications), behind-the-scenes infrastructure, supporting infrastructure.

Source: 01-AS - What Users Actually Do.

Hivemapper. A decentralized mapping network where contributors collect imagery to update a shared global map. Aliases: Hivemapper (mapping network).

Source: 05-IN - Everyday Examples of Distributed Systems.

HMAC. A standard hash-based MAC construction that uses inner and outer keyed hashes to authenticate messages safely. Aliases: Hash-based Message Authentication Code, hash-based message authentication code.

Source: 02-CP - Aside: HMAC.

Holder. The party that stores a verifiable credential and presents it when proof is needed. Aliases: Holder (verifiable credentials), credential holder.

Source: 01-AS - Identity Case: Verifiable Credentials.

Holding address. An address where money has been gathered and left still. A trace that reaches one has nowhere further to go until the money moves. Aliases: holding wallet.

Source: 03-FO - This trace stopped at a wallet, not a service.

Homomorphic encryption. Encryption that allows computation on encrypted data without first decrypting it. Aliases: FHE, fully homomorphic encryption.

Source: 04-DS - Transparency vs. Privacy.

Homomorphic evaluation. The operation of applying a function to ciphertexts so the encrypted output corresponds to the function's plaintext result. Aliases: Eval, homomorphic Eval.

Source: 05-CP - FHE at a Glance.

Honest supermajority. An assumption that enough stake follows the protocol, commonly two-thirds or more for finality decisions. Aliases: Honest supermajority (PoS assumption), honest 2/3 stake, honest majority.

Source: 03-CM - Safety and Liveness — PoS Invariants.

Honest work. Proof-of-work contributed by miners following the protocol, which Bitcoin assumes outweighs attacker work. Aliases: Honest work (Bitcoin security), honest majority work, majority of honest work.

Source: 01-BF - Security Model & Threats.

Hop. One move of funds from one address to the next in a trace. Each hop is written with the heuristic that supports it and a confidence word. Aliases: hops, trace hop.

Source: 03-FO - The ransom moved four times in two days.

Hosted wallet. FinCEN's name for a custodial wallet: a third party such as an exchange holds the keys and executes transactions for the user. Aliases: hosted wallets.

Source: 05-BF - Whoever holds the key controls the coins.

Hot wallet. A cryptocurrency wallet whose private keys are kept on an internet-connected system for operational use. Aliases: Hot wallet (exchange forensics), central hot wallet, exchange hot wallet, hot wallets.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Hub-Building. Attracting firms, capital, and infrastructure to make a jurisdiction a regional or global center. Aliases: crypto hub-building, regulated hub-building.

Source: 03-AS - Gulf States: Overview.

Huione Group. A Cambodia-based crypto and payments group whose parts, Huione Pay, Huione Crypto and Haowang Guarantee, FinCEN found operating as one and laundering at least $4 billion between August 2021 and January 2025. Aliases: Haowang Guarantee, Huione, Huione Crypto, Huione Pay.

Source: 05-FO - Huione laundered at least $4 billion.

Hybrid cryptographic system. A transition design that combines existing cryptography with post-quantum methods to reduce migration risk. Aliases: Hybrid cryptographic system (quantum transition), hybrid cryptographic systems, hybrid cryptography.

Source: 05-AS - What Can Be Done.

Hybrid P2P model. A design that combines centralized services such as indexing or login with direct peer-to-peer data transfer. Aliases: hybrid peer-to-peer model.

Source: 01-DS - Case Study: Napster.

Hybrid Posture. A strategy combining market participation with deliberate state steering, taxation, supervision, and constraints. Aliases: state-guided hybrid posture.

Source: 03-AS - India: Overview.

Hybrid signature. A migration technique that combines a classical signature with a post-quantum signature so both protections can be checked during rollout. Aliases: Hybrid signature (post-quantum migration), hybrid deployment, hybrid signatures.

Source: 05-CP - PQC Migration in Blockchain Systems.

Hybrid trust system. A design that combines a global reputation baseline with local, context-aware personalization. Aliases: hybrid design.

Source: 03-DS - PeerTrust vs. EigenTrust: Summary.

Hydra. The world's largest darknet market when Treasury sanctioned it in April 2022. It sold ransomware-as-a-service, hacking tools, stolen data and drugs, and was a route to Russian exchanges. Aliases: Hydra Market.

Source: 03-FO - A ransom cashes out at a service.

Hyperledger. A family of enterprise-oriented blockchain technologies often associated with permissioned network designs. Aliases: Hyperledger (blockchain framework).

Source: 01-IN - Every lab ends in an answer to an instructor.

Hyperledger Fabric. A permissioned blockchain framework that needs no native cryptocurrency. Aliases: Fabric, HLF, Hyperledger Fabric (permissioned ledger platform), Hyperledger Fabric (permissioned ledger).

Source: 02-IN - A blockchain is a record, not a currency.

I

ICD 203. The analytic standard that asks analysts to state their confidence. The course uses it only to standardize its confidence words, not to write formal memos.

Source: 11-FO - How sure: what a code read rests on.

Idempotent operation. An operation that can be safely repeated without causing duplicate or unintended effects. Aliases: idempotency, idempotent operations.

Source: 01-DS - Fallacy #1: The Network Is Reliable.

Identity and membership. The mechanisms that establish who a participant is and what standing they have in the network. Aliases: membership identity.

Source: 01-PC - Permissioned: the operator decides who writes.

Identity model. The protocol assumption about who can participate and how participants are recognized. Aliases: Identity model (consensus), membership model.

Source: 01-CM - Open chains buy openness with slower finality.

Identity verification. The repeated process of checking that a Fabric action is backed by a valid identity and a signature that satisfies the relevant policy. Aliases: Identity verification (Hyperledger Fabric), identity check, signature verification.

Source: 02-PC - The Big Picture: End-to-End Fabric Path.

Immutability. The practical resistance of accepted blockchain history to unnoticed revision because later blocks depend on earlier hashes. Aliases: Immutability (Ethereum contracts), Immutability (blockchain), Immutability (distributed ledgers), contract immutability, immutable, immutable contract, immutable history, strict immutability.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Immutable chain. A hash-linked history where changing an earlier transaction changes its block's Merkle root and header and breaks every later block's link. Aliases: Immutable chain (Bitcoin), tamper-evident chain.

Source: 02-BF - The Merkle root commits to every transaction.

Implementation bug. A flaw in the code that enforces a protocol, even when the abstract design rule is sound. Aliases: implementation flaw, software bug.

Source: 04-BF - Implementation vs. Design.

Improvement proposal. A formal proposal for changing a permissionless protocol, such as a Bitcoin Improvement Proposal or Ethereum Improvement Proposal. Aliases: BIP, BIPs, EIP, EIPs, Improvement proposal (permissionless governance).

Source: 04-DS - Governance Models.

Inactivity leak. A mechanism that gradually reduces the balances of offline validators during extended non-participation so the chain can recover liveness. Aliases: Inactivity leak (Ethereum finality recovery), Inactivity leak (PoS penalty), inactivity leak penalty, inactivity penalty.

Source: 03-CM - Nothing-at-Stake and Economic Security.

Incentive. A reward or cost structure that encourages participants to behave in ways that support the system. Aliases: Incentive (information sources), Incentive (network design), economic incentives, incentives, source incentive.

Source: 03-IN - P2P Normalizes Decentralization.

Incentive design. The design of rewards, penalties, governance rights, and market pressures that shape participant behavior. Aliases: Incentive design (blockchain systems), incentives shape outcomes.

Source: 05-AS - Key Lessons.

Incentives. Protocol rewards and fees that make honest block production more profitable than attacking or abandoning the system. Aliases: Incentives (Bitcoin security), economic incentives, miner incentives.

Source: 01-BF - Bitcoin, Simplified.

Inclusion list. A proposed mechanism for requiring proposers to include specified pending transactions. Aliases: Inclusion list (Ethereum censorship mitigation), inclusion lists.

Source: 02-EF - Censorship.

Inclusion proof. A short set of sibling hashes that lets a verifier confirm an item is included under a known Merkle root. Aliases: Inclusion proof (Merkle tree), Merkle proof, lightweight client proof, membership proof.

Source: 04-AS - Why the Merkle Root Matters.

Incoming viewing key. A Zcash read-only key that shows the payments coming in to an address, but not payments out or change. Aliases: Zcash incoming viewing key.

Source: 07-FO - A view key shows payments in, not spends.

Independence. A relationship where the outcome of one event does not change the probability of another event. Aliases: Independence (probability), independent events.

Source: 01-CP - Probability Refresher: Independence & Random Variables.

Independent line of evidence. Support for an assessment that comes from a different kind of evidence than the rest. Counting these kinds, not names or labels, sets the confidence word. Aliases: independent line, kind of evidence, kinds of evidence, line of evidence.

Source: 11-FO - How sure: what a code read rests on.

Independent validation. The practice of checking transactions and blocks against Bitcoin rules on one's own node instead of trusting another party. Aliases: Independent validation (Bitcoin), local validation.

Source: 03-BF - What Is a Bitcoin Node?.

Independent verification. The ability for participants to check the record and its changes for themselves rather than trusting another party's private assertion. Aliases: Independent verification (distributed ledgers), independently verifiable, verification.

Source: 04-DS - The Shared Notebook Problem.

Indexed field. An event parameter stored as a log topic so off-chain clients can search or filter by it. Aliases: Indexed field (Solidity event), indexed, indexed parameter.

Source: 04-EF - Events and Logs.

Indexer. An off-chain tool that reads blockchain data, such as Transfer logs, and organizes it for search, display or analysis. Aliases: Indexer (blockchain application), Indexer (blockchain data), indexers, indexing service.

Source: 05-EF - A token transfer shows up as a log.

Indicator. Something that, if detected, would change a judgment. ICD 203 asks analysts to name them. Aliases: indicators.

Source: 14-FO - When sources disagree, say so.

Indirect trust. Trust extended through recommendations from other peers rather than through personal experience. Aliases: hearsay, recommendation-based trust.

Source: 03-DS - Direct vs. Indirect Trust.

Indistinguishability. A property where no efficient algorithm can reliably tell one distribution, such as PRG output, from another, such as true randomness. Aliases: Indistinguishability (cryptography), computational indistinguishability, indistinguishable.

Source: 01-CP - Probability Refresher: Events and Outcomes.

Inferno Drainer. A scam-as-a-service drainer that Group-IB studied from November 2022 to November 2023, finding 16,000 domains imitating more than 100 crypto brands and at least $80 million stolen. Aliases: Inferno.

Source: 09-FO - A drainer kit is phishing sold as a service.

Inflation. A rise in general prices that reduces a currency's purchasing power.

Source: 04-IN - The Problem of Value: The Specter of Inflation.

Inflation hedging. Using a stablecoin or other asset to preserve purchasing power when a local currency is rapidly losing value. Aliases: Inflation hedging (stablecoin), inflation hedge.

Source: 02-AS - Major Stablecoins and Use Cases.

Information Asymmetry. A situation where one side of a transaction knows more about quality or risk than the other side.

Source: 04-IN - The "Market for Lemons": When Trust Breaks Down.

Information isolation. A condition where each participant sees only the messages sent directly to them rather than the whole system view.

Source: 05-IN - Byzantine Generals Problem.

Infrastructure. In the Diamond Model, the structures used to deliver a capability and take the result. On chain these are addresses, contracts, mixers and bridges. Aliases: infrastructure vertex.

Source: 12-FO - The Diamond: four features of one event.

Infrastructure centralization. A concentration of important network services in a small number of operators or implementations. Aliases: Infrastructure centralization (Ethereum security), builder centralization, relay centralization.

Source: 02-EF - Censorship.

Infrastructure chokepoint. A concentrated dependency where many users or apps rely on one provider, creating availability, censorship, or visibility risk. Aliases: centralized dependency, chokepoint, service dependency.

Source: 01-AS - RPC Providers: The Access Layer.

Infrastructure concentration. A condition where users, applications, or value rely heavily on a small number of providers, keys, bridges, or data feeds. Aliases: concentrated infrastructure.

Source: 06-EF - Centralization Chokepoints.

Infrastructure-scale adoption. A future where blockchain-style systems become part of major settlement, payment, and institutional infrastructure. Aliases: Infrastructure-scale adoption (blockchain futures), global infrastructure, infrastructure-scale blockchain adoption.

Source: 05-AS - A Spectrum of Futures.

Initcode. The contract creation code executed during deployment, including constructor logic, before returning runtime bytecode. Aliases: Initcode (Ethereum), init code, initialization code.

Source: 04-EF - Contract Deployment.

Initial block download. The first synchronization process in which a full node downloads and verifies blockchain history from the beginning. Aliases: IBD, Initial block download (Bitcoin).

Source: 03-BF - Setting Up a Full Node.

Initial Coin Offering bubble. A period of speculative token fundraising in which many projects raised money without durable products or value. Aliases: ICO bubble, ICO speculation.

Source: 05-AS - When Blockchain Systems Broke.

Initialization vector. A public value supplied with ciphertext so repeated plaintexts encrypt differently under modes that require freshness. Aliases: IV.

Source: 03-CP - Padding and Initialization Vectors.

Insider. A participant who appears to belong to the group but secretly acts against the group's shared objective. Aliases: Insider (adversarial participant), malicious insider.

Source: 05-IN - Opening Exercise: The Insider.

Institutional control. The influence of banks, firms, regulators, exchanges, and other organized actors over how blockchain systems are used and governed. Aliases: Institutional control (blockchain adoption), control and regulation, institutional adoption.

Source: 05-AS - What Actually Happened.

Institutional skepticism. A cautious or critical view from regulators and financial institutions that blockchain may offer limited net benefit or unacceptable risk. Aliases: Institutional skepticism (blockchain adoption), skeptical institutional critique.

Source: 05-AS - Future 3: Decline or Fragmentation.

Institutional Trust. Confidence that financial institutions, regulators, and intermediaries are accurately managing risk and honoring obligations.

Source: 04-IN - Case Study: The 2008 Global Financial Crisis.

Integer factorization. The problem of decomposing a large composite number into its prime factors. Aliases: Integer factorization (public-key cryptography), factor large integers, factoring.

Source: 05-CP - Shor's Algorithm (1994): Breaking RSA and ECC.

Integer overflow. A programming error where a number exceeds the range a field can represent and wraps into an unintended value. Aliases: Integer overflow (software implementation), overflow.

Source: 04-BF - Bitcoin Overflow Bug.

Integrity. A security property that lets users detect whether data or messages have been changed. Aliases: Integrity (distributed ledgers), data integrity, message integrity.

Source: 01-CP - What Is Cryptography?.

Intelligence-based clustering. Clustering that draws on data leaks, court documents and data partnerships rather than on-chain heuristics. A vendor tag built this way may be a court record you have already counted.

Source: 14-FO - An echo is not a second line.

Intent. A liquidity bridge design, such as Across: the user deposits on chain A, a relayer pays them on chain B from its own tokens, and the protocol repays the relayer later. Aliases: intents.

Source: 08-EF - A bridge moves a claim, not the coin.

Interactive consistency. A Byzantine agreement goal requiring loyal actors to reach the same plan despite faulty or malicious actors.

Source: 05-IN - The Byzantine Generals Problem (formal).

Interference. The quantum effect algorithms use to amplify useful answers and suppress wrong paths in certain computations. Aliases: Interference (quantum computing), quantum interference.

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Intermediary. A middle party such as a bank, card network, government, exchange, or platform that helps verify and enforce a transaction. Aliases: intermediaries, trusted middle party, trusted third parties, trusted third party.

Source: 03-IN - Digital Trust: The Centralized Model.

Internal call. A call one contract makes to another contract or address while running a transaction, which can send ETH; it is not a new transaction and shows only in a trace. Aliases: internal ETH transfer, internal calls.

Source: 01-EF - Contracts move value the transaction won't show.

Internal chain. The branch of an HD wallet account that makes change addresses; change level 1 in the derivation path. Aliases: change chain.

Source: 05-BF - One seed grows a whole tree of keys.

Internal transfer. A payment between two customers of the same exchange that the platform records on its own books and that may never reach the chain. Aliases: internal transfers, off-chain transfer.

Source: 05-BF - Whoever holds the key controls the coins.

Interoperability. The ability of a ledger system to work with external systems, institutions, or standards. Aliases: Interoperability (blockchain systems), Interoperability (ledger design), Interoperability (settlement infrastructure), cross-chain interoperability, external interoperability, interconnected systems, interoperable systems.

Source: 04-DS - Case Study B — Public Sector: EBSI & Estonia (KSI){.smaller}.

InterPlanetary File System. A peer-to-peer storage and sharing network where files are located by cryptographic hash rather than by one central server location. Aliases: IPFS, InterPlanetary File System (NFT storage).

Source: 05-IN - Everyday Examples of Distributed Systems.

Intrusion event. In the Diamond Model, one step in which an adversary uses a capability over infrastructure against a victim to produce a result. Not every feature needs to be known to record one. Aliases: Diamond event.

Source: 12-FO - The Diamond: four features of one event.

Invalid transaction. A transaction that remains in the block log for audit but does not update world state because validation failed. Aliases: Invalid transaction (Hyperledger Fabric), invalidated transaction, marked invalid.

Source: 02-PC - Phase B: Ordering, Validation, and Commit.

Invariant. A condition that should remain true after every valid contract call. Aliases: Invariant (smart contracts), accounting invariant, contract invariant.

Source: 04-EF - Example: Coin (Code, Part 1).

Inventory message. A Bitcoin protocol message used to announce that a node has a transaction or block available. Aliases: Inventory message (Bitcoin network), Inventory message (Bitcoin protocol), inv, inv message, inv(tx).

Source: 03-BF - Communication Protocol Basics.

Invoke. A client-requested transaction flow that proposes a chaincode action, gathers endorsements, submits for ordering, and reaches validation and commit. Aliases: Fabric invoke, Invoke (Hyperledger Fabric), transaction invoke.

Source: 02-PC - One Invoke: Proposal to Commit.

Irregular state change. The DAO fork at block 1,920,000, which moved the funds to a recovery address without rolling back transactions.

Source: 10-FO - The DAO: one split called many times over.

Irreversible history. Blockchain history that cannot be reverted without large-scale slashable validator misconduct. Aliases: Irreversible history (PoS finality), finalized history, irreversibility.

Source: 03-CM - Finalization: Irreversible History.

Irreversible transaction. A transaction that cannot normally be undone once accepted by the network. Aliases: Irreversible transaction (blockchain risk), irreversible settlement, irreversible transactions.

Source: 01-AS - Fraud and Scams.

isBlackListed. USDT's public mapping holding one true or false per address; anyone can query it, and transfer refuses a sender it marks true. Aliases: isBlackListed mapping.

Source: 05-EF - Tether's blacklist: three owner-only functions.

Issuer. The party that creates and signs a verifiable credential. Aliases: Issuer (digital money), Issuer (verifiable credentials), credential issuer, issuers.

Source: 01-AS - Identity Case: Verifiable Credentials.

Issuer action. Something a token issuer does to an address, such as a freeze. It shows that someone with records acted on this address on this date, but rarely says why. Aliases: issuer actions.

Source: 14-FO - The chain says what moved; others say who.

Iterative propagation. Repeatedly passing trust values through the network until the scores converge and stabilize. Aliases: trust propagation.

Source: 03-DS - EigenTrust: From Local Opinions to Global Scores.

J

JSON-RPC endpoint. A network address that accepts JSON-RPC requests for chain data such as balances, blocks, logs, or transaction submission. Aliases: API endpoint, JSON-RPC endpoint (blockchain access), JSON-RPC endpoints, endpoint.

Source: 01-AS - RPC Providers: The Access Layer.

Justification. The state reached when a supermajority of total stake votes for a link between checkpoint epochs. Aliases: Justification (PoS finality), justified link, supermajority vote link.

Source: 03-CM - Justification: Supermajority Vote Links.

Justified checkpoint. A checkpoint that has received at least two-thirds of total stake voting for the required source-to-target link. Aliases: Justified checkpoint (Ethereum finality), justification, justified.

Source: 02-EF - Gasper: Fork Choice + Finality.

K

Kademlia. A DHT design family that uses prefix-like distance logic for efficient peer and content lookup.

Source: 02-DS - The Blockchain Link: Efficient Routing.

Keccak-256. Ethereum's SHA-3-related hash function using pre-standardization Keccak padding rather than standardized SHA3-256 padding. Aliases: Keccak, Keccak sponge construction, Keccak-256 (Ethereum).

Source: 02-CP - Sponge Construction: How SHA-3 is Built.

Kelp DAO exploit. The 18 April 2026 attack in which a forged LayerZero message, claiming to come from Unichain, released rsETH on Ethereum and about $292 million was drained. Aliases: Kelp DAO, Kelp DAO hack.

Source: 08-EF - One sequencer orders each rollup's trades.

Key compromise. A failure where an attacker gains control of private keys or administrator credentials used to authorize ledger actions. Aliases: Key compromise (ledger operations), compromised keys, credential compromise.

Source: 04-DS - Threats to Distributed Ledgers (cont.).

Key Encapsulation Mechanism. A public-key mechanism for securely establishing shared key material, often used as a building block in modern encrypted protocols. Aliases: KEM, key encapsulation mechanisms.

Source: 05-CP - Post-Quantum Cryptography: The New Toolbox.

Key handling. The operational discipline of generating, storing, using, rotating, and protecting cryptographic keys. Aliases: key management, private-key handling.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Key image. A value derived from the real output's private key and published when a Monero output is spent; spending the same output twice repeats the key image, so the network refuses the second spend. Aliases: key images.

Source: 07-FO - A ring hides the spender among 15 decoys.

Key management. The operational practice of securely distributing, storing, using, and replacing symmetric keys. Aliases: Key management (blockchain operations), Key management (blockchain users), Key management (symmetric cryptography), Key management (wallet usability), key distribution, key rotation, key-management, private key management, private-key management, wallet recovery.

Source: 03-CP - Why Symmetric Crypto.

Key mixing. A block-cipher step that combines data with a round key, commonly using XOR. Aliases: round-key mixing.

Source: 03-CP - Building Blocks of Block Ciphers.

Key pair. A private key and the public key computed from it. Aliases: Key pair (asymmetric cryptography), Key pair (permissionless identity), key pairs, public-private key pair, public/private key pair.

Source: 04-CP - A private key is a number that controls coins.

Key path spend. A Taproot spend with one key and one signature, which shows nothing of what stood behind that key. Aliases: Taproot key path, key path, key-path spend.

Source: 06-BF - Taproot multi-sig can look like one key.

Key space. The set of possible keys or seeds an attacker would need to search. Aliases: keyspace.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Key store. A storage mechanism intended to hold cryptographic keys with access controls and platform protection. Aliases: OS-protected keystore, key store, keystore.

Source: 05-CP - Case Study: CryptoDefense — Keys Left in Plain Sight.

Keyless Signature Infrastructure. Estonia's hash-linked integrity infrastructure for proving that government and healthcare records have not been altered. Aliases: Estonia KSI, KSI.

Source: 04-DS - Case Study B — Public Sector: EBSI & Estonia (KSI){.smaller}.

Keynesian Economics. An economic framework that supports active government and central-bank intervention to manage recessions and demand.

Source: 04-IN - The Ideological Divide: Austrian vs. Keynesian Economics.

Keystream. A sequence of key-derived bits or bytes combined with plaintext, often by XOR, to produce ciphertext. Aliases: pseudorandom stream.

Source: 03-CP - Pseudo-Random Generators.

Know Your Customer. A compliance process that verifies a customer's real-world identity before allowing access to regulated financial services. Aliases: KYC, Know Your Customer (crypto compliance), Know Your Customer (identity), identity checks, mandatory ID checks.

Source: 01-FO - Decentralized Exchanges (DEXs).

KYC/AML. Compliance processes that collect or use customer identity and transaction information at financial services. Aliases: Anti-Money Laundering, KYC/AML (forensics and compliance), Know Your Customer.

Source: 04-AS - Attribution.

KYC/AML compliance. Identity and financial-crime controls that require institutions to know customers and monitor for illicit finance risks. Aliases: AML, Anti-Money Laundering, KYC, Know Your Customer.

Source: 03-IN - Digital Trust: The Centralized Model.

L

L2BEAT. A public site that rates rollups by how far they depend on people rather than code.

Source: 08-EF - L2BEAT stages: who can still change the rules.

L2BEAT Stages. L2BEAT's rating of a rollup from Stage 0 (fully controlled by few entities) through Stage 1 (mostly code, with a Security Council) to Stage 2 (fully controlled by code). Aliases: Stage 0, Stage 1, Stage 2, rollup stages.

Source: 08-EF - L2BEAT stages: who can still change the rules.

Label. A name attached to an address or cluster saying who controls it, such as an exchange; it is only as good as its source, and one label with an unknown basis is Low confidence. Aliases: address label, attribution label, open label set.

Source: 01-FO - A label is only as good as its source.

Lamport. SOL's smallest unit. Aliases: lamports.

Source: 08-EF - Solana: programs own the accounts.

Last-revealer bias. The small ability of the final randomness contributor to withhold its reveal and marginally influence the resulting seed. Aliases: Last-revealer bias (RANDAO), last revealer advantage.

Source: 02-EF - Global Randomness: RANDAO + VDF Beacons.

Latency. The time it takes for information to travel across a network or complete a protocol step. Aliases: Latency (distributed systems), Latency (ledger performance), network delay, propagation delay, transaction latency.

Source: 01-DS - Fallacy #2: Latency Is Zero.

Lattice. A repeating high-dimensional grid of points generated by basis vectors. Aliases: Lattice (cryptography), lattices.

Source: 05-CP - Lattices: Intuition Without the Math.

Lattice-based cryptography. A post-quantum cryptography family based on hard problems in high-dimensional lattices. Aliases: lattice cryptography.

Source: 05-CP - Post-Quantum Cryptography: The New Toolbox.

Laundering Chain. A sequence of services, assets, and transfers used to obscure the origin of funds before use or cash-out. Aliases: laundering infrastructure, laundering network.

Source: 03-AS - North Korea: Impact Signals.

Laundering playbook. The shape North Korea's laundering repeats across cases: out of freezable tokens, mix or swap, cross chains, then cash out. The services change as each is designated or shut; the shape stays. Aliases: North Korea's playbook.

Source: 13-FO - One playbook across North Korea's cases.

Layer 1. The underlying blockchain that provides final settlement, data publication, or dispute verification for higher-layer systems. Aliases: L1, Layer 1 (scaling), base chain, base layer.

Source: 01-AS - Scaling and Interoperability.

Layer 2. A separate execution environment built on top of a Layer 1 that processes transactions off the main chain and posts data or proofs back. Aliases: L2, L2s, Layer 2 (Ethereum scaling), Layer 2 system, Layer 2s.

Source: 01-EF - Most users reach the chain through a wallet.

Layer 2 network. A system that processes many user transactions away from the base chain while depending on that chain for settlement, data, or verification. Aliases: L2, L2 network, Layer 2, Layer 2 network (scaling), rollup.

Source: 01-AS - Scaling and Interoperability.

LayerZero. A message-verified bridge protocol; in the Kelp DAO exploit a forged LayerZero message released rsETH on Ethereum.

Source: 08-EF - Message bridges trust a message: Kelp DAO.

Lazarus Group. A North Korea-linked hacking organization associated with major exchange and bridge thefts. Aliases: DPRK-linked Lazarus Group.

Source: 03-AS - North Korea: Impact Signals.

Lazy mint. A minting pattern where a signed off-chain voucher authorizes token creation only when a buyer submits it on-chain. Aliases: lazy minting, signature-based mint.

Source: 06-EF - Minting on Ethereum.

Leader election. The process of selecting one node to coordinate a round or period of consensus activity.

Source: 05-IN - Classical Consensus Approaches.

Leaf node. A lower-capacity peer that connects to a supernode rather than participating heavily in network-wide flooding. Aliases: Leaf node (P2P network), leaf, leaf nodes.

Source: 01-DS - Gnutella's Fix: Supernodes & "Smart Flooding".

Leaf set. A Pastry node's set of numerically nearby peers used to finish routing once a lookup reaches the target neighborhood. Aliases: Leaf set (Pastry).

Source: 02-DS - DHT Example: Pastry.

Learning With Errors. A lattice-related problem based on solving noisy linear equations that are easy to generate but hard to recover. Aliases: LWE.

Source: 05-CP - Lattices: Intuition Without the Math.

Least significant byte. The final byte of a binary value, used in the toy PRG as the emitted output for each round. Aliases: LSB_8, last byte.

Source: 03-CP - Simple Pseudorandom Generator.

Ledger. An append-only record with added mechanisms for maintaining integrity, ordering, and agreement across participants. Aliases: Ledger (blockchain data structure), ledger history, ledgers, shared ledger.

Source: 04-DS - From Logs to Ledgers.

Ledger convergence. The process by which honest nodes settle on the same accepted ledger history. Aliases: chain convergence, history convergence.

Source: 01-CM - Consensus decides which history we all accept.

Ledger split. A condition where different parts of the network temporarily see incompatible versions of the ledger. Aliases: Ledger split (Bitcoin), conflicting ledger views, network split.

Source: 04-BF - What Can Consensus Attacks Achieve?.

Legacy Bank Rail. Traditional financial infrastructure such as SWIFT messaging and correspondent banking used to move and settle value. Aliases: legacy bank rails, legacy banking rails.

Source: 03-AS - Competing Financial Infrastructures.

Legal tender. Money that must be accepted for debts or payments under the law of an issuing jurisdiction.

Source: 02-AS - CBDCs vs Cryptocurrencies.

Lender of Last Resort. An institution that can provide emergency liquidity to prevent bank panics or wider financial collapse.

Source: 04-IN - The Modern Financial Architect: Central Banks.

Lending pool. A DeFi pool that holds deposits others borrow against collateral, and decides how much to lend from the collateral's value, often a price it reads from elsewhere. Aliases: lending pools.

Source: 10-FO - A DeFi pool is a contract anyone can call.

Length-extension attack. An attack on some Merkle-Damgard-based constructions where a known digest can be used as a starting state to append data. Aliases: length extension.

Source: 02-CP - Aside: Length-Extension Attack.

Liberty Reserve. A centralized digital money system whose operators were prosecuted for evading AML rules, and whose service ended.

Source: 05-FO - A firm that ignores the BSA can be shut down.

Light client. A blockchain participant that does not download the full chain and instead queries a full node for needed network data. Aliases: Light client (blockchain network), light clients, light node, light nodes.

Source: 01-DS - The Blockchain Link.

Light client (Bitcoin). A Bitcoin client that downloads block headers and uses proofs from full nodes instead of storing and validating every block. Aliases: SPV client, lightweight client.

Source: 03-BF - Bitcoin's Peer-to-Peer Network.

Lightweight client. A client that verifies selected blockchain facts using compact proofs and headers instead of storing every transaction. Aliases: Lightweight client (blockchain verification), light client, lightweight clients.

Source: 04-AS - Why the Merkle Root Matters.

Limits-of-tracing note. The lab's written product: where a trace stops, why, what still shows and how sure the analyst is. Aliases: limits of tracing note.

Source: 06-FO - A limits-of-tracing note: where it stops.

Linear cost. A scaling pattern where message cost grows roughly in proportion to the number of peers. Aliases: Linear cost (network scalability), O(N), linear message cost.

Source: 02-DS - Scalability Challenges{.smaller}.

Linux.Encoder. A ransomware case where AES keys were derived from predictable clock-seeded randomness. Aliases: Linux.Encoder.1.

Source: 05-CP - Case Study: Linux.Encoder RNG Failure.

Liquidation. The closing of an undercollateralized loan position, for which a lending protocol pays the liquidator a reward. Aliases: Liquidation (DeFi), liquidations.

Source: 03-EF - The mempool is public, so pending trades show.

Liquidation bonus. The discount a liquidator gets on a bad account's collateral; the Euler attacker claimed one larger than their donation.

Source: 10-FO - Euler, March 2023: a loan, a gift, a liquidation.

Liquidity. The availability of buyers, sellers, and trading depth that lets an asset be exchanged without large price disruption. Aliases: Liquidity (crypto markets), market liquidity.

Source: 04-BF - Lessons: Small Community Risks.

Liquidity bridge. A bridge where someone on the far chain pays you from their own funds, so assets are swapped rather than copies minted. Aliases: liquidity and intents, liquidity bridges, liquidity network, liquidity networks, third-party bridge.

Source: 08-EF - Optimistic waits a week; ZK waits for a proof.

Liquidity pool. A smart-contract reserve of tokens used to execute swaps and determine prices in an automated market maker. Aliases: Liquidity pool (AMM), pool, token pool.

Source: 01-FO - Decentralized Exchanges (DEXs).

Liquidity provider. Anyone who deposits both of a pool's tokens into it; they earn the pool's trading fee and can withdraw their share of the reserves. Aliases: LP, liquidity providers.

Source: 07-EF - A DEX pool is the other side of the trade.

Litecoin. A transparent cryptocurrency whose mining payouts are public, unlike Monero's.

Source: 08-FO - Most cryptojacking mines Monero, on CPUs.

Liveness. The guarantee that the system continues making progress and is not permanently stalled. Aliases: Liveness (Ethereum consensus), Liveness (consensus property), Liveness (consensus), availability, block-production liveness, consensus liveness, progress.

Source: 05-IN - Consensus as the Core Goal.

Liveness failure. A failure mode where the network cannot make progress, often because the ordering service stalls or loses quorum. Aliases: Liveness failure (Hyperledger Fabric), liveness, ordering-service stall.

Source: 02-PC - Where Consensus Can Still Fail.

LMD-GHOST. Ethereum's fork-choice rule that follows the branch with the greatest weight from validators' latest attestations. Aliases: LMD-GHOST (Ethereum fork choice), Latest-Message-Driven Greedy Heaviest-Observed SubTree, fork-choice rule.

Source: 02-EF - Forks in Proof of Stake — Causes and Resolution.

Local MSP configuration. The MSP material a peer, orderer, or client uses locally to identify itself and verify trusted identities. Aliases: Local MSP configuration (Hyperledger Fabric), local MSP, local trust material.

Source: 02-PC - What is an MSP?.

Local satisfaction ratio. A peer's direct record of successful versus failed interactions with another peer. Aliases: local satisfaction ratios, net satisfaction.

Source: 03-DS - EigenTrust: From Local Opinions to Global Scores.

Location addressing. A storage addressing model where the address names a server location, so the operator can change or remove the content behind it. Aliases: Location addressing (storage), location-addressed storage.

Source: 06-EF - Metadata, IPFS, and the Pointer Problem.

Lock-and-mint bridge. A bridge that locks or burns the real coin on chain A and mints a wrapped copy on chain B; burning the copy releases the coin. Aliases: burn-and-mint, lock and mint, lock-and-mint.

Source: 08-EF - A deposit and a withdrawal touch both chains.

Locking script. The script condition attached to an output that defines what must be provided later to spend that output. Aliases: Locking script (Bitcoin Script), Locking script (Bitcoin), script condition, scriptPubKey.

Source: 02-BF - Every output is locked by a small script.

Log. A time-ordered record of events or actions that preserves history by adding new entries rather than rewriting old ones. Aliases: Log (Ethereum), Log (distributed ledgers), audit trail, event log, log entry, logs.

Source: 04-DS - From Logs to Ledgers.

Logarithmic cost. A scaling pattern where each network doubling adds only a small number of extra routing hops. Aliases: Logarithmic cost (network scalability), O(log N), logarithmic routing cost.

Source: 02-DS - Scalability Challenges{.smaller}.

Logarithmic verification. The scaling pattern where Merkle proof length and verification work grow with tree depth rather than total dataset size. Aliases: Logarithmic verification (Merkle proof), O(log n) verification, logarithmic proof size.

Source: 02-CP - Why Proofs Are Efficient.

Logic contract. The contract that contains executable code used by a proxy while the proxy keeps the persistent state. Aliases: Logic contract (proxy pattern), implementation, implementation contract, logic.

Source: 04-EF - Upgradeability and Proxy Patterns.

Logical clock. An abstract ordering mechanism that helps distributed systems reason about event sequence without relying on one perfect physical clock. Aliases: logical clocks.

Source: 05-IN - Timing Faults.

Logical ring. Chord's circular identifier space where peers and keys are placed according to hash-derived IDs. Aliases: Logical ring (Chord), clock, ring.

Source: 02-DS - DHT Example: Chord.

Long-range attack. An attack that uses old credentials or stake to construct an alternative version of distant ledger history. Aliases: Long-range attack (PoS), Long-range attack (Proof of Stake attack), Long-range attack (distributed ledgers), fabricated alternative history, history rewrite attack, long-range attacks, long-range history rewrite, old-key attack.

Source: 04-DS - Threats to Distributed Ledgers (cont.).

Longest chain rule. The convention that nodes accept the valid chain with the most cumulative proof-of-work as the current ledger history. Aliases: Longest chain rule (Bitcoin), chain selection rule, most cumulative proof-of-work rule.

Source: 01-BF - How Double-Spending is Solved.

Longest Proof-of-Work chain. The valid chain backed by the most accumulated proof-of-work, which Bitcoin nodes treat as the agreed history. Aliases: Longest Proof-of-Work chain (Bitcoin), highest-work chain, longest PoW chain, longest chain.

Source: 01-BF - A Practical Solution to Byzantine Generals.

Longest valid chain. The protocol-selected branch of valid blocks that represents the most accepted accumulated work under the chain-selection rule. Aliases: Longest valid chain (proof of work), heaviest chain, heaviest valid chain, longest chain.

Source: 02-CM - The PoW Rule at a Glance.

Longest-chain selection. The rule of favoring the valid chain with the greatest accumulated proof-of-work. Aliases: Longest-chain selection (Bitcoin consensus), longest chain, most accumulated work.

Source: 04-AS - Bitcoin's Core Design Synthesis.

Look-alike address. An address the attacker generates so its first and last characters match a real payee's, the part most wallets show. Aliases: look-alike.

Source: 09-FO - Four ways out of a wallet look different.

LOOKS. The token LooksRare paid as trading rewards, with each user's daily share set by their volume divided by the platform's volume. Aliases: LOOKS token.

Source: 11-FO - Wash trading: the seller on both sides.

LooksRare. An NFT marketplace that paid trading rewards in LOOKS as a share of total platform volume, excluding private sales, so trading with yourself earned tokens. The rewards ended as of September 2023.

Source: 11-FO - Wash trading: the seller on both sides.

Low Time Preference. A preference for saving and long-term investment over immediate consumption.

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

LP token. In Uniswap v2, the ERC-20 token a liquidity provider receives as a share of a pool's reserves; burning it takes that share of both tokens back out. Aliases: LP tokens, liquidity token, pool token, pool tokens.

Source: 07-EF - A DEX pool is the other side of the trade.

Lure. The bait that gets a victim to act, such as a fake airdrop or giveaway on a drainer site, or a look-alike transfer planted in the victim's history. Aliases: lures.

Source: 09-FO - An approval lets someone else spend your tokens.

M

m-of-n. The multi-sig rule that m signatures from a set of n public keys are needed to spend, such as 2-of-3; m is the minimum number of signatures that must match a key. Aliases: 2-of-3, m of n, multi-sig threshold.

Source: 06-BF - A multi-sig needs m of n keys to spend.

Machine-readable provenance chain. A structured history of content or credential changes that software systems can inspect rather than relying only on human judgment. Aliases: provenance chain.

Source: 05-AS - Opportunity: Authenticity and Provenance.

Majority attack. An attack where one actor or coalition gains enough validating influence to control recent ledger history or approve fraudulent activity. Aliases: 51% attack, Majority attack (distributed ledgers), collusion attack, majority hash attack.

Source: 04-DS - Threats to Distributed Ledgers.

Majority attack economics. The security idea that attacking Bitcoin requires controlling enough work to outpace the honest network, making many attacks economically difficult. Aliases: Majority attack economics (Bitcoin), majority attack, majority-work economics.

Source: 01-BF - Lessons from Precursors.

Majority hash-power attack. An attack where dominant mining power can outcompete honest miners and replace recent chain history. Aliases: 51 percent attack, Majority hash-power attack (Proof of Work), majority hash attack.

Source: 04-AS - Common PoW Attack Themes.

Majority voting. A decision rule where the value supported by most participants is accepted as the group's result. Aliases: majority decision.

Source: 05-IN - Strategies for Handling Suspicion.

Malicious actor. A participant that intentionally tries to mislead, disrupt, or subvert the network. Aliases: adversary, malicious participant.

Source: 05-IN - Why Suspicion Matters.

Malicious smart contract. A smart contract intentionally designed to deceive users, drain assets, or trigger harmful approvals or transfers. Aliases: malicious contract, scam contract.

Source: 01-AS - Fraud and Scams.

Malware campaign. A set of malware samples and intrusions run by one operator, which the same wallet across many samples can group together. Aliases: campaign, mining campaign.

Source: 08-FO - Proceeds run victim, pool, wallet, exit.

Malware report. A threat-intelligence write-up of a malware sample, which for a miner names the wallet and pool found in its settings, command line or network traffic. Aliases: CTI report.

Source: 08-FO - Proceeds run victim, pool, wallet, exit.

Managed Anonymity. A privacy model where small transactions may receive limited privacy while larger or suspicious flows remain traceable to authorities. Aliases: anonymity for small value and traceable for high value.

Source: 03-AS - China: Impact Signals.

Managed membership. A participation model where admitted identities are known and governed instead of allowing open anonymous entry. Aliases: Managed membership (permissioned ledgers), curated membership, known identities.

Source: 02-PC - How Fabric Avoids Public-Chain Consensus Costs.

Mapping. A key-value lookup table commonly used for balances, permissions, registries, and other sparse associations. Aliases: Mapping (Solidity), key-value lookup, mapping(Key => Value).

Source: 04-EF - Types: Reference Types.

Market churn. Rapid creation, disappearance, or turnover of crypto assets that makes discovery, due diligence, liquidity, and risk assessment harder. Aliases: Market churn (crypto assets), cryptocurrency churn, token churn.

Source: 01-AS - Finance (High-Level Overview).

Market for Lemons. A market failure where uncertainty about quality drives good participants away and leaves worse options behind. Aliases: lemons problem.

Source: 04-IN - The "Market for Lemons": When Trust Breaks Down.

Market resolution. The process that determines which event outcome won and how positions should be paid. Aliases: Market resolution (prediction markets), outcome resolution, resolution.

Source: 01-AS - Prediction Markets Case: Polymarket.

masterCopy. The value in slot 0 of Bybit's Safe cold wallet. The cold wallet delegatecalled a contract that rewrote it to an address the thieves chose, after which they controlled the wallet. Aliases: master copy.

Source: 13-FO - Bybit: the way in was a wallet's website, not the contract.

Mathematical security. Security based on precise definitions, assumptions, and proofs rather than on hiding how the system works. Aliases: security through mathematics.

Source: 01-CP - Cryptography: From Obscurity to Mathematical Security.

Max fee per gas. The sender's ceiling on the total per-gas price they are willing to pay for a transaction. Aliases: Max fee per gas (Ethereum fee market), max fee, maxFeePerGas, max_fee_per_gas.

Source: 03-EF - A pending transaction already shows the trade.

Max priority fee per gas. The sender's maximum per-gas tip, paid to the block's fee recipient for including the transaction. Aliases: Max priority fee per gas (Ethereum fee market), maxPriorityFeePerGas, max_priority_fee_per_gas, priority fee, tip.

Source: 03-EF - A pending transaction already shows the trade.

Maximal Extractable Value. Value a block builder or proposer can gain from transaction ordering, inclusion, or exclusion choices. Aliases: MEV, Maximal Extractable Value (Ethereum block building), Maximal Extractable Value (Ethereum block production), Maximal Extractable Value (block proposal), maximal extractable value.

Source: 03-CM - The Block Proposer: Selection and Building.

mBridge. A shared CBDC ledger, compatible with the Ethereum Virtual Machine, on which each founding central bank ran a validating node; the BIS handed the project to its partners in October 2024. Aliases: Project mBridge, mBridge (CBDC platform), mBridge Ledger.

Source: 01-PC - A CBDC is a ledger a central bank runs.

Mean Time To Failure. The average time a system runs before failing. Aliases: MTTF.

Source: 01-DS - Availability, MTTF, and MTTR.

Mean Time To Repair. The average time needed to recover from a failure and restore service. Aliases: MTTR.

Source: 01-DS - Availability, MTTF, and MTTR.

Median-time-past. A timestamp rule that compares a new block's time against the median of recent block times to limit manipulation. Aliases: MTP, Median-time-past (blockchain timestamps).

Source: 02-CM - Retargeting: Technical Implementation.

Medium of Exchange. A function of money that lets people trade through a commonly accepted instrument instead of direct barter.

Source: 04-IN - What is Money? A Social Technology.

Membership NFT. An NFT used as a key for access to content, events, community spaces, or other member privileges. Aliases: membership NFTs.

Source: 06-EF - Three Patterns: DAOs, Membership NFTs, and Social Tokens.

Membership Service Provider. A Fabric component or role that connects participants to verifiable organizational identities, often using certificates. Aliases: MSP, MSPs, Membership Service Provider (Hyperledger Fabric), membership service provider.

Source: 04-DS - Permissioned Ledgers.

Memo. On Stellar, a note sent with a payment that tells an exchange which account a payment to its single shared deposit address belongs to; a muxed address does the same job. Aliases: Stellar memo.

Source: 04-FO - Shared deposit addresses link their senders.

Memory. The temporary workspace available during one call and cleared afterward. Aliases: Memory (Solidity data location), temporary memory.

Source: 04-EF - Types: Reference Types.

Memory-hard puzzle. A PoW puzzle designed to require substantial memory as well as computation, making some hardware specialization harder. Aliases: memory-hard PoW.

Source: 02-CM - Proof-of-work Puzzles.

Mempool. A node's waiting area for valid transactions that have been seen by the network but not yet included in a block. Aliases: Mempool (blockchain network), memory pool, pending transaction pool, transaction pool.

Source: 02-IN - Every payment takes the same five steps.

Mempool (Bitcoin). A node's local collection of known pending transactions waiting to be included in a block. Aliases: memory pool, pending transaction pool.

Source: 01-BF - Primitive 3: Consensus via Proof-of-Work.

Mempool (Ethereum). The set of pending transactions relayed through the network before a validator includes them in a block. Aliases: Mempool (Ethereum transactions), memory pool, pending transaction pool, transaction pool.

Source: 01-EF - Pending, then included, then final.

Merged mining. A proof-of-work arrangement where miners can reuse the same mining work to try to secure more than one compatible chain. Aliases: AuxPoW, Merged mining (proof of work), auxiliary proof of work, merge mining, merged-mined.

Source: 04-BF - Bitcoin-Like PoW Chains.

Merkle internal node. A non-leaf hash computed from child hashes inside a Merkle tree. Aliases: internal node, parent node.

Source: 02-CP - What Is a Merkle Tree?.

Merkle leaf. A hash of an individual data item at the bottom layer of a Merkle tree. Aliases: leaf, leaf hash, leaves.

Source: 02-CP - What Is a Merkle Tree?.

Merkle proof. A compact proof that a data item is included in a Merkle tree with a particular root. Aliases: Merkle inclusion proof, Merkle proof (allowlist), inclusion proof, membership proof.

Source: 02-CP - Merkle Proofs (Membership).

Merkle root. The top hash in a Merkle tree that commits to all leaves below it. Aliases: Merkle root (Bitcoin), root, root hash, transaction root.

Source: 02-CP - Merkle Trees: Motivation.

Merkle tree. A tree of hashes whose root commits to a collection of data items and supports compact membership verification. Aliases: Merkle tree (Bitcoin), Merkle trees, hash tree, tree of hashes.

Source: 02-CP - Merkle Trees: Motivation.

Merkle-Damgard construction. An iterative hash design that processes message blocks through a compression function and carries the resulting state forward. Aliases: Merkle-Damgård, Merkle–Damgård, Merkle–Damgård construction.

Source: 02-CP - Merkle–Damgård: How SHA-256 is Built.

Merkle-Patricia trie. Ethereum's hash-linked key-value data structure used to commit compactly to ordered protocol data such as state, transactions, and receipts. Aliases: MPT, Merkle Patricia trie, Merkle-Patricia trie (Ethereum), trie.

Source: 02-EF - Execution Payload: Header Structure.

Mersenne Twister. A fast general-purpose PRNG useful for simulations but unsuitable for generating cryptographic secrets. Aliases: Mersenne Twister (not for secrets), mt19937, std::mt19937.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Mesh topology. A network shape where nodes maintain multiple overlapping peer connections rather than relying on one central hub. Aliases: Mesh topology (Bitcoin network), peer mesh.

Source: 03-BF - Gossip and Peer Connections.

Message authentication code. A keyed tag used to verify both message integrity and knowledge of a shared secret. Aliases: MAC.

Source: 02-CP - Aside: Length-Extension Attack.

Message call. An invocation that asks an account or contract function to execute with specific input data, value, gas, and caller context. Aliases: Message call (EVM), call, contract call.

Source: 04-EF - Smart Contracts: Concept and Scope.

Message hash. The fixed-length hash of a transaction, which is what the private key actually signs. Aliases: Message hash (digital signatures), digest, hash, message digest.

Source: 04-CP - A signature binds the key to these exact bytes.

Message header. The part of a Bitcoin protocol message that identifies the command, payload length, and checksum. Aliases: Message header (Bitcoin protocol), protocol message header.

Source: 03-BF - Communication Protocol Basics.

Message integrity. The property that data has not been altered in transit or storage. Aliases: data integrity, integrity.

Source: 03-DS - Cryptographic vs. Social Trust.

Message payload. The message-specific data carried inside a Bitcoin protocol message. Aliases: Message payload (Bitcoin protocol), payload.

Source: 03-BF - Communication Protocol Basics.

Message-verified bridge. A bridge that passes messages between chains; chain B acts when its verifiers say a message about chain A is genuine. Aliases: message bridge, message bridges, message-verified.

Source: 08-EF - A bridge moves a claim, not the coin.

Meta-features. The Diamond Model's extra attributes of an event: timestamp, phase, result, direction, methodology and resources. The lesson uses timestamp and phase only. Aliases: meta-feature.

Source: 12-FO - The Diamond: four features of one event.

Meta-transaction. ERC-2771's design in which a user signs a request, a gas relay pays to send it, and a trusted forwarder passes on the signer's address, which the receiving contract treats as the caller. Aliases: ERC-2771, gas relay, meta-transactions, trusted forwarder.

Source: 03-EF - Paymasters and relayers pay gas, not the bill.

Metadata. Descriptive data about a token, such as its name or a link to its image; for an NFT, the optional tokenURI function points to it. Aliases: Metadata (NFT), Metadata (tokenization), metadata JSON, token metadata.

Source: 05-EF - An NFT is a token ID with one owner.

Metadata hash. A hash of the compilation's metadata that the compiler appends to the bytecode. Changing any source byte, even a comment, changes it, which makes it the fingerprint that separates an exact match from a match. Aliases: compilation metadata hash.

Source: 11-FO - Verified source: a match someone checked.

Metadata pointer problem. The integrity gap created when an on-chain token points to off-chain metadata or media that may be mutable, unavailable, or controlled by another system. Aliases: pointer problem.

Source: 06-EF - Metadata, IPFS, and the Pointer Problem.

Metallic Standard. A monetary form based on scarce metals such as gold or silver, often issued as coins. Aliases: metallic standards.

Source: 04-IN - A Brief History of Money's Form.

MEV relay. An off-chain coordination service that checks built blocks and forwards blinded bids from builders to validators. Aliases: MEV relay (Ethereum MEV), relay, relays.

Source: 03-EF - Searchers, builders and relays set the order.

MEV-Boost. Middleware that lets validators receive builder-created blocks through relays without directly building every block themselves. Aliases: MEV-Boost (Ethereum MEV), MEV-Boost (Ethereum block building), maximal extractable value boost.

Source: 02-EF - Censorship.

MiCA. The European Union's unified regulatory framework for crypto issuers, exchanges, stablecoins, and service providers. Aliases: Markets in Crypto-Assets, Markets in Crypto-Assets regulation.

Source: 03-AS - EU: Overview.

Milk Sad. A wallet seed generation failure in Libbitcoin Explorer where weak randomness made supposedly large key spaces enumerable. Aliases: CVE-2023-39910, Milk Sad (wallet entropy case), Milk Sad vulnerability.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Miner. A participant that packs waiting transactions into blocks and is paid the block subsidy plus the fees through the block's coinbase transaction. Aliases: Miner (Bitcoin network), Miner (Bitcoin), Miner (blockchain role), miners, mining node.

Source: 02-IN - Bitcoin spends whole coins and makes change.

Miner fee. The value left unassigned to transaction outputs, calculated as total inputs minus total outputs and claimed by the miner. Aliases: Miner fee (Bitcoin), fee, transaction fee.

Source: 02-BF - A trace is read one transaction at a time.

Minimum viable product. A usable early-stage system suitable for broader testing but not yet a mature production deployment. Aliases: MVP.

Source: 02-AS - mBridge So Far.

Mining. The Proof-of-Work process through which participants compete to add blocks or history to the Bitcoin ledger. Aliases: Bitcoin mining, Mining (Bitcoin), Mining (Proof of Work), mine, miners.

Source: 03-IN - The Synthesis: Bitcoin.

Mining concentration. A condition where a small number of large mining pools or operators control a large share of block production. Aliases: Mining concentration (Bitcoin network), mining centralization.

Source: 03-BF - Network Growth and Concerns.

Mining malware. Malware that installs a mining program on a server or endpoint and keeps mining until someone finds and removes it. Aliases: malware mining.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

Mining pool. A group of miners who combine hash power and share rewards according to a payout rule. Aliases: Mining pool (Bitcoin mining), Mining pool (Bitcoin), mining pools, pool, pooled mining.

Source: 02-CM - Mining Pools, Variance, and Payouts.

Mining proxy. A relay some cryptojacking campaigns put between their infected machines and the pool, so one wallet does not show thousands of connections.

Source: 08-FO - Proceeds run victim, pool, wallet, exit.

Mint account. The Solana account that defines one token type; it can name a freeze authority. Aliases: mint accounts, token mint.

Source: 08-EF - Solana keeps each token in its own account.

Mint event. The log a Uniswap v2 pool writes when liquidity is added. Aliases: Mint log.

Source: 07-EF - A swap leaves Swap and Sync logs.

Minter. An authorized account that can create new token-like units in the example contract. Aliases: Minter (token contract), minting role.

Source: 04-EF - Example: Coin (Code, Part 1).

Minting. Creating new tokens under a contract's rules; ERC-20 and ERC-721 record a mint as a Transfer from address zero. Aliases: Minting (NFT), Minting (token), Minting (tokens), mint, mint NFTs, minting transaction, mints, token minting.

Source: 05-EF - Coin or Token? Start With the Balance Record.

MITRE AADAPT. MITRE's matrix of adversary techniques against digital assets, modeled after ATT&CK and reported on 14 July 2025. As of 1 November 2026 it holds ten ATT&CK tactics plus a new Fraud tactic, and 33 techniques and 33 sub-techniques numbered ADT. Aliases: AADAPT, Adversarial Actions in Digital Asset Payment Technologies.

Source: 12-FO - AADAPT: ATT&CK's shape for digital assets.

MITRE ATT&CK. MITRE's catalog of attacker techniques, each with a name and an ID, which threat-intelligence reports use to label what an intrusion did. Aliases: ATT&CK, ATTACK framework.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

Mixed earnings. Pool earnings for one wallet that may include legitimate mining as well as mining on hijacked machines.

Source: 08-FO - Characterize: for, against, and alternatives.

Mixer. A service or protocol that pools funds from multiple users to obscure direct transaction links between sources and destinations. Aliases: Mixer (blockchain forensics), Mixer (blockchain obfuscation), crypto mixer, mixer pool, mixers, mixing service, tumbler, tumblers.

Source: 01-FO - Step 4 ended Low: now name the service.

ML-DSA. NIST's post-quantum digital signature standard, published as FIPS 204 on 13 Aug 2024. Aliases: FIPS 204.

Source: 04-CP - Quantum threatens the curve, not yet the coins.

ML-KEM. NIST's post-quantum standard for setting up a shared key, published as FIPS 203 on 13 Aug 2024. Aliases: FIPS 203.

Source: 04-CP - Quantum threatens the curve, not yet the coins.

Mode of operation. A rule set that applies a fixed-size block cipher safely to longer messages or data streams. Aliases: Mode of operation (block cipher), block-cipher mode, mode, modes of operation.

Source: 03-CP - Modes of Operation Visual.

Monero. A privacy-focused cryptocurrency that uses ring signatures, stealth addresses, and RingCT to hide sender, receiver, and amount information. Aliases: Monero (privacy coin), XMR.

Source: 04-AS - Privacy Coins.

Monetary Hierarchy. The existing ordering of monetary and payment power among states, currencies, institutions, and infrastructures. Aliases: monetary hierarchies, payment hierarchy.

Source: 03-AS - Closing the Geopolitical Discussion.

Monetary Policy. Central-bank action, such as setting interest rates, used to influence money conditions and the broader economy. Aliases: Monetary policy (Bitcoin), issuance policy, supply rule.

Source: 04-IN - The Modern Financial Architect: Central Banks.

Monetary sovereignty. A state's ability to issue and govern its currency and transmit monetary policy through its financial system.

Source: 02-AS - Why Governments Care.

Money. A social coordination tool that works by serving as a medium of exchange, unit of account, and store of value.

Source: 04-IN - What is Money? A Social Technology.

Money Flower. A BIS taxonomy that classifies forms of money by issuer, form, access, and technology. Aliases: BIS Money Flower.

Source: 04-IN - A Taxonomy for the Future: The "Money Flower".

Money laundering. The process of moving illicit funds through layers of transactions or services to hide their criminal source. Aliases: Money laundering (blockchain forensics), crypto laundering, laundering.

Source: 01-FO - The chain shows a service; records show who.

Money services business. The U.S. category of business, which includes money transmitters, that must register with FinCEN, run an AML program, file reports and keep records under the Bank Secrecy Act. Aliases: MSB, MSBs.

Source: 05-FO - A VASP moves crypto for others.

Money transmitter. FinCEN's term for a business that moves value for others; since 2013 it has included exchangers and administrators of virtual currency. Aliases: money transmission.

Source: 05-FO - A VASP moves crypto for others.

Mortgage-Backed Security. A financial security created by bundling mortgage loans into an investable product. Aliases: MBS, mortgage-backed securities.

Source: 04-IN - Case Study: The 2008 Global Financial Crisis.

msg.sender. The Solidity context variable that identifies the immediate caller of the current call frame. Aliases: immediate caller, msg.sender (Solidity).

Source: 04-EF - Example: Coin (Code, Part 1).

msg.value. The Solidity context variable that reports how much ETH was attached to the current call. Aliases: msg.value (Solidity).

Source: 04-EF - Upgradeability and Proxy Patterns.

Multi-signature authorization. A spending condition that requires multiple valid signatures from a defined set of public keys. Aliases: Multi-signature authorization (Bitcoin Script), multi-signature, multisig, multisignature.

Source: 03-BF - Why Bitcoin Script Exists.

Multi-signature wallet. A control mechanism requiring multiple approved signers before an administrative action can execute. Aliases: Multi-signature wallet (upgrade governance), multi-sig, multisig.

Source: 04-EF - Upgradeability and Proxy Patterns.

Multi-version concurrency control. A commit-time check that asks whether the keys read during simulation still have the same versions when the transaction is validated. Aliases: MVCC, MVCC check, Multi-version concurrency control (Hyperledger Fabric), state-version check.

Source: 02-PC - Peers: Two Jobs, One Node Type.

Multisig. A control arrangement where a threshold of multiple keys must approve an action such as moving funds or administering infrastructure. Aliases: Multisig (bridge security), multi-sig, multisignature, multisignature wallet.

Source: 06-EF - Centralization Chokepoints.

Multivariate cryptography. A post-quantum family based on the difficulty of solving systems of multivariate polynomial equations. Aliases: multivariate cryptography.

Source: 05-CP - Post-Quantum Cryptography: The New Toolbox.

Munition. A legal classification that treated strong cryptographic software like export-controlled military technology. Aliases: Munition (cryptography export control), munitions classification.

Source: 03-IN - The Crypto Wars: U.S. Gov Position (1990s).

MuSig2. An n-of-n multi-signature scheme that combines n keys into one Taproot key, so every key must sign and the output looks like a single signer's. Aliases: BIP 327, BIP327.

Source: 06-BF - Taproot multi-sig can look like one key.

Mutual suspicion. A design posture where each participant verifies others because they may be faulty, compromised, or hostile.

Source: 05-IN - Why Suspicion Matters.

N

Nakamoto consensus. A permissionless consensus style where scarce work and the heaviest valid chain guide nodes toward one history over time. Aliases: Nakamoto-style consensus, PoW consensus, Proof-of-Work consensus.

Source: 01-CM - Open chains buy openness with slower finality.

Named identity. An identity tied to a known participant rather than only to a pseudonymous key or economic stake. Aliases: Named identity (permissioned ledger), authenticated identity, recognized identity.

Source: 01-PC - Permissioned: the operator decides who writes.

Napster. An early P2P file-sharing system that used centralized servers for metadata search while transferring files directly between users. Aliases: Napster (case study).

Source: 01-DS - Case Study: Napster.

NAT traversal. A networking technique that helps peers communicate even when routers or firewalls hide their local addresses.

Source: 01-DS - Communications Application: Skype's P2P Telephony.

Native bridge. A rollup's own bridge: assets deposited into its contract on Ethereum are minted on the rollup, and withdrawals complete from that contract; it is a lock-and-mint bridge. Aliases: native rollup bridge, rollup bridge contract.

Source: 08-EF - A deposit and a withdrawal touch both chains.

Native coin. An asset built into a chain's base ledger that pays network fees and may support staking or mining incentives, such as ETH on Ethereum. Aliases: ETH, native asset, native coins.

Source: 05-EF - Coin or Token? Start With the Balance Record.

Native SegWit. The address format whose single-key addresses (P2WPKH) begin bc1q; in a derivation path, purpose 84' selects it. Aliases: P2WPKH, native SegWit address.

Source: 05-BF - One seed grows a whole tree of keys.

Negligible advantage. An advantage over random guessing so small that it is treated as practically unusable by any efficient adversary. Aliases: negligible probability.

Source: 01-CP - Probability Refresher: Events and Outcomes.

Neighbor capture. A Sybil tactic in which fake identities occupy a target's peer connections so the attacker can shape what the target sees. Aliases: capturing neighbor slots.

Source: 03-DS - Threats: Sybil Attacks.

Neighbor list. The set of peers a node currently knows about or connects to for sending and receiving network information. Aliases: Neighbor list (P2P node), peer list, peer table.

Source: 02-DS - The Eclipse Attack.

Nested service. A service that operates inside an exchange's cluster. Under co-spending alone it looks like the exchange itself. Aliases: nested exchange, nested services.

Source: 12-FO - Six typologies you have already met.

Network congestion. A condition where heavy demand for network capacity slows activity or raises the cost of getting transactions processed. Aliases: Network congestion (blockchain).

Source: 03-IN - Promises vs. Reality.

Network deanonymization. The process of linking a Bitcoin transaction to the node, IP address, user, or service that likely created it. Aliases: Network deanonymization (Bitcoin), deanonymization, transaction-origin deanonymization.

Source: 04-BF - Mitigating Network Deanonymization.

Network latency. Delay in message delivery across a network. Aliases: latency.

Source: 05-IN - Why Decentralization Is Harder.

Network partition. A communication split where parts of a distributed network cannot reliably exchange messages with each other. Aliases: Network partition (Bitcoin), Network partition (distributed ledgers), network partitions, partition, partitioned network, partitioning, partitions.

Source: 05-IN - Centralized vs. Distributed Architectures.

Network verification. The process by which participating nodes independently check transactions and blocks against Bitcoin's rules. Aliases: Network verification (Bitcoin), network validation.

Source: 01-BF - Lessons from Precursors.

Niche ecosystem. A future where blockchain remains useful inside limited communities or markets without replacing mainstream systems. Aliases: Niche ecosystem (blockchain futures), bounded niche, specialized ecosystem.

Source: 05-AS - A Spectrum of Futures.

Nixon Shock. President Nixon's August 15, 1971 suspension of the U.S. dollar's convertibility to gold.

Source: 04-IN - The Turning Point: Ending Bretton Woods.

Node. An individual computer running blockchain software and participating in relaying, checking, or storing network data. Aliases: Node (blockchain network), Node (distributed network), Node (distributed systems), nodes, participant, peer, peer node, peers.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Node (Bitcoin). A computer running Bitcoin software that receives, relays, checks, or stores transactions and blocks. Aliases: Bitcoin node, Node (Bitcoin network), node, nodes.

Source: 01-BF - Primitive 2: The Public Ledger via P2P Networking.

Node (Permissioned Ledgers). A software instance that keeps a copy of the ledger and answers queries about it. Aliases: Node (permissioned ledger), network node, software instance.

Source: 01-PC - Tron works like Ethereum, with T-addresses.

Node consolidation. A reduction in independently operated full nodes that can weaken network diversity and resilience. Aliases: Node consolidation (Bitcoin network).

Source: 03-BF - Network Growth and Concerns.

Node operator. A person or organization that runs and configures a Bitcoin node. Aliases: Node operator (Bitcoin), operator.

Source: 03-BF - Node Operator Responsibilities.

Non-faulty process. A participant that follows the protocol rules and is not exhibiting the fault behavior under discussion. Aliases: correct node, honest node, honest participant.

Source: 01-CM - Consensus decides which history we all accept.

Non-fungible token. A token that is one numbered item with one owner rather than an amount; on Ethereum, defined by the ERC-721 standard. Aliases: NFT, NFTs, Non-fungible token (tokenization), non-fungible tokens.

Source: 05-EF - An NFT is a token ID with one owner.

Non-interactive proof. A proof format where the prover publishes one proof that anyone can verify without back-and-forth challenge rounds. Aliases: Non-interactive proof (zero-knowledge proofs), non-interactive ZK, non-interactive proof.

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

Non-repudiation. A property that makes it difficult for someone to deny having authorized a message or transaction.

Source: 03-IN - Exercise.

Nonce. A one-time value used to provide freshness and prevent dangerous reuse in cryptographic operations. Aliases: Nonce (Proof of Work), Nonce (block header), extra-nonce, nonce field, nonces.

Source: 01-CP - What Is Randomness?.

Nonce (Bitcoin). A value miners change while repeatedly hashing a candidate block header to search for a valid proof-of-work. Aliases: Nonce (Bitcoin mining), nonces.

Source: 01-BF - Primitive 3: Consensus via Proof-of-Work.

Nonce (Cryptography). A value that must not repeat with the same key in modes such as CTR and GCM. Aliases: unique nonce.

Source: 03-CP - Counter-Mode Pseudorandom Generator (PRG).

Nonce (Ethereum). A per-account counter used in Ethereum state, including in address derivation and transaction sequencing. Aliases: Nonce (Ethereum account), account nonce, sequence number.

Source: 01-EF - A key or code controls each account.

Nonpayable. The default Solidity behavior where a function reverts if ETH is attached to the call. Aliases: Nonpayable (Solidity), nonpayable function.

Source: 04-EF - Visibility, Mutability, and Payable.

Normalization. The step that converts a peer's positive local experiences into bounded trust values whose total influence is limited. Aliases: Normalization (EigenTrust), normalized local trust.

Source: 03-DS - EigenTrust: From Local Opinions to Global Scores.

Not Turing-complete. A design constraint where the language avoids unbounded computation such as arbitrary loops. Aliases: Not Turing-complete (Bitcoin Script), non-Turing-complete.

Source: 03-BF - Introduction to Bitcoin Script.

Nothing-at-stake. The incentive problem where validators can cheaply sign multiple competing forks unless penalties make that behavior costly. Aliases: Nothing-at-stake (PoS attack incentive), Nothing-at-stake (PoS risk), Nothing-at-stake (Proof of Stake attack), nothing at stake, nothing-at-stake incentives.

Source: 03-CM - Nothing-at-Stake.

Nouns DAO. An NFT-native DAO where one new Noun is auctioned roughly daily and each Noun acts as membership and one governance vote. Aliases: Nouns, Nouns DAO (NFT governance).

Source: 06-EF - Case Study: Nouns DAO.

O

Obfuscation. Techniques or services used to make blockchain tracing and attribution more difficult. Aliases: Obfuscation (blockchain tracing), obfuscation service, obfuscation services.

Source: 04-AS - Obfuscation and Countermeasures.

Observability. The practice of using logs, block history, and world-state queries to determine which Fabric transaction stage succeeded or failed. Aliases: Observability (Hyperledger Fabric), live observation, troubleshooting evidence.

Source: 02-PC - Observability: What to Watch Live.

Odd-leaf rule. A published convention for handling a tree level with an odd number of hashes. Aliases: Odd-leaf rule (Merkle tree), carry it up, duplicate last hash, odd number of leaves.

Source: 02-CP - Building a Merkle Tree.

OFAC. The U.S. Treasury office whose sanctions list shaped the Tornado Cash censorship episode discussed in the lesson. Aliases: OFAC (policy), Office of Foreign Assets Control.

Source: 02-EF - Censorship.

Off-chain. Dependent on systems, institutions, or files outside the blockchain's consensus process. Aliases: Off-chain (blockchain), off chain.

Source: 06-EF - On-Chain vs. Off-Chain.

Off-chain blind spot. An area of crypto activity recorded inside private systems such as exchanges or OTC desks rather than on a public blockchain. Aliases: Off-chain blind spot (blockchain forensics), off-chain ledger, private ledger blind spot.

Source: 01-FO - Challenges in Forensic Analysis.

Off-chain evidence. Information outside the blockchain, such as exchange records, devices, accounts, OSINT, or institutional records, used to support attribution. Aliases: Off-chain evidence (forensics), external evidence, off-chain identity links, off-chain links.

Source: 04-AS - Attribution.

Off-chain identifier. A link from a service to a person found off the chain, such as an alias, an IP analysis or a forum post. Aliases can be shared or planted. Aliases: off-chain identifiers.

Source: 14-FO - The chain says what moved; others say who.

Offset. The position in a pad or generated stream where encryption or decryption begins. Aliases: Offset (keystream position), counter position.

Source: 03-CP - Pseudo-Random Generators.

Offshore VASP. A VASP run from outside a jurisdiction, whose operators that jurisdiction struggles to identify and whose risk it struggles to reduce. Aliases: offshore VASPs.

Source: 05-FO - Rules push money to firms that ignore them.

Omission fault. A failure where a node remains active but fails to send or receive some messages. Aliases: message omission, omission faults.

Source: 05-IN - Omission Faults.

Omnibus account. A wallet where an exchange holds many customers' coins together, so on chain they appear as one pooled balance. Aliases: omnibus accounts, omnibus holding, omnibus wallet, pooled wallet.

Source: 05-BF - Whoever holds the key controls the coins.

On-chain. Recorded or enforced through blockchain consensus, such as token ownership, IDs, transfer history, and contract execution. Aliases: On-chain (blockchain), on chain.

Source: 06-EF - On-Chain vs. Off-Chain.

On-chain analysis. The examination of blockchain transactions, addresses, contracts, and flows using data recorded directly on a public ledger. Aliases: On-chain analysis (blockchain forensics), ledger analysis, on-chain ledger analysis.

Source: 01-FO - The chain shows a service; records show who.

On-chain attestation. A verifiable on-chain record or token asserting a credential, reputation claim, membership status, or other statement. Aliases: On-chain attestation (credential systems), on-chain attestations.

Source: 06-EF - Analyst Implications.

On-chain community. A community coordinated through blockchain accounts, tokens, smart contracts, or governance mechanisms. Aliases: on-chain communities, token community.

Source: 05-AS - Future 2: Specialized Crypto Ecosystem.

On-chain linkability. The ability to connect addresses, inputs, outputs, or transactions using patterns visible on the public ledger. Aliases: On-chain linkability (Bitcoin), linkability.

Source: 04-BF - Mitigating Network Deanonymization.

On-chain message. A message sent between addresses on the chain itself, which Euler and its attacker used as a channel for negotiation. Aliases: messages on chain, on-chain messages.

Source: 10-FO - After the exploit, the money moves on.

On-chain state. The record a contract keeps on chain, such as token balances and owners, which the system treats as authoritative. Aliases: On-chain state (token architecture), canonical state, contract state.

Source: 05-EF - A token transfer shows up as a log.

On-chain token. A balance that a smart contract keeps for each address on an existing chain; the contract's code decides who can create, move or freeze it. Aliases: blockchain token, token.

Source: 05-EF - A token is a balance a contract keeps.

On-off attack. An attack pattern where a participant behaves well long enough to gain trust, then switches to harmful behavior before disappearing. Aliases: on-off exploitation, reputation milking attack.

Source: 03-DS - Threats: On–Off (Reputation Milking).

One-time output key. The key that appears on the Monero ledger in place of a recipient's address; the sender derives it from the recipient's public address so no one else can link it to that address. Aliases: one-time key, output key.

Source: 07-FO - Monero hides sender, receiver and amount.

One-time pad. An encryption method that XORs plaintext with truly random key material of equal length that is never reused. Aliases: OTP, random pad.

Source: 03-CP - Infinite-length One Time Pad.

One-way function. A function that is easy to compute in the forward direction but infeasible to invert with available algorithms and resources. Aliases: One-way function (public-key cryptography), mathematical one-way function, one-way, one-way property.

Source: 01-CP - Sets and Functions.

onlyOwner. A Solidity modifier that lets only the contract's owner key call a function, as with USDT's addBlackList, removeBlackList and destroyBlackFunds. Aliases: owner-only function.

Source: 05-EF - Tether's blacklist: three owner-only functions.

OP Stack. The software behind a family of optimistic rollups, such as OP Mainnet. Aliases: OP Mainnet, OP Stack chains.

Source: 08-EF - Optimistic waits a week; ZK waits for a proof.

OP_CHECKLOCKTIMEVERIFY. A Bitcoin Script opcode that prevents spending until an absolute block height or timestamp condition is satisfied. Aliases: CLTV, OP_CHECKLOCKTIMEVERIFY (Bitcoin Script).

Source: 03-BF - Script Primitives.

OP_CHECKMULTISIG. A Bitcoin Script opcode that validates multiple signatures against a set of public keys. Aliases: OP_CHECKMULTISIG (Bitcoin Script).

Source: 03-BF - Script Primitives.

OP_CHECKSEQUENCEVERIFY. A Bitcoin Script opcode that enforces relative time-lock conditions between transactions. Aliases: CSV, OP_CHECKSEQUENCEVERIFY (Bitcoin Script).

Source: 03-BF - Script Primitives.

OP_CHECKSIG. A Bitcoin Script opcode that verifies a single digital signature against a public key. Aliases: OP_CHECKSIG (Bitcoin Script).

Source: 03-BF - Script Primitives.

OP_CHECKSIGADD. The tapscript opcode that adds one to a running count for each key that signed; a final OP_NUMEQUAL checks the count against m.

Source: 06-BF - Taproot multi-sig can look like one key.

OP_HASH160. A Bitcoin Script opcode that hashes data with SHA-256 and then RIPEMD-160. Aliases: HASH160, OP_HASH160 (Bitcoin Script).

Source: 03-BF - Script Primitives.

Opcode. A primitive instruction in Bitcoin Script that performs a small operation such as checking a signature, hashing data, or comparing values. Aliases: Opcode (Bitcoin Script), Script opcode, operation code.

Source: 03-BF - Script Primitives.

Open Blockchain Rail. Public-chain, stablecoin, or exchange-linked infrastructure used to move value outside purely legacy or state-run systems. Aliases: crypto rail, crypto rails, open blockchain rails.

Source: 03-AS - Competing Financial Infrastructures.

Open membership. A participation model where many independent or unknown actors can join the network. Aliases: open network.

Source: 05-IN - Why Decentralization Is Harder.

Open membership enumeration. The ability to read a token-gated community's membership list from public on-chain ownership data. Aliases: Open membership enumeration (token community), membership enumeration.

Source: 06-EF - Analyst Implications.

Open participation. A model where nodes can join or leave without prior admission by a central membership authority. Aliases: Open participation (consensus), open membership, permissionless participation.

Source: 01-CM - Open chains buy openness with slower finality.

Open-source intelligence. Intelligence gathered from publicly available sources such as websites, forums, social media, records, and other open data. Aliases: OSINT, open source intelligence.

Source: 01-FO - A label is only as good as its source.

Open-source review. The practice of publicly inspecting source code so flaws can be found, discussed, and fixed by a broader community. Aliases: open review, public code review.

Source: 04-BF - Implementation vs. Design.

Operational complexity. The coordination, monitoring, key rotation, upgrade, and incident-response work required to run a ledger across organizations. Aliases: Operational complexity (distributed ledgers), operational burden.

Source: 04-DS - Common Pitfalls (cont.).

Operational finality. The practical finality Fabric gives after a valid commit under normal managed-membership and crash-fault assumptions. Aliases: Fabric finality, Operational finality (Hyperledger Fabric), immediate finality.

Source: 02-PC - How Fabric Avoids Public-Chain Consensus Costs.

Optimistic rollup. A rollup design that assumes posted results are valid unless challenged with a fraud proof during a dispute period. Aliases: Optimistic rollup (Ethereum scaling), optimistic rollups.

Source: 04-AS - Layer 2s, Rollups, and Blobs.

Optional privacy. Privacy the user has to choose, as with Zcash's shielded pool, in contrast to Monero, where every transaction is private by default. Aliases: opt-in privacy.

Source: 07-FO - Most Zcash stayed public; round trips leak.

Oracle. An external system that brings off-chain facts on-chain so smart contracts can use them deterministically. Aliases: Oracle (blockchain external data), Oracle (blockchain), Oracle (smart contract dependency), data feed, oracle network, oracles.

Source: 01-EF - A contract sees only what is on chain.

Oracle manipulation. Moving the price a lending pool reads, usually with a large trade paid for by a flash loan, then borrowing more than the collateral is worth and repaying the loan.

Source: 10-FO - A price oracle can be moved with borrowed money.

Order book. The way exchanges match trades between buyers and sellers, which an automated market maker replaces by pricing trades from a pool's reserves. Aliases: order books.

Source: 07-EF - A DEX pool is the other side of the trade.

Orderer / sequencer. A participant or service role that establishes a consistent order for accepted transactions. Aliases: Orderer / sequencer (permissioned ledger role), orderer, ordering role, sequencer.

Source: 01-PC - The trace stops at the operator's edge.

Orderer cluster. The Fabric service that establishes one agreed transaction sequence for a channel and delivers ordered blocks to peers. Aliases: Orderer cluster (Hyperledger Fabric), orderer, orderers, ordering service.

Source: 02-PC - From Permissioned Concepts to Fabric Terms.

Ordering. The agreed sequence in which ledger entries are placed, especially when conflicting or simultaneous actions depend on who went first. Aliases: Ordering (distributed ledgers), event ordering, shared ordering.

Source: 04-DS - From Logs to Ledgers.

Ordering service. A service or small cluster that places accepted requests into one consistent sequence for validation and commit. Aliases: Ordering service (Hyperledger Fabric), Ordering service (permissioned ledger), leader-based cluster, orderer, ordering cluster, ordering services.

Source: 01-PC - The trace stops at the operator's edge.

Organization. An admitted member of a permissioned network; in Hyperledger Fabric each organization has its own certificate authority. Aliases: Organization (permissioned ledger identity), consortium member, member organization.

Source: 01-PC - Permissioned: the operator decides who writes.

Orphaned branch. A competing branch that loses the consensus race and is no longer treated as the accepted ledger history. Aliases: Orphaned branch (blockchain), orphaned fork, pruned fork, stale branch.

Source: 01-CM - A stale block is routine, not an attack.

OSINT. Publicly available information used to connect blockchain activity with real-world behavior or identities. Aliases: OSINT (forensics), open-source intelligence.

Source: 04-AS - Attribution.

OTC desk. A desk that takes large amounts off-chain; on chain it shows as many small deposits gathered into wholesale amounts. Aliases: OTC, OTC aggregation, OTC broker, OTC desks, over-the-counter broker, over-the-counter desk.

Source: 07-EF - Where funds can leave a chain.

Outbound connection. A peer connection that a Bitcoin node initiates to another node. Aliases: Outbound connection (Bitcoin node), outbound peer connection.

Source: 03-BF - Joining the Bitcoin Network.

Output descriptor. A short text that tells a wallet which scripts and addresses to make, such as wsh(sorted-multi(2,K1,K2,K3)) for a sorted SegWit 2-of-3. Aliases: descriptor, descriptor wallet, descriptor wallets, descriptors.

Source: 06-BF - Key order in the script carries no meaning.

Overlay maintenance. The recurring protocol work peers perform to repair neighbor links, refresh routing information, and preserve data availability. Aliases: background maintenance, self-healing.

Source: 02-DS - The "Self-Healing" Solutions.

Overlay network. A logical network built on top of the physical internet that defines which peers know about each other and how application messages move. Aliases: logical topology, overlay.

Source: 02-DS - What is an "Overlay Network?".

Own-wallet move. One person moving an NFT between their own wallets. It shows a Transfer log with no payment and no marketplace call, so no sale, price or volume was recorded. Aliases: own-wallet moves.

Source: 11-FO - Look-alikes that are not wash trades.

Owner's list. A service's own published list of its wallets, such as Binance's. It is the owner's word for the exact address, but it covers only what the owner chose to list, usually hot and cold wallets rather than customer deposit addresses. Aliases: owner's published wallet list, published wallet list, transparency page.

Source: 14-FO - The chain says what moved; others say who.

P

P2P exchanger. Typically a person in the business of buying and selling crypto with others found through ads, forums, websites or word of mouth. The seller sends the crypto once the buyer's off-chain payment arrives. Aliases: peer-to-peer exchanger.

Source: 04-FO - P2P markets: two people trade, an escrow holds.

P2P overlay. A logical network of peer connections built on top of the underlying internet or transport network. Aliases: overlay network, peer-to-peer overlay.

Source: 01-DS - Centralized vs. P2P.

P2P trading platform. A platform that lets buyers and sellers of crypto find each other, and sometimes stands between them by holding the crypto in escrow. Aliases: P2P market, peer-to-peer market, peer-to-peer platform.

Source: 04-FO - P2P markets: two people trade, an escrow holds.

P2SH. A legacy wrapper whose addresses start with 3; when the output is spent, the input's scriptSig carries the signatures and the redeem script that holds the rule. Aliases: pay to script hash, pay-to-script-hash.

Source: 06-BF - The rule is written in the script.

P2SH-P2WSH. SegWit nested inside P2SH: the address starts with 3, the scriptSig holds only a SegWit program, and the witness holds the signatures and the script. Aliases: SegWit nested in P2SH, nested SegWit, wrapped SegWit.

Source: 06-BF - One rule, three wrappers on chain.

P2WSH. The native SegWit script wrapper: a 62-character bc1q address whose output holds the SHA-256 hash of a witness script, which the witness reveals on spend. Aliases: native SegWit script hash, pay-to-witness-script-hash.

Source: 06-BF - One rule, three wrappers on chain.

Pad reuse. The unsafe use of the same one-time-pad segment or keystream position for more than one encryption. Aliases: keystream reuse, offset reuse.

Source: 03-CP - Infinite-length One Time Pad.

Padding. Extra bytes added so plaintext fits the exact block size required by a block cipher mode. Aliases: PKCS#7 padding, Padding (block cipher).

Source: 03-CP - Padding and Initialization Vectors.

Panoramix. The decompiler built into Etherscan's decompiler tab; it decompiles complex contracts only partly. The course names it for awareness, not for exercises.

Source: 11-FO - A decompiler rebuilds structure, not intent.

parent_root. A beacon-block field committing to the parent block that the current beacon block extends. Aliases: parent root, parent_root (beacon block field).

Source: 02-EF - Beacon Blocks and Execution Payloads.

Parity multisig. A widely used multisignature wallet system whose 2017 bugs caused major ETH theft and later froze funds. Aliases: Parity multisig (Ethereum incident), Parity multisig library, Parity multisig wallet.

Source: 02-EF - Security Risk: Smart Contract Code Reuse.

Parks case. A 2025 federal case in which Parks used more than $3.5 million of cloud computing, obtained by deceiving two cloud providers, to mine Ether, Litecoin and Monero worth nearly $1 million. Aliases: Parks.

Source: 08-FO - Most cryptojacking mines Monero, on CPUs.

Partial failure. A failure mode where some components or links are broken while others continue running, creating inconsistent views of system health. Aliases: partial outage.

Source: 01-DS - Distributed Failures.

Partition. A condition where groups of nodes cannot reliably exchange messages with each other. Aliases: Partition (distributed network), network partition, sub-partition.

Source: 02-EF - Balance Attack.

Partition resistance. A design goal of keeping blockchain peers connected enough that the network avoids isolated islands and inconsistent chain views. Aliases: Partition resistance (blockchain networking), partition-resistant.

Source: 02-DS - The Blockchain Link.

Partitioning attack. An attack that attempts to split the network into disconnected regions so information and consensus updates are delayed. Aliases: Partitioning attack (Bitcoin network), network partition attack, routing partition.

Source: 03-BF - Fault Tolerance.

Passphrase. An optional extra word or phrase that changes the seed made from a seed phrase; every passphrase opens a valid wallet, but only the right one opens the funded wallet. Aliases: BIP39 passphrase, seed passphrase.

Source: 05-BF - A seed phrase is the whole wallet as words.

Pastry. A structured overlay that routes toward a target by matching more digits of the target identifier at each hop. Aliases: Pastry (DHT).

Source: 02-DS - DHT Example: Pastry.

Path-dependent adoption. A pattern where blockchain's future role depends on earlier choices by institutions, users, regulators, and developers. Aliases: Path-dependent adoption (blockchain futures), path dependency.

Source: 05-AS - What Does the Future Hold?.

Paxful. An online peer-to-peer virtual currency platform that kept an escrow wallet, whose key it controlled, to hold customers' crypto while a trade was pending.

Source: 04-FO - P2P markets: two people trade, an escrow holds.

Paxos. A classical consensus algorithm that uses leader election and majority logic to reach agreement despite crashes or lost messages.

Source: 05-IN - Classical Consensus Approaches.

Pay-Per-Last-N-Shares. A pool payout model where block rewards are distributed across the recent window of submitted shares when a block is found. Aliases: PPLNS.

Source: 02-CM - Pool Payout Model: Pay-Per-Last-N-Shares (PPLNS).

Pay-Per-Share. A pool payout model where miners receive a fixed payment for each valid share regardless of when the pool finds blocks. Aliases: PPS.

Source: 02-CM - Pool Payout Model: Pay-Per-Share (PPS).

Pay-to-PubKey-Hash. A standard Bitcoin script pattern requiring a public key that matches a stored hash and a valid signature from that key. Aliases: P2PKH, Pay-to-PubKey-Hash (Bitcoin Script), pay to public key hash.

Source: 03-BF - Bitcoin Script Basics.

Payable. A Solidity annotation that permits a function or address to receive ETH. Aliases: Payable (Solidity), payable function.

Source: 04-EF - Visibility, Mutability, and Payable.

PayJoin. A protocol for two parties to build a CoinJoin during a payment between them: the receiver adds an input of their own, so the co-spend rule becomes unreliable. Aliases: BIP 78.

Source: 06-FO - PayJoin hides a co-spend inside a payment.

Paymaster. Under ERC-4337, a contract that pays the gas for a user's operation instead of the user. Aliases: paymasters.

Source: 03-EF - Paymasters and relayers pay gas, not the bill.

Payment Censorship. The blocking of financial transactions by banks, payment processors, or other intermediaries for policy, political, or business reasons.

Source: 04-IN - The Problem of Control: Chokepoints & Censorship.

Payment Chokepoint. A point in a payment system where access, surveillance, or blockage can be applied. Aliases: financial chokepoint, payment bottleneck.

Source: 03-AS - What We're Analyzing Today.

Payment infrastructure layer. The rails and intermediaries that move money, distinct from the currency unit being moved. Aliases: Payment infrastructure layer (state control), payment layer.

Source: 02-AS - The Permission Layer: Three Challenges to State Control.

Payment leg. The part of a P2P trade where the buyer pays the seller in cash, bank transfer or gift card. It happens off-chain. Aliases: fiat leg.

Source: 04-FO - P2P markets: two people trade, an escrow holds.

Payment messaging. The transmission of payment instructions about who should pay whom, separate from the final movement of value. Aliases: messaging.

Source: 02-AS - Global Payment Infrastructure: Overview.

Payment portal. The criminals' Tor site, reached from the ransom note, where they set the price and give the victim the address to pay. Aliases: Tor portal, portal chat, ransom portal.

Source: 03-FO - A note leads to a portal, a price, an address.

Payment rails. The messaging, settlement, and intermediary systems that move money between institutions and across borders. Aliases: payment infrastructure, rails.

Source: 02-AS - Global Payment Infrastructure: Overview.

Payment Stablecoin. A stablecoin designed for payments and backed by reserves under supervisory standards. Aliases: payment stablecoins, regulated payment stablecoin.

Source: 03-AS - United States: Impact Signals.

Payment system modernization. Efforts to make payment systems faster, cheaper, more resilient, and better suited to digital commerce.

Source: 02-AS - Why Governments Care.

Pectra upgrade. The Ethereum upgrade that activated on mainnet on 7 May 2025, at epoch 364032, and brought EIP-7702. Aliases: Pectra.

Source: 09-FO - EIP-7702 lets a plain account run code.

Pedersen commitment. The commitment RingCT uses to hide Monero amounts while still letting every node check that inputs equal outputs plus the fee. Aliases: Pedersen commitments.

Source: 07-FO - Monero hides sender, receiver and amount.

Peel chain. A laundering pattern where small amounts are peeled off through a sequence of transactions while the remaining balance continues onward. Aliases: Peel chain (crypto laundering), peel chains, peeling chain.

Source: 01-FO - Step 4 ended Low: now name the service.

Peeling Chain. A repeated UTXO pattern where small amounts are peeled off to destinations while most funds continue to a new change address. Aliases: Peeling Chain (Bitcoin forensics), Peeling chain (forensics pattern), peel chain, peeling chains.

Source: 02-FO - A peel chain: a small peel, the rest rolls on.

Peer. A participant in a peer-to-peer network that can request, provide, rate, or verify resources without relying on a central server. Aliases: Peer (Hyperledger Fabric), Peer (P2P network), node, participant, peers.

Source: 03-DS - Intro: “The Bad Download”.

Peer and orderer logs. Runtime records from peers and orderers that show which transaction stage is happening and where a failure occurred. Aliases: Fabric logs, Peer and orderer logs (Hyperledger Fabric), container logs.

Source: 02-PC - Observability: What to Watch Live.

Peer discovery. The process a peer uses to find other peers it can connect to in a decentralized network. Aliases: Peer discovery (Bitcoin network), Peer discovery (P2P networking), discovery, node discovery, peer finding.

Source: 02-DS - From P2P Models to P2P Design.

Peer diversity. The practice of maintaining varied peer connections so one attacker, region, or failure mode is less likely to dominate a node’s view. Aliases: Peer diversity (Bitcoin network), Peer diversity (P2P security), address diversity, connection diversity, diverse peers, route diversity.

Source: 02-DS - The Eclipse Attack.

Peer table. A node’s local record of known or connected peers used to decide which network connections to maintain. Aliases: Peer table (Bitcoin network), address table, peer list.

Source: 04-BF - Sybil Attacks.

Peer-to-peer design. The engineering of how peers discover each other, route messages, maintain membership, and resist manipulation without a central coordinator. Aliases: P2P design, P2P network design.

Source: 02-DS - From P2P Models to P2P Design.

Peer-to-peer electronic cash. A payment system where parties can transact directly online without routing payment authority through a financial institution. Aliases: P2P electronic cash, Peer-to-peer electronic cash (Bitcoin), peer-to-peer cash.

Source: 01-BF - Bitcoin: The Mission Statement.

Peer-to-peer gossip. A propagation pattern where Ethereum nodes forward transactions or blocks through direct peer connections. Aliases: P2P gossip, Peer-to-peer gossip (Ethereum), gossip.

Source: 03-EF - The mempool is public, so pending trades show.

Peer-to-peer network. A network design where nodes exchange information directly with peers instead of depending on one central server. Aliases: P2P, P2P net, P2P network, P2P networking, Peer-to-peer network (Bitcoin), Peer-to-peer network (blockchain layer), Peer-to-peer network (blockchain pillar), peer-to-peer, peer-to-peer file sharing network, peer-to-peer networking.

Source: 02-IN - Nodes check a payment before passing it on.

PeerTrust. A context-aware reputation model that calculates trust from weighted feedback, rater credibility, transaction scope, and contextual factors.

Source: 03-DS - PeerTrust: How Trust Is Calculated.

Penalty calibration. The process of setting rewards and punishments strong enough to deter attacks without making honest operation too risky. Aliases: Penalty calibration (PoS protocol design), penalty tuning.

Source: 03-CM - Nothing-at-Stake and Economic Security.

Perfect secrecy. A security property where ciphertext alone reveals no information about the plaintext.

Source: 03-CP - Infinite-length One Time Pad.

Permission layer. The practical control layer that determines who may issue, access, transfer, freeze, or route money. Aliases: Permission layer (state control).

Source: 02-AS - The Permission Layer: Three Challenges to State Control.

Permissioned chain. A blockchain network where participation, validation, or access is restricted to approved entities. Aliases: permissioned blockchain, permissioned chains.

Source: 01-IN - Every lab ends in an answer to an instructor.

Permissioned consensus. A consensus model where known participants use roles, signatures, quorums, and policy to agree on ledger updates. Aliases: known-membership consensus.

Source: 01-PC - Permissioned: the operator decides who writes.

Permissioned ledger. A blockchain design where access, transaction submission, or validation is limited to approved entities. Aliases: consortium blockchain, consortium ledger, permissioned blockchain, permissioned distributed ledger, permissioned ledgers, permissioned system, private blockchain.

Source: 02-IN - A blockchain is a record, not a currency.

Permissioned System. A system where access depends on approval, eligibility, or compliance with a gatekeeper's rules. Aliases: Permissioned System (financial access), Permissioned system (CBDC), Permissioned system (blockchain participation), consortium ledger, permissioned blockchain, permissioned by definition, permissioned financial system, permissioned infrastructure, permissioned ledger.

Source: 04-IN - The Problem of Control: Chokepoints & Censorship.

Permissionless finance. Financial activity on open blockchain networks where users can create wallets and transact without prior approval from a central gatekeeper. Aliases: Permissionless finance (blockchain), permissionless access, permissionless systems.

Source: 01-FO - Decentralized Exchanges (DEXs).

Permissionless ledger. A distributed ledger where anyone can participate under protocol rules without approval from a central authority. Aliases: open ledger, permissionless blockchain, permissionless ledgers, permissionless system, public blockchain.

Source: 04-DS - Permissionless Ledgers.

Permissionless Participation. The ability to use or join a system without prior approval from a gatekeeper. Aliases: Permissionless participation (Bitcoin), open participation, permissionless access.

Source: 04-IN - Design Trade-Offs: Efficiency vs. Autonomy.

Permissionless system. A blockchain system where anyone can run a node, validate, and transact without prior authorization. Aliases: Permissionless system (blockchain governance), Permissionless system (blockchain

participation), Permissionless system (blockchain), open blockchain, permissionless blockchain, permissionless network, permissionless systems, public blockchain.

Source: 02-AS - Types of Digital Money.

Permit. An ERC-2612 function that sets a token allowance from a message the owner signs off chain, so the owner sends no transaction and the spender puts the signature on chain. Aliases: ERC-2612, permit signature.

Source: 09-FO - A permit is an approval signed off chain.

Permit2. A permit-style signature that, like an ERC-2612 permit, gives token permission without a transaction from the owner; Scam Sniffer counts the two together.

Source: 09-FO - A permit is an approval signed off chain.

Permutation box. A bit-reordering step in a block cipher that spreads influence across positions to create diffusion. Aliases: P-box, permutation.

Source: 03-CP - Building Blocks of Block Ciphers.

Persistent state. Contract data that remains in Ethereum state after execution and can affect future calls. Aliases: Persistent state (smart contract), Persistent state (smart contracts), contract state, contract storage, on-chain state, persistent storage.

Source: 01-EF - A contract is code plus storage at an address.

PGP. Strong encryption software released by Phil Zimmermann in 1991 that became a flashpoint in debates over public access to cryptography. Aliases: Pretty Good Privacy.

Source: 03-IN - Framing.

Phishing. A deceptive message, website, or workflow that tricks users into revealing secrets or approving harmful actions. Aliases: Phishing (crypto fraud), fake website, phishing attack.

Source: 01-AS - Fraud and Scams.

Physical entropy. Unpredictability gathered from real-world physical processes such as thermal noise, user input, or network jitter. Aliases: entropy source, physical entropy source.

Source: 01-CP - Sources of Randomness.

Physical topology. The underlying internet connectivity beneath an overlay, such as routers, links, and physical network paths. Aliases: physical network topology.

Source: 02-DS - What is an "Overlay Network?".

Pig butchering. A scam type whose proceeds, Chainalysis reported as of January 2026, were moving through centralized exchanges less and less. Aliases: crypto investment scam, investment scam, pig butchering scam, pig-butchering, pig-butchering scam.

Source: 07-EF - Where funds can leave a chain.

Pivoting. Taking one data element, such as an address, and using it with other data sources to find related elements. Each edge of the Diamond is a pivot to try, and each pivot gives a lead to test, not a conclusion. Aliases: Diamond pivot.

Source: 12-FO - Pivot along an edge to the next lead.

Plaintext. The original readable data before encryption or after successful decryption. Aliases: message.

Source: 03-CP - Infinite-length One Time Pad.

Policy and governance. The rules and decision processes that define who may act, approve, administer, and change the network. Aliases: governance policy.

Source: 01-PC - Permissioned: the operator decides who writes.

Policy Friction. Deliberate regulatory or tax burdens that shape activity without fully banning it. Aliases: regulatory friction, tax friction.

Source: 03-AS - India: Impact Signals.

Policy layer. A set of explicit rules that define who may read, submit, approve, govern, or change the network. Aliases: Policy layer (permissioned ledger), policy controls, policy layers.

Source: 01-PC - Permissioned: the operator decides who writes.

Policy literacy. The ability to understand how technical design choices connect to law, regulation, governance, and public debate.

Source: 03-IN - Why This History Matters.

Policy Spectrum. A range of recurring state approaches rather than a single pro-crypto or anti-crypto classification. Aliases: crypto policy spectrum.

Source: 03-AS - Strategic Approaches to Cryptocurrency.

Policy threshold. An endorsement rule requiring a specified number of organizations or roles from a larger set to sign a transaction result. Aliases: OutOf, OutOf policy, Policy threshold (endorsement), threshold policy.

Source: 02-PC - Endorsement Policies as Trust Logic.

Political system. A way of understanding blockchain as a debate over decentralization, control, governance, and censorship resistance. Aliases: Political system (blockchain), political project.

Source: 05-AS - What Is Blockchain Really?.

Polynomial time. A runtime that grows by a polynomial function of input size and is treated as the practical side of cryptographic feasibility. Aliases: efficient algorithm.

Source: 01-CP - Desirable Properties of PRGs.

Pool ban. A mining pool's refusal to keep serving a wallet reported for cryptojacking, which can push the campaign to another pool.

Source: 08-FO - The wallet in the sample is the pivot.

Pool login. The identifier a miner sends when it connects to a mining pool, usually the wallet the pool pays. Aliases: Stratum login, login parameter, wallet as login.

Source: 08-FO - Proceeds run victim, pool, wallet, exit.

Pool reserves. The amounts of the two tokens a DEX pool holds; their ratio is the pool's price, and the other reserve caps what any sale can take out. Aliases: pool reserve, reserve0, reserve1.

Source: 07-EF - The pool's reserves set its price.

Pool statistics. The figures many mining pools publish for each wallet, such as the last share, the current hashrate and the total paid to the wallet.

Source: 08-FO - The wallet in the sample is the pivot.

Post-mix. What a user does with outputs after a mix. Spending mixed outputs together, or paying one to an address already seen, can undo the mix. Aliases: post-mix co-spend, post-mixing.

Source: 06-FO - Before and after the mix still leak.

Post-mortem. A protocol's own account of an attack, or an analysis such as CertiK's, that explains the sequence of calls in the exploit transaction. Aliases: case write-up, write-up.

Source: 10-FO - A DeFi pool is a contract anyone can call.

Post-order invalidation. A transaction failure after ordering when committing peers mark the transaction invalid during validation. Aliases: Post-order invalidation (Hyperledger Fabric), commit-time invalidation, post-order failure.

Source: 02-PC - Where Consensus Can Still Fail.

Post-quantum cryptography. Cryptography designed to replace schemes that a large quantum computer could break. Aliases: PQC, post-quantum algorithms, post-quantum crypto, quantum-safe cryptography.

Source: 04-CP - Quantum threatens the curve, not yet the coins.

Post-quantum migration. The coordinated transition from vulnerable cryptographic assumptions to quantum-resistant alternatives in live systems. Aliases: PQC migration, Post-quantum migration (blockchain governance), cryptographic migration.

Source: 04-AS - Cryptographic Challenges Moving Forward.

Pragma. A source-code directive that constrains the Solidity compiler version and expected language behavior. Aliases: Pragma (Solidity), compiler directive, pragma solidity.

Source: 04-EF - Solidity: File Structure, From Top to Bottom.

Pragmatic Adoption. Uptake driven by immediate usefulness rather than abstract commitment to a technology or ideology. Aliases: practical adoption, utility-driven adoption.

Source: 03-AS - Latin America: Overview.

Pre-image resistance. The property that, given a digest, it should be computationally infeasible to find any input that hashes to it. Aliases: one-way property, preimage resistance.

Source: 02-CP - Hash Function Properties.

Pre-mix. What a user does before a mix. In Whirlpool, a TX0 transaction first splits a fresh input into pool-sized outputs, spending from the cluster being traced. Aliases: TX0, TX0 transaction, pre-mixing.

Source: 06-FO - Before and after the mix still leak.

Pre-order rejection. A transaction failure before ordering caused by issues such as insufficient endorsements, invalid identity, or policy mismatch. Aliases: Pre-order rejection (Hyperledger Fabric), pre-order failure.

Source: 02-PC - Where Consensus Can Still Fail.

Pre-signed transaction. A transaction a signer approves in advance to be sent later. At Drift, pre-signed transactions hid admin actions and handed over admin control on the day of the theft. Aliases: pre-approved transaction, pre-signed transactions.

Source: 13-FO - Drift: a week of preparation, twelve minutes of theft.

Pre-trusted anchor. A trusted starting peer used to ground a global reputation calculation and constrain how trust propagates. Aliases: anchor, anchor node, pre-trusted peer, trusted reference point.

Source: 03-DS - EigenTrust: Anchors and Collusion Resistance.

Predictable monetary policy. A supply schedule whose issuance rules and cap are visible in the protocol rather than changed by discretionary authority. Aliases: Predictable monetary policy (Bitcoin), transparent issuance.

Source: 01-BF - Benefits of Bitcoin's Design.

Prediction market. A market where users trade positions tied to real-world outcomes, so prices can reflect collective expectations. Aliases: event market, prediction markets.

Source: 01-AS - Prediction Markets Case: Polymarket.

Prefix matching. A routing strategy where each hop chooses a peer whose identifier shares a longer prefix with the target. Aliases: Prefix matching (DHT routing), prefix-based routing.

Source: 02-DS - DHT Example: Pastry.

Preimage resistance. The property that it is computationally hard to recover an input from only its hash digest. Aliases: pre-image resistance.

Source: 04-AS - Hash Properties That Matter for Blockchain.

Previous block hash. The header field that points to the prior block and links a Bitcoin block into the chain. Aliases: Previous block hash (Bitcoin block header), prev block hash.

Source: 04-AS - Bitcoin Block Structure.

Price impact. How far a trade moves a pool's price: the larger the trade compared with the pool, the worse the price the trader gets.

Source: 07-EF - A bigger trade gets a worse price.

Price manipulation. Moving a thin pool's price with a trade, which is easy when other contracts read that pool as a price feed. Aliases: price-manipulation attack.

Source: 07-EF - Arbitrage holds a deep pool near the market.

Price oracle. An oracle that brings a price to a contract, such as a lending pool reading an exchange pool to value collateral. Aliases: onchain oracle.

Source: 10-FO - A price oracle can be moved with borrowed money.

Price volatility. Large and unpredictable changes in crypto-asset prices that can undermine trust in their use as money or savings. Aliases: Price volatility (crypto assets), high price volatility, volatility.

Source: 05-AS - External Constraints.

Primary money laundering concern. The finding Treasury makes under Section 311 about a foreign institution, which opens the way to special measures against it. Aliases: of primary money laundering concern.

Source: 05-FO - Section 311 cuts a firm off from U.S. banks.

Primary source. A source close to the original event, dataset, protocol decision, or official action being described. Aliases: Primary source (crypto research), primary sources.

Source: 01-AS - Key Sources.

Priority fee. The per-gas tip paid to the block's fee recipient: the proposer, or an outside builder that pays the proposer in a last transaction. Aliases: Priority fee (Ethereum fee market), max priority fee, tip.

Source: 03-EF - Fees set the queue: base fee burned, tip paid.

Privacy. Controls that limit who can see sensitive ledger data while still preserving appropriate verification. Aliases: Privacy (distributed ledgers), confidentiality.

Source: 04-DS - Transparency vs. Privacy.

Privacy coin. A cryptocurrency designed to provide stronger transaction privacy than transparent ledgers such as Bitcoin. Aliases: Monero, Privacy coin (blockchain design), XMR, ZEC, Zcash, privacy coins, privacy-preserving blockchain.

Source: 03-IN - Why This History Matters.

Privacy controls. Mechanisms that separate who can participate from who can see particular data or proofs. Aliases: Privacy controls (permissioned ledger), selective disclosure, selective visibility.

Source: 01-PC - The trace stops at the operator's edge.

Privacy through cryptography. The use of cryptographic tools to protect personal autonomy and private activity in digital systems. Aliases: cryptographic privacy.

Source: 05-AS - The Original Vision.

Privacy-aware design. A system design goal that protects user anonymity and data while still preserving enough accountability for trust decisions. Aliases: privacy-aware.

Source: 03-DS - Key Considerations in System Design.

Privacy-enhancing asset. A cryptoasset, such as Monero or Zcash, whose privacy design hides parts of the transaction record, so less public evidence is visible than on Bitcoin's transparent ledger. Aliases: Privacy-enhancing asset (cryptoasset), privacy asset, privacy coin.

Source: 04-BF - Other PoW Chains Change the Questions.

Private audit. A compliance workflow where an organization proves rules were satisfied without revealing sensitive transaction details. Aliases: Private audit (zero-knowledge), private audits.

Source: 05-CP - Zero-Knowledge in Blockchain Practice.

Private data channel. A permissioned-ledger mechanism that limits sensitive data visibility to selected participants. Aliases: Private data channel (permissioned ledgers), private channel, private channels.

Source: 04-DS - Permissioned Ledgers.

Private data collection. A Fabric privacy mechanism that restricts sensitive transaction data to a defined subset of participants. Aliases: PDC, Private data collection (Hyperledger Fabric), collections, private collections, private data collections.

Source: 04-DS - Case Study A — Hyperledger Fabric (Permissioned).

Private key. The secret credential a user controls to authorize blockchain actions such as signing a transaction. Aliases: Private key (Bitcoin ownership), Private key (wallet control), private keys, private signing key, secret key.

Source: 02-IN - Every payment takes the same five steps.

Private transaction submission. Sending a transaction to a private service, such as a private mempool, instead of broadcasting it through the public mempool. Aliases: Private transaction submission (Ethereum), direct submission, private submission.

Source: 03-EF - The mempool is public, so pending trades show.

Private-key compromise. A failure state where an attacker obtains a private key and can act as the key holder. Aliases: compromised private key, key compromise.

Source: 04-CP - Keys fail through people and bad randomness.

Probabilistic Finality. A settlement model where confidence that a transaction will not be reversed increases over time rather than becoming immediately absolute. Aliases: Probabilistic finality (Bitcoin), Probabilistic finality (Proof of Work), probabilistic confirmation, probabilistic security, probabilistic settlement.

Source: 04-IN - The Institutional Counterpoint: The BIS Critique.

Probabilistic Lead. An inference that helps guide an investigation but does not by itself prove ownership, intent, or identity. Aliases: Probabilistic Lead (blockchain forensics), heuristic lead, investigative lead.

Source: 02-FO - Two outputs aren't always payment and change.

Probability space. A mathematical model of possible outcomes and events used to reason about uncertainty. Aliases: sample space.

Source: 01-CP - Probability Refresher: Events and Outcomes.

Procedure. One adversary's specific implementation of a technique, such as a group sending proceeds through Tornado Cash in 100 ETH lots. Aliases: procedures.

Source: 12-FO - ATT&CK: tactics are why, techniques how.

Process. An independently running participant in a distributed consensus protocol. Aliases: Process (consensus), node, participant.

Source: 01-CM - A ledger is a shared log no one can quietly edit.

Programmable claim. A claim whose creation, ownership, transfer, or state changes are governed by code rather than manual recordkeeping.

Source: 06-EF - What Is a Token?.

Programmable coordination. The use of shared rules, consensus, verification, and smart contracts to coordinate activity among participants. Aliases: Programmable coordination (blockchain technology).

Source: 05-AS - What Is Blockchain Really?.

Programmable money. Money or payment instruments that can execute rules, conditions, or automated logic as part of transfer and settlement. Aliases: programmable payments.

Source: 05-AS - Future 1: Global Settlement Layer.

Programmable spending condition. A rule encoded in a Bitcoin output that specifies what must be provided before that output can be spent. Aliases: Programmable spending condition (Bitcoin Script), spending condition.

Source: 03-BF - Introduction to Bitcoin Script.

Programmable state. Shared blockchain state that can be changed by on-chain logic rather than only by simple payments. Aliases: Programmable state (Ethereum), general-purpose programmable state.

Source: 04-AS - Ethereum's Goals.

Project Aber. A UAE-Saudi central-bank experiment testing whether a shared wholesale digital currency could improve cross-border interbank settlement. Aliases: Aber.

Source: 03-AS - Gulf States: Impact Signals.

Project mBridge. A multi-central-bank platform for testing faster and more direct cross-border settlement using shared digital rails. Aliases: mBridge.

Source: 03-AS - Gulf States: Impact Signals.

Proof of Stake. A consensus approach that uses staked value and protocol rules to help select validators and discourage rewriting history. Aliases: PoS, Proof of Stake (Ethereum consensus), Proof of Stake (Ethereum post-Merge consensus), Proof of Stake (consensus cost), Proof of Stake (consensus mechanism), Proof of Stake (consensus), Proof-of-Stake, proof-of-stake.

Source: 01-CP - Example: The Double-Spend Problem.

Proof of Work. A consensus approach where participants prove eligibility to add a block by performing computational work. Aliases: PoW, Proof of Work (Bitcoin consensus), Proof of Work (Ethereum pre-Merge consensus), Proof of Work (consensus cost), Proof of Work (consensus mechanism), Proof of Work (consensus), Proof-of-Work, Proof-of-Work (Bitcoin consensus mechanism), proof of work, proof-of-work, work.

Source: 02-IN - A miner packs waiting payments into a block.

Proof-of-space puzzle. A resource-binding puzzle where participants prove they have committed storage rather than repeatedly burning computation. Aliases: proof of space, storage-based puzzle.

Source: 02-CM - Proof-of-work Puzzles.

Proof-of-useful-work. A PoW design goal where the expensive computation also produces externally useful scientific or computational output. Aliases: useful proof of work.

Source: 02-CM - Proof-of-work Puzzles.

Proof-of-work puzzle. A computational challenge that is expensive to solve but quick for other nodes to verify. Aliases: PoW puzzle.

Source: 02-CM - Proof-of-work Puzzles.

Propagation asymmetry. Uneven message delivery that gives different validators different views of blocks or attestations at the same time. Aliases: Propagation asymmetry (Ethereum network), attestation delay, message delay.

Source: 02-EF - Balance Attack.

Proposal. A signed client request sent to endorsing peers asking them to simulate a chaincode action. Aliases: Proposal (Hyperledger Fabric), signed proposal, transaction proposal.

Source: 02-PC - One Invoke: Proposal to Commit.

Proposal simulation. The pre-order execution step where endorsers run chaincode against current state to compute a proposed result without committing it. Aliases: Proposal simulation (Hyperledger Fabric), chaincode simulation, simulation.

Source: 02-PC - Chaincode: What It Is and How It Runs.

Proposer boost. A fork-choice weight bonus for the current slot's block when validators see it on time. Aliases: Proposer boost (Ethereum fork choice), proposer boost timing.

Source: 02-EF - Balance Attack.

Proposer-builder separation. A block production design that separates the validator duty of proposing a block from the specialized work of building it. Aliases: PBS, Proposer-builder separation (Ethereum MEV).

Source: 03-EF - Searchers, builders and relays set the order.

proposer_index. A beacon-block field identifying the validator selected to propose that block. Aliases: proposer index, proposer_index (beacon block field).

Source: 02-EF - Beacon Blocks and Execution Payloads.

Protocol coordination. Aligning many independent participants through shared rules and incentives instead of central management. Aliases: protocol-governed coordination.

Source: 01-AS - Infrastructure as a Use Case.

Protocol governance friction. The difficulty of changing Bitcoin's protocol because broad agreement is needed across a decentralized ecosystem. Aliases: Protocol governance friction (Bitcoin), governance friction, slow adoption.

Source: 01-BF - Limitations and Trade-Offs.

Protocol migration strategy. A planned process for moving a blockchain network, wallets, clients, exchanges, and users to new cryptographic or protocol rules. Aliases: Protocol migration strategy (quantum transition), migration strategy, staged upgrade.

Source: 05-AS - What Can Be Done.

Provenance. The history of an asset's ownership and transfers; for an NFT, readable from the Transfer logs that name its token ID. Aliases: Provenance (supply chain), Provenance (tokenization), asset history, asset provenance, origin history, ownership history, product history.

Source: 05-EF - An NFT is a token ID with one owner.

Proxy pattern. An upgrade design that keeps state and the user-facing address in a proxy while delegating execution to replaceable logic code. Aliases: Proxy pattern (smart contract upgradeability), proxy contract, upgradeable proxy.

Source: 04-EF - Upgradeability and Proxy Patterns.

PSBT. A partially signed Bitcoin transaction: a file (.psbt) that multi-sig signers pass between them, through the roles creator, updater, signer, combiner, finalizer and extractor, until it is broadcast. Aliases: .psbt, BIP 174, BIP174, partially signed Bitcoin transaction.

Source: 06-BF - A PSBT carries a half-signed spend.

Pseudonymity. A privacy model where public addresses appear on-chain instead of real names, but activity can still be linked and analyzed. Aliases: Pseudonymity (Bitcoin), Pseudonymity (blockchain forensics), Pseudonymity (blockchain privacy), Pseudonymity (blockchain), ownership without identity, pseudonymous, pseudonymous address, pseudonymous identity.

Source: 02-IN - The chain shows every payment, but no names.

Pseudonymous. Identified by addresses or keys rather than real-world names, while activity may still be visible on-chain. Aliases: Pseudonymous (blockchain identity), pseudonymity.

Source: 02-AS - Types of Digital Money.

Pseudonymous coordination. Decision-making and value transfer among participants identified by addresses or handles rather than verified legal identities. Aliases: Pseudonymous coordination (token governance), pseudonymous coordination at scale.

Source: 06-EF - Analyst Implications.

Pseudonymous identity. Identity represented by cryptographic keys rather than verified real-world names. Aliases: Pseudonymous identity (permissionless ledgers), pseudonymity, pseudonymous keys.

Source: 04-DS - Permissioned vs. Permissionless: Comparison.

Pseudorandom function. A keyed function whose outputs should be computationally indistinguishable from random for anyone without the key. Aliases: PRF, keyed pseudorandom function.

Source: 03-CP - Counter-Mode Pseudorandom Generator (PRG).

Pseudorandom generator. An algorithm that expands a short random seed into a longer sequence that should appear random to efficient observers. Aliases: PRG, PRNG, pseudorandom number generator.

Source: 01-CP - Sets and Functions.

Pseudorandom number generator. A deterministic generator that produces random-looking outputs from a seed but may not be secure against prediction. Aliases: PRNG, Pseudorandom number generator (implementation failure), general-purpose PRNG.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Public auditability. The property that anyone can inspect the public ledger history to check recorded transactions and validation outcomes. Aliases: Public auditability (Bitcoin), publicly auditable transactions.

Source: 04-DS - Case Study C — Bitcoin Ledger (Permissionless).

Public Blockchain. A blockchain network that is broadly accessible and not operated as a closed internal ledger by one institution. Aliases: open blockchain, public chain.

Source: 03-AS - How Blockchain Changes State Power.

Public goods. Work or infrastructure that benefits an ecosystem broadly but is difficult to monetize through ordinary markets. Aliases: Public goods (Gitcoin), public-goods funding.

Source: 06-EF - Case Study: Gitcoin.

Public key. The shareable half of a key pair: on secp256k1, a point on the curve computed from the private key, used to verify signatures. Aliases: Public key (Bitcoin ownership), Public key (Bitcoin), public keys, public verification key, verification key.

Source: 04-CP - A private key is a number that controls coins.

Public key hash. The hash of a public key that a P2PKH locking script requires the spender to match. Aliases: Public key hash (P2PKH), pubKeyHash.

Source: 03-BF - Bitcoin Script Basics.

Public Key Infrastructure. The certificate and identity-management infrastructure used to bind participants or organizations to cryptographic credentials. Aliases: PKI, Public Key Infrastructure (permissioned ledgers).

Source: 04-DS - Common Pitfalls.

Public ledger. The publicly propagated record of transactions and blocks that nodes use to agree on one history. Aliases: Public ledger (Bitcoin), global history, shared ledger, shared public record, transparent ledger.

Source: 01-BF - Bitcoin, Simplified.

Public legitimacy. The social and political acceptance needed for a technology to be trusted, permitted, and adopted at scale. Aliases: Public legitimacy (blockchain adoption), legitimacy.

Source: 05-AS - External Constraints.

Public mempool. The default state in which pending Ethereum transactions can be observed by anyone running appropriate node infrastructure. Aliases: Public mempool (Ethereum), public pending transaction pool.

Source: 03-EF - A DEX trade is visible before it lands.

Public mint. A minting pattern where anyone can call the mint function, usually constrained only by price, timing, or supply cap. Aliases: open mint.

Source: 06-EF - Minting on Ethereum.

Public verifiability. The property that validator assignments, votes, and punishment evidence can be checked by other participants from public protocol data. Aliases: Public verifiability (PoS), publicly verifiable.

Source: 03-CM - Orientation: Proof of Work vs Proof of Stake.

Public-key cryptography. A cryptographic model that separates a private signing key from public material used to verify signatures. Aliases: asymmetric crypto, asymmetric cryptography.

Source: 04-AS - Asymmetric Crypto and Digital Signatures.

Pump and arbitrage. The name for a second bZx flash-loan attack, on 15 February 2020, the same week as the oracle-manipulation attack.

Source: 10-FO - bZx, Feb 2020: borrow, move a price, borrow.

Pump-and-dump. A scheme pattern in which a token's price is pushed up and then sold into; Solidus Labs found about 98% of over 7 million Pump.fun tokens showed it (January 2024 to March 2025). Aliases: pump and dump, pump-and-dumps.

Source: 08-EF - Cheap Solana tokens draw pump-and-dumps.

Pump.fun. A Solana token launchpad on which over 7 million tokens with at least five trades launched between January 2024 and March 2025. Aliases: Pump fun.

Source: 08-EF - Cheap Solana tokens draw pump-and-dumps.

Purchasing Power. The amount of goods or services a given amount of money can buy.

Source: 04-IN - The Problem of Value: The Specter of Inflation.

Q

Quadratic funding. A matching-fund mechanism where broad support from many contributors can receive more weight than a few large donations.

Source: 06-EF - Case Study: Gitcoin.

Quantum computing. A computing model that uses quantum effects to speed up certain structured problems that underlie cryptographic security. Aliases: Quantum computing (cryptography), quantum computers.

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Quantum error correction. Techniques for protecting fragile quantum computations from noise long enough to run useful algorithms. Aliases: error correction.

Source: 05-CP - Shor's Algorithm (1994): Breaking RSA and ECC.

Quantum risk. The risk that future quantum algorithms could undermine current public-key systems more severely than hash functions. Aliases: Quantum risk (cryptography), post-quantum risk, quantum threat.

Source: 04-AS - Cryptographic Challenges Moving Forward.

Qubit. A quantum information unit that can participate in superposition and entanglement rather than storing only a classical 0 or 1. Aliases: qubits.

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Quorum. A required amount of participant support before a protocol treats a proposal or decision as valid. Aliases: Quorum (consensus), Quorum (permissioned consensus), quorum rule, quorum threshold, threshold.

Source: 01-CM - Open chains buy openness with slower finality.

Quorum Certificate. A compact proof formed by aggregating enough validator attestations to show that a quorum approved a block or checkpoint. Aliases: QC, Quorum Certificate (PoS vote aggregation), Quorum certificate (consensus), aggregate signature, aggregated signatures, quorum certificate.

Source: 03-CM - The Committee: Attestation and Validation.

Quorum disruption. An attack or outage that prevents enough authorized approvers from participating to meet a quorum rule. Aliases: approval disruption, preventing quorum.

Source: 01-PC - Fabric demo: watch the gatekeeper work.

Quorum rule. A validation rule that accepts a result only after enough independent participants agree. Aliases: Quorum rule (volunteer computing), quorum rules.

Source: 01-DS - Volunteer Computing Pipeline: BOINC.

Quorum system. A protocol rule requiring a minimum number of consistent responses before treating a decision as authoritative. Aliases: quorum, quorums.

Source: 05-IN - Strategies for Handling Suspicion.

Quoted value. A holding multiplied by the last price, as a wallet or price site shows it; it ignores the pool's depth, so it can be far more than any sale could raise. Aliases: quoted at.

Source: 07-EF - Quoted value is not what can be sold.

R

Raft. A classical consensus algorithm designed to make leader-based replicated consensus easier to understand and implement. Aliases: Raft (Hyperledger Fabric ordering), Raft consensus, Raft ordering.

Source: 05-IN - Classical Consensus Approaches.

Rail Competition. Rivalry among different systems for moving and settling value at scale. Aliases: financial rail competition, payment rail competition.

Source: 03-AS - Competing Financial Infrastructures.

Rainbow table. A precomputed table of candidate inputs and hashes used to speed up guessing against unsalted hashes. Aliases: precomputed lookup table, rainbow table attack.

Source: 02-CP - Aside: Hash Salting.

RANDAO. Ethereum's production mechanism for building shared randomness from proposer reveals over time. Aliases: RANDAO (Ethereum randomness), RANDAO seed, rolling randomness.

Source: 02-EF - Validator Selection & Randomness.

randao_reveal. A proposer's BLS signature included in a beacon block and mixed into Ethereum's rolling randomness state. Aliases: RANDAO reveal, randao reveal, randao_reveal (beacon block field).

Source: 02-EF - Beacon Blocks and Execution Payloads.

Random variable. A numeric value assigned to the outcome of a random process. Aliases: random variables.

Source: 01-CP - Probability Refresher: Independence & Random Variables.

Randomness. Unpredictability with no discernible pattern from the perspective of an observer or attacker. Aliases: true randomness.

Source: 01-CP - Probability Refresher: Events and Outcomes.

RandomX. Monero's mining algorithm since 30 November 2019, designed to run best on general-purpose CPUs.

Source: 08-FO - Most cryptojacking mines Monero, on CPUs.

Ransom address. The address the criminals tell the victim to pay. The victim knows it, so a ransom trace starts from a known payment instead of a guess. Aliases: ransom payment address.

Source: 03-FO - The ransom moved four times in two days.

Ransom note. The message ransomware criminals leave on a victim's systems, pointing to a Tor site and demanding a price in cryptocurrency. Aliases: ransomware note.

Source: 03-FO - A note leads to a portal, a price, an address.

Ransomware affiliate. The party that runs a ransomware attack using a ransomware-as-a-service group's tools, paying the group's developers a share of each ransom. Aliases: affiliate.

Source: 03-FO - Developer and affiliate split it on-chain.

Ransomware extortion. A criminal payment demand made after malware disrupts or threatens a victim's systems or data. Aliases: Ransomware extortion (illicit finance), ransom demand, ransomware payment.

Source: 03-FO - Ransom: a payment you can watch from the start.

Ransomware-as-a-service. A ransomware model in which developers supply the malware and affiliates run the attacks, with the developers taking a share of each ransom. Aliases: RaaS.

Source: 03-FO - Developer and affiliate split it on-chain.

Rater credibility. The trustworthiness of the peer providing a rating, used to decide how much that rating should affect the final score. Aliases: reviewer credibility.

Source: 03-DS - PeerTrust: How Trust Is Calculated.

Raydium. A Solana platform with liquidity pools, about 93% of which showed soft-rug signs in Solidus Labs' study.

Source: 08-EF - Cheap Solana tokens draw pump-and-dumps.

Reachable node. A Bitcoin node that accepts inbound network connections from other peers. Aliases: Reachable node (Bitcoin network), listening node, publicly reachable node.

Source: 03-BF - Elements of the Bitcoin Network.

Reactor. A commercial blockchain analytics platform whose clustering, per the D.C. Circuit's Sterlingov opinion, rests on co-spend, a behavioral heuristic and intelligence-based clustering. Aliases: Chainalysis Reactor.

Source: 14-FO - An echo is not a second line.

Read-write set. The proposed transaction output listing which ledger keys were read, which versions were observed, and which values should be written. Aliases: RW set, Read-write set (Hyperledger Fabric), read write set, read-write set.

Source: 02-PC - Chaincode: What It Is and How It Runs.

Rebranding attack. An attack where a peer with a bad reputation abandons its identity and rejoins as a new participant. Aliases: clean slate attack, identity reset.

Source: 03-DS - Threats: Rebranding Attacks.

receipts_root. The header commitment to execution results such as status, gas used, and emitted logs. Aliases: receipts root, receipts_root (Ethereum execution payload).

Source: 02-EF - Execution Payload: Header Structure.

Receive function. A special Solidity function that handles plain ETH transfers sent without calldata. Aliases: Receive function (Solidity), receive.

Source: 04-EF - ABI, Encoding, and Function Selectors.

Recency weighting. A scoring method that gives recent behavior more influence than older behavior. Aliases: recency decay.

Source: 03-DS - Threats: On–Off (Reputation Milking).

Recommendation 15. The part of the Financial Action Task Force's anti-money-laundering standard that, since 2019, covers virtual assets and VASPs. Aliases: FATF Recommendation 15, R.15.

Source: 05-FO - A VASP moves crypto for others.

Recommender credibility. A measure of how much weight to give a peer's recommendation based on that peer's own trustworthiness. Aliases: recommender quality.

Source: 03-DS - Direct vs. Indirect Trust.

Red flag indicator. A sign FinCEN lists that a bank may see on the victim's side of a ransom payment, such as a customer using Tor or an unidentified web portal to talk to the person they pay in cryptocurrency. Aliases: FinCEN red flags, red flags.

Source: 03-FO - The portal and chat are leads; ask for them.

Redeem script. In P2SH, the script that holds the spending rule, such as OP_2, three public keys, OP_3, OP_CHECKMULTISIG; the spender reveals it last in the scriptSig. Aliases: redeem scripts, redeemScript.

Source: 06-BF - The rule is written in the script.

Redemption. Exchanging a token with its issuer for the asset behind it, such as dollars for a fiat-backed stablecoin. Aliases: Redemption (real-world asset token), redeem, redemption gates.

Source: 05-EF - Same Token Standard, Different Promise.

Reduction. A logical transformation showing that breaking one scheme would also solve another problem believed to be hard. Aliases: Reduction (cryptographic proof), security reduction.

Source: 01-CP - Proofs and Reasoning.

Redundancy. The use of extra components or copies so a system can keep operating when one component fails.

Source: 05-IN - Crash Faults.

Reentrancy. A vulnerability pattern where an external call lets another contract call back into the original contract before the first execution finishes safely. Aliases: Reentrancy (smart contract risk), Reentrancy (smart contract security), reentrancy bug, reentrant call, reentry.

Source: 04-EF - From Bitcoin Script to Solidity.

Reentrancy vulnerability. A bug where external calls let an attacker reenter contract logic before state such as balances is safely updated. Aliases: Reentrancy vulnerability (smart contract security), recursive withdraw bug, reentrancy.

Source: 02-EF - Real-World Ethereum Security Examples.

Reference type. A Solidity type category for larger data structures whose behavior depends on where the data is stored. Aliases: Reference type (Solidity), reference types.

Source: 04-EF - Types: Reference Types.

Regulation and compliance. The legal rules and required controls that shape who may build, issue, custody, transact, or scale blockchain systems. Aliases: Regulation and compliance (blockchain adoption), compliance, regulatory compliance.

Source: 05-AS - External Constraints.

Regulatory chokepoint. A service or interface where rules can be enforced because crypto activity touches banks, exchanges, bridges, or other identifiable operators. Aliases: Regulatory chokepoint (crypto policy), chokepoint, compliance chokepoint, regulated chokepoint.

Source: 01-FO - Decentralized Exchanges (DEXs).

Regulatory classification. The legal categorization of an asset or service, such as security or commodity, that determines which rules apply. Aliases: Regulatory classification (digital assets), asset classification, legal classification.

Source: 01-AS - Regulation and Uncertainty.

Regulatory uncertainty. Unsettled or uneven legal treatment that makes user access, platform behavior, and market outcomes harder to predict. Aliases: Regulatory uncertainty (crypto markets), legal uncertainty, unclear rules.

Source: 01-AS - Regulation and Uncertainty.

Relay. Infrastructure that passes builder-created candidate blocks to validators, often through MEV-Boost. Aliases: MEV relay, Relay (Ethereum block building), relays.

Source: 02-EF - Censorship.

Relay policy. A node's local rules for which valid transactions it will keep, filter, rate-limit, or forward to peers. Aliases: Relay policy (Bitcoin node), mempool policy, policy rules.

Source: 03-BF - Decisions Nodes Make.

Relayer. In an intents bridge such as Across, the party that pays the user on chain B using its own tokens and is repaid later by the protocol. Aliases: relayers.

Source: 08-EF - Liquidity bridges: someone pays you there.

Remote Procedure Call. A request-response interface that wallets and apps use to ask blockchain nodes for data or to submit transactions. Aliases: RPC, Remote Procedure Call (blockchain access).

Source: 01-AS - Walking a Transaction Step-by-Step.

Reorg risk. The risk that a recent accepted branch will be replaced by another valid branch with more cumulative work. Aliases: Reorg risk (Bitcoin), chain reorg risk, reorganization risk.

Source: 01-BF - Security Model & Threats.

Reorganization. A change in the canonical chain where nodes replace one recent branch with a longer or heavier competing branch. Aliases: Reorganization (blockchain), chain reorg, reorg, reorgs.

Source: 02-CM - Attacks: Double Spend.

Replacement transaction. A transaction that reuses the same account nonce with a higher fee to supersede an earlier pending transaction. Aliases: Replacement transaction (Ethereum), replace-by-fee, transaction replacement.

Source: 03-EF - The mempool is public, so pending trades show.

Replication. The practice of maintaining multiple copies or instances so answers can be compared and service can continue after failures. Aliases: Replication (P2P storage), Replication (distributed ledgers), Replication (distributed systems), data replication, replica, replicas, replicate, replicated components, replicated copy, replicated state.

Source: 05-IN - Commission Faults.

Representative Money. Money that represents a redeemable claim on an underlying asset held elsewhere, such as gold or silver in a vault.

Source: 04-IN - A Brief History of Money's Form.

Reputation. A summary of a peer's past behavior used to estimate whether that peer is likely to behave correctly in the future. Aliases: Reputation (P2P trust).

Source: 03-DS - What Is "Trust" in P2P?.

Reputation milking. A strategy where a peer builds a good track record on low-risk actions and then exploits that reputation for a more damaging failure. Aliases: milking reputation.

Source: 03-DS - PeerTrust: Handling Manipulation.

Reputation system. A scoring system that aggregates evidence about peer behavior so participants can choose whom to trust. Aliases: trust system.

Source: 03-DS - Properties of Reputation Systems.

Reputation-based selection. Choosing peers for an interaction based on established track records and trust scores.

Source: 03-DS - Trust but Verify: The Complete Trust Loop.

Require check. A Solidity guard that stops execution and reverts when a required condition is not true. Aliases: Require check (Solidity), precondition check, require.

Source: 04-EF - Example: Coin (Code, Part 1).

Required approvals. The policy-defined set of organizations or nodes that must approve a proposed action. Aliases: Required approvals (transaction policy), approval requirements, organizational signoffs.

Source: 01-PC - Permissioned: the operator decides who writes.

Reserve. The assets or collateral meant to support confidence that a stablecoin can be redeemed at its target value. Aliases: Reserve (stablecoin), backing assets, reserves.

Source: 02-AS - Major Stablecoins and Use Cases.

Reserve Currency. A currency widely held and used by governments and institutions for international settlement and reserves. Aliases: world reserve currency.

Source: 04-IN - The Turning Point: Ending Bretton Woods.

Resource Hijacking. The MITRE ATT&CK technique, T1496, for an attacker using a victim's resources for their own ends; cryptojacking falls under it. Aliases: ATT&CK T1496, T1496.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

Retail CBDC. A public-facing central bank digital currency intended for citizens, businesses, and everyday payments. Aliases: retail CBDCs, retail central bank digital currency, retail central-bank digital currency.

Source: 02-AS - Central Bank Digital Currencies (CBDCs).

Retry. A resilience pattern where a failed or uncertain operation is attempted again, usually with limits or backoff. Aliases: Retry (distributed systems), retries, retrying.

Source: 01-DS - Availability, MTTF, and MTTR.

Reversibility. The ability of a trusted institution to undo or correct a transaction, especially when fraud occurs.

Source: 03-IN - Digital Trust: The Centralized Model.

Revert. The rollback of state changes from a failed EVM execution while gas already spent remains charged. Aliases: Revert (EVM execution), Revert (EVM), Revert (Ethereum execution), reversion, reverted, reverted transaction, reverts, state revert, transaction revert.

Source: 01-EF - Gas prices the work; a failed call still pays.

Reward farming. Wash trading to collect tokens a marketplace pays per unit of trading volume. Trading with yourself raises your share of volume, so it earns rewards. Aliases: farm rewards, volume farming.

Source: 11-FO - Wash trading: the seller on both sides.

Ring signature. A signature technique that proves one member of a group signed without revealing which member it was. Aliases: Ring signature (Monero privacy), ring signatures.

Source: 04-AS - Privacy Coins.

Ring size. The number of possible inputs in a Monero ring: the real one plus its decoys. It has been 16 since the August 2022 upgrade, and was 11 before that. Aliases: ring of 16.

Source: 07-FO - Monero hides sender, receiver and amount.

RingCT. Monero's confidential transaction mechanism for hiding transaction amounts while preserving validity checks. Aliases: Ring Confidential Transactions, RingCT (Monero privacy).

Source: 04-AS - Privacy Coins.

Robustness. The ability of a reputation system to resist manipulation by fake identities, collusion, slander, and other attacks. Aliases: Robustness (reputation systems).

Source: 03-DS - Properties of Reputation Systems.

Rollback. A coordinated move to abandon a chain segment and reorganize onto an earlier valid branch. Aliases: Rollback (blockchain incident response), chain rollback, rollback past a block.

Source: 04-BF - Emergency Response to the Overflow Bug.

Rollup. A Layer 2 design that batches many transactions and posts compressed data or commitments back to Ethereum. Aliases: Rollup (Ethereum scaling), rollups.

Source: 04-AS - Layer 2s, Rollups, and Blobs.

Rollup batch. A bundle of a rollup's transaction data that its sequencer posts to Ethereum, as calldata or in blobs. Aliases: batch of transactions, rollup batches.

Source: 08-EF - A rollup posts its batches to Ethereum.

Rough consensus. An informal adoption process where independent operators converge on a protocol change without being forced by a central authority. Aliases: Rough consensus (permissionless governance), operator consensus.

Source: 04-DS - Governance Models.

Round. One repeated step of a block cipher that combines operations such as substitution, permutation, and key mixing. Aliases: Round (block cipher), rounds.

Source: 03-CP - Building Blocks of Block Ciphers.

Round amounts. Transfers of exact, round values sent close together in time, a pattern that stands out against a service's normal traffic. Aliases: round Bitcoin amounts, round-amount transfers.

Source: 03-FO - Silk Road: stolen coins left in round amounts.

Round key. A per-round key value derived from the main secret key for use inside a block cipher. Aliases: round keys.

Source: 03-CP - Building Blocks of Block Ciphers.

Round trip. A request-and-response exchange whose delay accumulates when a protocol requires many sequential steps. Aliases: Round trip (network protocol), back-and-forth exchange, round trips.

Source: 01-DS - Fallacy #2: Latency Is Zero.

Round-trip heuristic. Linking a Zcash shielding and a deshielding of the same unusual amount made within a few blocks of each other. Aliases: Zcash round trip, round-trip match.

Source: 07-FO - Most Zcash stayed public; round trips leak.

Route leak. An Internet routing incident where routes are incorrectly propagated, causing traffic to take unintended paths. Aliases: routing leak.

Source: 04-BF - Partitioning Attacks.

Routing bias. A connection-selection flaw that makes peers over-connect within one region or group and under-connect to the rest of the network.

Source: 02-DS - Partitions & Healing.

Routing state. The local neighbor and shortcut information a peer maintains so it can forward messages toward useful next hops. Aliases: Routing state (P2P overlay), routing information, routing table state.

Source: 02-DS - Structured Overlay Networks.

Royalty signaling. A contract-level declaration of royalty recipient and basis points for secondary sales, without guaranteeing marketplace enforcement. Aliases: EIP-2981, NFT royalty signaling, Royalty signaling (NFT).

Source: 06-EF - Minting on Ethereum.

RPC provider. A service that lets wallets and dApps read from and send transactions to Ethereum nodes through an API. Aliases: RPC provider (Ethereum infrastructure), RPC provider (access layer), RPC providers, node provider, node providers, remote procedure call provider.

Source: 06-EF - Centralization Chokepoints.

RSA. A public-key cryptosystem whose security relies on the difficulty of factoring large composite numbers. Aliases: RSA (quantum vulnerability).

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Rug pull. A project-driven exit scam or abandonment in which operators collect user funds or value and then disappear, drain assets, or break promised utility. Aliases: Rug pull (token project), exit scam, rug pulls.

Source: 06-EF - The Frosties Rug Pull.

Runtime bytecode. The bytecode that remains stored at a contract address after deployment and is executed by future calls. Aliases: Runtime bytecode (Ethereum), runtime code.

Source: 04-EF - Contract Deployment.

S

Safety. The guarantee that honest nodes do not finalize conflicting values or contradictory system states. Aliases: Safety (consensus property), Safety (consensus), consensus safety, consistency, safety property.

Source: 05-IN - Consensus as the Core Goal.

Safe{Wallet}. The wallet web interface Bybit's signers used. A developer's machine was compromised, malicious JavaScript was placed in its storage bucket, and the signers' browsers loaded it, which altered the transaction they signed. Aliases: Safe Wallet.

Source: 13-FO - Bybit: the way in was a wallet's website, not the contract.

Sale-restricted token. A token whose code lets anyone buy but blocks selling except by the creator, so buyers hold a balance with a price and no exit. Aliases: honeypot, honeypot token.

Source: 07-EF - Thin pools let creators pull the rug.

Salt. An added random value that makes otherwise similar inputs produce different cryptographic results. Aliases: Salt (hashing), hash salting, salting, salts.

Source: 01-CP - What Is Randomness?.

Samourai Whirlpool. Samourai's CoinJoin design: 5 inputs by 5 outputs, later up to 8 by 8, all of one pool size of 0.001, 0.01, 0.05 or 0.5 BTC. Aliases: Samourai, Samourai Wallet, Whirlpool.

Source: 06-FO - Equal outputs mark a CoinJoin on sight.

Sanctions. Legal restrictions that bar interaction with designated entities, actors, contracts, or infrastructure. Aliases: Sanctions (crypto policy), designation, economic sanctions, financial sanctions, sanctions enforcement.

Source: 01-FO - A label is only as good as its source.

Sanctions exposure. Funds or addresses linked to a sanctioned party, for example an address that shares a wallet with a listed one. It is a risk signal, not proof of a violation. Aliases: exposure, exposure statement, sanctions risk.

Source: 05-FO - The SDN list names some addresses, not all.

Sanctions power. The ability to restrict financial activity by controlling access to payment networks, settlement systems, or banking relationships. Aliases: Sanctions power (payment infrastructure), sanctions enforcement, sanctions leverage.

Source: 02-AS - Global Payment Infrastructure: Overview.

Sanctions Resilience. The ability to keep operating economically despite external restrictions on payments, banking, reserves, or trade. Aliases: sanctions resistance.

Source: 03-AS - How Blockchain Changes State Power.

Sandwich attack. An MEV strategy that places transactions before and after a user's swap to profit from the induced price movement. Aliases: Sandwich attack (MEV), front/back-run, sandwiching.

Source: 03-EF - A DEX trade is visible before it lands.

Satoshi. The smallest bitcoin unit, equal to 0.00000001 BTC. Aliases: Satoshi (Bitcoin), sat.

Source: 02-BF - Fees are priced per virtual byte, not per BTC.

Satoshi Nakamoto. The pseudonymous author who released the Bitcoin whitepaper and early software, with the real identity still unknown. Aliases: Nakamoto.

Source: 01-BF - Satoshi Nakamoto's Whitepaper.

Scalability. A network's ability to process many transactions quickly and cheaply at large scale. Aliases: Scalability (Bitcoin), Scalability (blockchain systems), Scalability (blockchain), Scalability (reputation systems), scaling, throughput.

Source: 04-IN - The Engineer's Dilemma: The Blockchain Trilemma.

Scalar multiplication. Multiplying the curve's fixed point G by the private key k to get the public key point, $K = k \times G$. Aliases: Scalar multiplication (ECC), kP, point multiplication.

Source: 04-CP - A private key is a number that controls coins.

Scaling layer. An additional protocol or system that improves transaction throughput or cost while depending on another chain or trust model. Aliases: L2, Scaling layer (blockchain), layer 2, scaling layers.

Source: 05-AS - Technical Constraints.

Scam cash-out. The typology in which scam proceeds are routed to laundering networks rather than to exchanges that can freeze them. On chain, victim payments are gathered into a few addresses, sent in many small transfers into an aggregating service, then moved out in a few large ones. Aliases: pig-butchering cash-out, scam cash-outs.

Source: 12-FO - Scam cash-out: away from what can freeze.

Scam token. A token built to take buyers' money, for example through a rug pull; a 2021 study found over 10,000 listed on Uniswap, roughly half the tokens it studied. Aliases: scam tokens.

Source: 07-EF - Thin pools let creators pull the rug.

Scarcity. The design property that bitcoin issuance is limited and tied to costly work rather than unlimited creation. Aliases: Scarcity (Bitcoin), costliness.

Source: 01-BF - Lessons from Precursors.

Script execution stack. The stack on which Bitcoin nodes evaluate unlocking data and locking-script operations to decide whether a spend is valid. Aliases: Script execution stack (Bitcoin Script), execution stack, stack.

Source: 03-BF - Bitcoin Script Basics.

Script path spend. A Taproot spend that reveals a script; a multi-sig there shows m-of-n with OP_CHECKSIG, OP_CHECKSIGADD and a final count. Aliases: Taproot script path, script path, script-path spend.

Source: 06-BF - Taproot multi-sig can look like one key.

SDN list. OFAC's list of Specially Designated Nationals. U.S. persons may not deal with anyone on it, directly or indirectly, and since 2018 some entries list known virtual currency addresses. Aliases: Specially Designated Nationals and Blocked Persons List, Specially Designated Nationals list.

Source: 05-FO - OFAC designates; every U.S. person blocks.

Searcher. An off-chain actor that scans pending transactions and market state for profitable MEV opportunities. Aliases: MEV searcher, Searcher (MEV), searchers.

Source: 03-EF - The mempool is public, so pending trades show.

Second pre-image resistance. The property that, given one known input, it should be infeasible to find a different input with the same digest. Aliases: second-preimage resistance.

Source: 02-CP - Hash Function Properties.

Second-preimage resistance. The property that it is hard to find a different input with the same digest as a chosen existing input. Aliases: second preimage resistance, second-pre-image resistance.

Source: 04-AS - Hash Properties That Matter for Blockchain.

Secret key. A private value used with a public algorithm to control encryption, authentication, signing, or other cryptographic behavior. Aliases: key, secret keys.

Source: 01-CP - From Secrecy to Integrity.

Section 311. The USA PATRIOT Act authority that lets Treasury find a foreign institution to be of primary money laundering concern and impose special measures, up to barring U.S. institutions from holding a correspondent account for it. Aliases: 311 action, Section 311 of the USA PATRIOT Act.

Source: 05-FO - Section 311 cuts a firm off from U.S. banks.

Security. The difficulty an attacker faces in compromising the network or reversing transactions. Aliases: Security (blockchain trilemma property).

Source: 04-IN - The Engineer's Dilemma: The Blockchain Trilemma.

Security bit-strength. A shorthand measure of the computational effort an attacker is expected to need, expressed as an equivalent number of key or digest bits. Aliases: bit-strength, security bits.

Source: 05-CP - Grover's Algorithm (1996).

Security budget. The ongoing rewards that sustain honest mining, mainly block subsidies and transaction fees. Aliases: PoW security budget, Security budget (PoS), Security budget (proof of work), chain security budget.

Source: 02-CM - Security Model and Economics.

Security Council. A council some rollups keep that can act in an emergency; at L2BEAT Stage 1, at least 75% of it can still block or forge a message to Ethereum. Aliases: L2 security council, security councils.

Source: 08-EF - One sequencer orders each rollup's trades.

Security margin. The gap between the best known attack cost and the level of effort a system is meant to withstand. Aliases: Security margin (cryptography), security margins.

Source: 05-CP - Cryptographic Assumptions Decay Over Time.

Security proof. A formal argument that a cryptographic claim follows from stated assumptions and logical rules. Aliases: proof, proofs.

Source: 01-CP - Proofs and Reasoning.

Security through obscurity. A fragile security model that depends on keeping the method itself secret rather than relying on openly analyzable assumptions and keys. Aliases: obscurity.

Source: 01-CP - Cryptography: From Obscurity to Mathematical Security.

Seed. The initial input that determines the output sequence of a pseudorandom generator. Aliases: Seed (PRG input), Seed (randomness), initial seed, random seed.

Source: 01-CP - Sources of Randomness.

Seed phrase. A sequence of words used to recover control of a wallet's keys. Aliases: Seed phrase (wallet recovery), backup phrase, recovery phrase.

Source: 01-AS - Usability Challenges.

Seizure. Legally taking control of assets connected to criminal, stolen, or sanctioned activity. Aliases: Seizure (asset recovery), asset seizure, fund recovery.

Source: 01-FO - The chain shows a service; records show who.

Selective disclosure. A privacy approach where people reveal identity or information only when it is necessary or desired. Aliases: Selective disclosure (ledger privacy).

Source: 03-IN - Privacy as a Prerequisite for Open Society.

Selector collision. Two different function signatures that produce the same 4-byte selector. Because four bytes leave room for collisions, a database may list several names for one selector. Aliases: four-byte selector collision, function selector collision.

Source: 11-FO - Looking up a selector: a name, not proof.

Self-Custody. Holding and controlling one's own assets instead of relying on a custodian such as a bank or exchange. Aliases: Self-custody (Bitcoin), Self-custody (key custody), key management, non-custodial control, non-custodial custody, self custody, self-custodial wallet.

Source: 04-IN - Design Trade-Offs: Efficiency vs. Autonomy.

Self-financed address. A buyer address funded by the selling address, or by the address that first funded the seller. A sale to it is a transfer within one group, not a sale to a separate party. Aliases: self-financed buyer, self-financed sale, self-financing.

Source: 11-FO - Self-financed: follow the buyer's money back.

Self-inverse. A property where applying the same XOR value twice cancels it out and restores the original input. Aliases: Self-inverse (XOR property), XOR reversibility.

Source: 03-CP - Exclusive OR (XOR) Logic.

Self-liquidation. An attacker liquidating their own insolvent account with a second contract they also control, taking its collateral at a discount.

Source: 10-FO - Euler, March 2023: a loan, a gift, a liquidation.

Self-policing. A design approach where decentralized participants and mechanisms provide moderation, storage, or computation without central enforcement.

Source: 03-DS - Key Considerations in System Design.

Selfish mining. A mining strategy where a coalition withholds found blocks and releases them strategically to earn more than its fair share. Aliases: Selfish mining (Bitcoin), Selfish mining (Proof of Work), block withholding, strategic withholding.

Source: 02-CM - Selfish Mining: Strategic Withholding.

Semantic security. A confidentiality property where repeated encryption should not reveal whether plaintexts are equal.

Source: 03-CP - Padding and Initialization Vectors.

Sequencer. The role or service that orders Layer 2 transactions, updates Layer 2 state, and periodically posts batch information to Layer 1. Aliases: L2 sequencer, Sequencer (Layer 2).

Source: 01-AS - Scaling and Interoperability.

Sequencing. The act of placing endorsed transactions into one agreed order for a specific channel. Aliases: Sequencing (Hyperledger Fabric ordering), block sequence, ordering, transaction sequence.

Source: 02-PC - Orderer Cluster and Raft Quorum.

Serial identity strategy. An identity attack pattern where an adversary uses one identity at a time and discards it when its reputation is damaged. Aliases: serial strategy.

Source: 03-DS - Threats: Rebranding Attacks.

Service cluster. A group of addresses inferred to belong to one service, entity, or platform. Aliases: Service cluster (exchange forensics), entity cluster, exchange cluster.

Source: 01-FO - Step 4 ended Low: now name the service.

Session key. A temporary symmetric key used to protect data after a secure session has been established. Aliases: session keys.

Source: 03-CP - Why Symmetric Crypto.

Set. A collection of distinct elements used to describe spaces such as all possible keys or all possible messages. Aliases: Set (mathematics), sets.

Source: 01-CP - Mathematics — The Language of Trust.

setApprovalForAll. The NFT (ERC-721) call that lets an operator manage all of the owner's tokens in a collection, not just one; it emits an ApprovalForAll event. Aliases: ApprovalForAll, ApprovalForAll event.

Source: 09-FO - An approval lets someone else spend your tokens.

Setter. A function that updates contract state. Aliases: Setter (Solidity), setter function.

Source: 04-EF - Example: SimpleStorage (Code).

Settlement. The final completion of a payment or transfer obligation through the actual movement of value. Aliases: Settlement (payment).

Source: 02-AS - Global Payment Infrastructure: Overview.

Settlement policy. An operational rule for deciding when a service treats a transaction as settled, based in this lesson on finalized checkpoints. Aliases: Settlement policy (Ethereum operations), finality-based settlement, settlement rule.

Source: 02-EF - Double Spending (Threat Model).

SHA-1. An older Secure Hash Algorithm with 160-bit outputs that is deprecated because practical collisions have been found.

Source: 02-CP - Collision Resistance.

SHA-256. A 256-bit member of the SHA-2 family widely used for integrity checks and Bitcoin hashing.

Source: 02-CP - What Is a Cryptographic Hash Function?.

SHA-3. A modern Secure Hash Algorithm family based on a sponge construction rather than Merkle-Damgard iteration. Aliases: SHA3.

Source: 02-CP - Collision Resistance.

Share. A partial proof of work meeting an easier pool-set difficulty, used to measure a miner's contributed work. Aliases: Share (mining pool), pool share, valid share.

Source: 02-CM - Pool Payout Model: Pay-Per-Share (PPS).

Shared deposit address heuristic. The rule that addresses sending funds to the same exchange deposit address are likely controlled by one entity, because a deposit address belongs to one exchange account. Aliases: deposit address clustering, deposit address reuse heuristic.

Source: 04-FO - Shared deposit addresses link their senders.

Shared funder. One address that funded the wallets on both sides of a sale, found by tracing each wallet's first incoming ETH. It ties buyer and seller to one party; an exchange that funds millions does not count, because the funder must be the seller's own. Aliases: common funder.

Source: 11-FO - Self-financed: follow the buyer's money back.

Shared history. The one record that many nodes each keep and check their own copy of, with no central keeper. Aliases: Shared history (blockchain), shared reality, shared truth.

Source: 02-IN - A blockchain is a record, not a currency.

Shared randomness. A common random seed that all honest consensus participants can compute and use for validator assignments. Aliases: Shared randomness (Ethereum consensus), deterministic seed, global randomness.

Source: 02-EF - Global Randomness: RANDAO + VDF Beacons.

Shared record. A common record that multiple organizations or participants rely on without assigning one party as the sole owner of truth. Aliases: Shared record (distributed ledgers), shared log, shared notebook.

Source: 04-DS - The Shared Notebook Problem.

Shared secret key. A secret value known to the communicating parties that enables both encryption and decryption. Aliases: secret key, shared key.

Source: 03-CP - The Shared Key.

Shared state. The common view of system data or action that honest nodes agree to treat as authoritative. Aliases: Shared state (blockchain systems), Shared state (blockchain), shared ledger state, single shared state.

Source: 05-IN - Consensus as the Core Goal.

SHAttered. A practical SHA-1 collision demonstration that showed two different files could share the same SHA-1 digest. Aliases: SHAttered attack.

Source: 05-CP - Hash Fragility: Collision Resistance Isn't Forever.

Shielded transaction. A transaction that proves validity while hiding details such as sender, receiver, or amount. Aliases: shielded transactions.

Source: 05-CP - Zero-Knowledge in Blockchain Practice.

Shielding. A Zcash transaction from a transparent address into the shielded pool; the ledger shows the sending address and the amount going in. Aliases: shielding transaction, t to z transaction.

Source: 07-FO - Zcash has two pools: one public, one shielded.

Shor's algorithm. A quantum algorithm that would break ECDSA, working out a private key from a public key, on a large enough quantum computer. Aliases: Shor algorithm, Shor's algorithm (quantum cryptanalysis), Shor's quantum algorithm, Shor's algorithm.

Source: 04-CP - Quantum threatens the curve, not yet the coins.

Short transaction ID. A compact identifier for a transaction used so peers can reconstruct a block from transactions already in their mempool. Aliases: Short transaction ID (compact block relay), short TX ID.

Source: 03-BF - Block Propagation.

Shortest Vector Problem. A lattice problem asking for the shortest nonzero vector in a lattice. Aliases: SVP.

Source: 05-CP - Lattices: Intuition Without the Math.

Sibling hash. A neighboring hash supplied at each tree level so a verifier can hash upward toward the Merkle root. Aliases: Sibling hash (Merkle proof), sibling hashes.

Source: 02-CP - Merkle Proofs (Membership).

Signature database. A database that maps 4-byte selectors back to the human-readable function signatures people have submitted. A hit gives the name that produces those bytes, not proof of what the contract's code does. Aliases: selector database, selector lookup, signature lookup.

Source: 11-FO - Looking up a selector: a name, not proof.

Signature phishing. Phishing that tricks a victim into signing a message, such as a permit or an EIP-7702 authorization, which the thief then uses to take funds. Aliases: malicious signature, malicious signatures.

Source: 09-FO - A permit is an approval signed off chain.

Signature verification. The public-key check that confirms whether a signature matches a transaction's hash and the signer's public key. Aliases: verification, verify.

Source: 04-CP - A signature binds the key to these exact bytes.

Signed approval. A cryptographically signed authorization from a known member that counts toward policy or consensus requirements. Aliases: Signed approval (permissioned consensus), approved signature, signed approvals.

Source: 01-PC - Permissioned: the operator decides who writes.

Signed message. An off-chain message signed by a wallet to prove key control without sending a transaction or paying gas. Aliases: Signed message (wallet authentication), signature challenge, signed challenge.

Source: 06-EF - How Token Gating Works.

Signed voucher. An off-chain authorization signed by a deployer or issuer that a buyer can submit to mint a token. Aliases: Signed voucher (lazy minting), mint voucher, voucher.

Source: 06-EF - Minting on Ethereum.

Signing key. The online key a validator uses to sign block proposals, attestations, and other consensus messages. Aliases: Signing key (PoS validator operations), validator signing key.

Source: 03-CM - Becoming a Validator.

Simplified Payment Verification. A lightweight Bitcoin verification method that checks block headers and Merkle proofs rather than validating the full blockchain. Aliases: SPV.

Source: 03-BF - Setting Up a Light Client.

Sinbad. A custodial Bitcoin mixer, sanctioned in 2023, that some industry experts believe succeeded the Blender.io mixer. Aliases: Sinbad mixer, Sinbad.io.

Source: 06-FO - A mixer pools coins and pays out others.

Single point of failure. A concentrated institution, server, or process whose compromise or shutdown can disrupt the whole system. Aliases: SPOF, central point of failure, chokepoint, single points of failure.

Source: 03-IN - Digital Trust: The Centralized Model.

Single-purpose DAO. A DAO organized around one concrete objective rather than ongoing protocol or community governance. Aliases: single purpose DAO.

Source: 06-EF - Case Study: ConstitutionDAO.

Siphon Funds. The AADAPT technique that covers peel chains, layering, anonymous wallets and money mules. Aliases: ADT3028.

Source: 12-FO - AADAPT names most of the typologies.

Slander. A coordinated attack that floods an honest peer with false negative ratings to damage its reputation. Aliases: Slander (P2P reputation threat), false negative feedback, smear campaign.

Source: 03-DS - Properties of Reputation Systems.

Slashing (Ethereum). A protocol penalty that burns validator stake when cryptographic evidence proves severe consensus misbehavior. Aliases: Slashing (Ethereum PoS penalty), slash, slashed.

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Slashing (Proof of Stake). A protocol penalty that removes validator stake when cryptographic evidence proves misbehavior. Aliases: Slashing (PoS penalty), Slashing (Proof of Stake penalty), slash, slashed, slashing penalty, stake penalty.

Source: 03-CM - Slashing — Making Misbehavior Expensive.

Slashing evidence. Cryptographic proof included in consensus-layer data that a validator committed slashable behavior. Aliases: Slashing evidence (Ethereum), attester slashings, proposer slashings, slashings.

Source: 02-EF - Beacon Blocks and Execution Payloads.

Slot. A specific position in the ordered history where a protocol chooses one value or block. Aliases: Slot (consensus log), height, position, slot (beacon block field), slot field.

Source: 01-CM - A stale block is routine, not an attack.

Slot (Ethereum). A 12-second Ethereum consensus time window in which at most one block may be proposed and a committee may attest. Aliases: Slot (Ethereum PoS time unit), slots.

Source: 02-EF - Slots and Epochs.

Slot (Proof of Stake). The basic PoS time unit in which a block proposer may be selected and a committee may attest to the proposed block. Aliases: Slot (PoS time unit), slots.

Source: 03-CM - Proof of Stake: The Full Lifecycle.

Smart contract. Program logic deployed to a blockchain that can execute rules or update state according to platform conditions. Aliases: Smart contract (application layer), on-chain contract, on-chain program, on-chain programs, smart contracts.

Source: 01-IN - By day 10 you can do nine things.

Smart contract auditing. Reviewing smart contract code to identify vulnerabilities, risky patterns, or unintended behavior before or after deployment. Aliases: contract auditing, smart-contract audit.

Source: 05-AS - Opportunities.

Smart contract code reuse. The deployment of many contracts from the same or closely related code patterns. Aliases: Smart contract code reuse (Ethereum security), contract code reuse, template reuse.

Source: 02-EF - Security Risk: Smart Contract Code Reuse.

Smart contract randomness. Randomness used by smart contracts for outcomes such as lotteries, games, or NFT reveals. Aliases: on-chain randomness.

Source: 05-CP - Case Study: Linux.Encoder RNG Failure.

Smart flooding. A more selective flooding strategy where stronger peers forward queries to other hubs instead of every weak participant. Aliases: Smart flooding (P2P search), smart flood.

Source: 01-DS - Gnutella's Fix: Supernodes & "Smart Flooding".

Smart wallet. A wallet design that reduces seed-phrase or key-management burden by adding programmable recovery or account features. Aliases: Smart wallet (wallet usability), smart wallets.

Source: 01-AS - Usability Challenges.

Smart-contract exploit. The use of a flaw in smart contract logic to drain assets, change system behavior, or undermine a decentralized application. Aliases: contract exploit, smart contract exploit.

Source: 05-AS - When Blockchain Systems Broke.

SNARK. A succinct non-interactive proof system often used to prove computation correctness with short proofs. Aliases: zk-SNARK, zkSNARK.

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

Social layer. The communication, culture, moderation, reputation, and norms that surround a token system but are not enforced by the chain. Aliases: Social layer (token community), off-chain social layer.

Source: 06-EF - Two-Layer Community Architecture.

Social Technology. A human-made coordination system whose usefulness depends on shared expectations and acceptance.

Source: 04-IN - What is Money? A Social Technology.

Social token. A fungible token used to coordinate identity, status, rewards, access, or participation around a community or creator. Aliases: social tokens.

Source: 06-EF - Three Patterns: DAOs, Membership NFTs, and Social Tokens.

Social trust. Trust based on observed honesty and reliability over time rather than on cryptographic identity alone. Aliases: behavioral trust, social (behavioral) trust.

Source: 03-DS - Cryptographic vs. Social Trust.

Socio-technical system. A system whose behavior depends on both technical mechanisms and human incentives, institutions, and coordination. Aliases: socio-technical design.

Source: 01-BF - And Now, Bitcoin.

Soft fork. A rule change that tightens what counts as valid without forcing the same kind of chain split as a hard fork. Aliases: Soft fork (blockchain protocol), soft-fork tightening.

Source: 04-BF - Lessons: Small Community Risks.

Soft rug pull. A quiet rug pull; Solidus Labs found signs of one in about 93% of Raydium liquidity pools (January 2024 to March 2025). Aliases: soft rug, soft rugs.

Source: 08-EF - Cheap Solana tokens draw pump-and-dumps.

Solana. A separate Layer 1, not an Ethereum L2, where everything is an account and code lives in programs that own accounts. Aliases: SOL.

Source: 08-EF - Solana: programs own the accounts.

Solana account. Solana's basic record: everything is an account with a 32-byte address, and only its owner program can change its data or debit its lamports. Aliases: Solana accounts.

Source: 08-EF - Solana: programs own the accounts.

Solana program. Solana's code; each account has an owner program, which alone can modify the account's data or debit its lamports. Aliases: Solana programs, owner program.

Source: 08-EF - Solana: programs own the accounts.

Solidified block. Tron's word for a final block: one at least 19 distinct active Super Representatives have each produced a block at or above. Aliases: solidified, solidified data.

Source: 01-PC - 27 elected producers; final after 19 confirm.

Solidity. A smart contract programming language that compiles into EVM bytecode. Aliases: Solidity (Ethereum programming), Solidity (Ethereum).

Source: 01-EF - Every node runs the same program: the EVM.

Solo mining. Mining independently without pooling hash power or sharing rewards with other miners. Aliases: independent mining.

Source: 02-CM - Mining Pools, Variance, and Payouts.

sortedmulti. The descriptor form of a multi-sig that sorts the keys before building the script. Aliases: sortedmulti().

Source: 06-BF - Key order in the script carries no meaning.

Soulbound Token. A token minted with transfer disabled so it is intended to remain bound to one wallet. Aliases: SBT, SBTs, soulbound token, soulbound tokens.

Source: 06-EF - Soulbound Tokens.

Sound Money. Money that reliably holds value over time, especially because its supply is predictable or difficult to manipulate.

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

Soundness. The property that a cheating prover cannot convince the verifier of a false statement except with negligible probability. Aliases: Soundness (zero-knowledge proofs).

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

Source checkpoint. The earlier checkpoint named in a validator vote that links one epoch boundary to another. Aliases: Source checkpoint (PoS finality vote), source epoch.

Source: 03-CM - Justification: Supermajority Vote Links.

Source-code verification. The off-chain process of matching published source code and compiler settings to deployed contract bytecode. Aliases: Source-code verification (Ethereum), contract verification, verified source.

Source: 04-EF - Contract Deployment.

Source-quality factors. ICD 203's questions for judging a source before it counts: accuracy and completeness, possible denial and deception, age and continued currency, source access and validation, and motivation and possible bias. A line that fails them does not move the confidence word. Aliases: ICD 203 source factors, source factors.

Source: 14-FO - Judge each source before you count it.

Sourcify. A contract verification service that recompiles published source and reports whether it is an exact match, a match, or no match for the bytecode at an address.

Source: 11-FO - Verified source: a match someone checked.

Sovereign Decentralized Settlement. Final transfer of value through a decentralized system without relying on banks or governments as trusted intermediaries. Aliases: decentralized settlement.

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

Sovereign participant. A node that independently chooses and enforces the rules and policies it accepts. Aliases: Sovereign participant (Bitcoin node).

Source: 03-BF - Decisions Nodes Make.

Sparrow. Open-source self-custody wallet software that creates a user's keys and addresses. Aliases: Sparrow wallet.

Source: 05-BF - Most people hold bitcoin one of two ways.

Sparse tree. A Merkle tree design for very large key spaces where most possible leaves are empty. Aliases: Sparse tree (Merkle tree), sparse Merkle tree, sparse trees.

Source: 02-CP - Design Choices & Variants.

Spear-phishing. Phishing aimed at a chosen person; in the Ronin case it compromised one employee and opened the way to the validator keys.

Source: 09-FO - Ronin: stolen keys, not broken code.

Special due diligence. A duty in the final Huione rule that U.S. institutions take care to keep Huione out indirectly, through the banks they deal with.

Source: 05-FO - Huione: what the rule does and doesn't do.

Special measures. The five measures Section 311 allows: measures one to four add recordkeeping and reporting duties, and measure five can bar U.S. institutions from holding a correspondent account for the named institution. Aliases: measure five, special measure five.

Source: 05-FO - Section 311 cuts a firm off from U.S. banks.

Speculation. Market behavior driven mainly by expected price changes rather than present utility or fundamentals. Aliases: Speculation (crypto markets), market speculation, speculative behavior.

Source: 01-AS - Fragmentation and Speculation.

Speculative layer. The visible trading and price-speculation activity around tokens, especially secondary-market NFT collections. Aliases: NFT speculative layer, Speculative layer (token markets).

Source: 06-EF - Hidden Infrastructure Growth.

Spender. The address an ERC-20 approval names, which may then move the owner's tokens with transferFrom, up to the approved amount.

Source: 09-FO - An approval lets someone else spend your tokens.

SPFS. Russia's financial messaging system built as an alternative to SWIFT after sanctions pressure. Aliases: SPFS (payment messaging), System for Transfer of Financial Messages.

Source: 02-AS - Global Payment Infrastructure: Overview.

Sponge capacity. The hidden portion of a sponge hash state that provides the construction's security margin. Aliases: capacity, capacity bits.

Source: 02-CP - Sponge Construction: How SHA-3 is Built.

Sponge construction. A hash design that absorbs input into a larger internal state and then squeezes output while keeping part of the state hidden. Aliases: sponge.

Source: 02-CP - Sponge Construction: How SHA-3 is Built.

Sponge rate. The portion of a sponge hash state that receives message blocks and from which output bits are extracted. Aliases: rate, rate bits.

Source: 02-CP - Sponge Construction: How SHA-3 is Built.

Spot price. A pool's current price before a trade, the ratio of its reserves; small trades fill close to it.

Source: 07-EF - A bigger trade gets a worse price.

Stabilize. A maintenance task where peers check nearby neighbors so successor links stay correct as membership changes. Aliases: Stabilize (Chord maintenance), stabilization, stabilize().

Source: 02-DS - The "Self-Healing" Solutions.

Stablecoin. A token that aims to keep a stable price against a reference asset such as the dollar; its type (fiat-backed, crypto-backed or algorithmic) decides who stands behind that dollar and who, if anyone, can freeze it. Aliases: stablecoins.

Source: 05-EF - Three ways a stablecoin holds its peg.

Stablecoin issuer. The company that creates a stablecoin and holds its contract's admin key, such as Tether or Circle; that key can call functions no other address can, such as mint, burn, blacklist and pause. Aliases: issuer layer, stablecoin issuers.

Source: 05-EF - The issuer sits above an open network.

Stack-based scripting language. A programming model where operations consume and produce values on a last-in, first-out stack. Aliases: Stack-based scripting language (Bitcoin Script), stack language.

Source: 03-BF - Introduction to Bitcoin Script.

Stake. Locked value that gives a validator weight in consensus and can be reduced if the validator violates protocol rules. Aliases: Stake (PoS), Stake (Proof of Stake), bonded stake, staked capital, staked collateral.

Source: 03-CM - Orientation: Proof of Work vs Proof of Stake.

Stake deposit. The locked funds a participant provides to enter the validator set and become economically accountable. Aliases: Stake deposit (PoS validator lifecycle), deposit stake, validator deposit.

Source: 03-CM - Proof of Stake: The Full Lifecycle.

Stake-weighted selection. A selection process where validator assignment is random but the chance of selection is proportional to stake. Aliases: Stake-weighted selection (PoS), stake-weighted lottery, stake-weighted randomness.

Source: 03-CM - The Block Proposer: Selection and Building.

Staking rewards. Protocol payments that compensate validators for participating honestly and performing assigned duties. Aliases: Staking rewards (PoS incentives), validator rewards.

Source: 03-CM - Nothing-at-Stake and Economic Security.

Stale block. A valid block that is not included in the ultimately canonical chain after a competing branch wins. Aliases: orphan block, stale/orphan block.

Source: 02-CM - Block Propagation and Stale/Orphan Blocks.

STARK. A transparent, scalable non-interactive proof family used for succinct verification of computation. Aliases: zk-STARK, zkSTARK.

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

State actor. An actor that steals for a government rather than for profit. Its patience, backing and scale change the trace: preparation can sit on chain days before a theft, and laundering runs at volume across many addresses. Aliases: state actors, state-sponsored actor, state-sponsored hacking group.

Source: 13-FO - A state actor steals to fund a state.

State Capacity. A government's practical ability to observe, regulate, enforce, and influence outcomes.

Source: 03-AS - How Blockchain Changes State Power.

State Digital Rail. Official digital payment or settlement infrastructure such as CBDCs and shared central-bank platforms. Aliases: central-bank digital rail, state digital rails.

Source: 03-AS - Competing Financial Infrastructures.

State overhead. The storage burden created by the trust data a system must keep, such as ratings, scores, histories, or matrices.

Source: 03-DS - Key Considerations in System Design.

State variable. A variable stored in contract storage whose value can persist between calls. Aliases: State variable (Solidity), state variables.

Source: 04-EF - Solidity: File Structure, From Top to Bottom.

State visibility. The degree to which government authorities can observe participants, balances, and transactions in a money system. Aliases: State visibility (digital money).

Source: 02-AS - Types of Digital Money.

State-Observable Payment Architecture. A payment design that gives state authorities practical visibility into activity under government-defined rules. Aliases: state-managed payment architecture, state-observable payments.

Source: 03-AS - China: Overview.

state_root. The header commitment to Ethereum's global state after the block's transactions execute. Aliases: state root, state_root (Ethereum execution payload).

Source: 02-EF - Execution Payload: Header Structure.

Stealth address. A one-time destination address technique that reduces visible links between payments and a recipient's public identity. Aliases: Stealth address (Monero privacy), stealth addresses.

Source: 04-AS - Privacy Coins.

Storage. The persistent on-chain data area where contract state remains across transactions. Aliases: Storage (Solidity data location), contract storage, persistent storage.

Source: 04-EF - From Bitcoin Script to Solidity.

Storage layout. The ordered arrangement of state variables in storage that upgradeable contracts must preserve across logic versions. Aliases: Storage layout (proxy pattern), storage layout discipline.

Source: 04-EF - Upgradeability and Proxy Patterns.

Storage slot. A numbered location in a contract's storage. Decompiled code shows which slots a function reads and writes, but you learn what a slot holds only by reading it on chain. Aliases: storage slots.

Source: 11-FO - A decompiler rebuilds structure, not intent.

Store of Value. A function of money that allows purchasing power to be saved for future use.

Source: 04-IN - What is Money? A Social Technology.

Strategic Autonomy. The ability to preserve key capabilities without excessive dependence on outside powers or platforms.

Source: 03-AS - EU: Overview.

Strategic disruption. An attack goal focused on causing instability, fear, or market movement rather than directly stealing coins. Aliases: economic disruption, price dislocation.

Source: 04-BF - Why Attack Bitcoin?.

Strategic Posture. A government's overall stance toward crypto and digital finance based on its interests, risks, and desired control points. Aliases: crypto posture, state posture.

Source: 03-AS - Strategic Approaches to Cryptocurrency.

Stratum protocol. The protocol miners use to talk to mining pools; the wallet travels in it as the login. Aliases: Stratum.

Source: 08-FO - Proceeds run victim, pool, wallet, exit.

Stream cipher. A symmetric cipher that generates a keystream and encrypts data bit by bit or byte by byte. Aliases: stream ciphers.

Source: 03-CP - Block vs Stream Ciphers.

Strict liability. OFAC's power to impose a civil penalty even when the violator had no knowledge or reason to know of the violation.

Source: 05-FO - OFAC designates; every U.S. person blocks.

Strong cryptography. Cryptographic tools that are difficult for outsiders, including governments or attackers, to break in practice. Aliases: strong encryption.

Source: 03-IN - The Cypherpunk Movement.

Struct. A custom grouped record type that combines multiple fields into one named data structure. Aliases: Struct (Solidity), structs.

Source: 04-EF - Types: Reference Types.

Structured overlay network. A P2P overlay that deliberately organizes peers and data so lookups can move toward a target instead of broadcasting everywhere. Aliases: structured overlay.

Source: 02-DS - Structured Overlay Networks.

Sub-technique. A more specific means under a technique, numbered after its parent, such as ADT3028.005 Peel Chains under Siphon Funds. A sub-technique takes its parent's tactics. Aliases: sub-techniques.

Source: 12-FO - ATT&CK: tactics are why, techniques how.

Subprime Mortgage. A higher-risk home loan made to a borrower considered less likely to repay. Aliases: subprime mortgage loans.

Source: 04-IN - Case Study: The 2008 Global Financial Crisis.

Substitution box. A nonlinear mapping in a block cipher that replaces input values to create confusion. Aliases: S-box, substitution.

Source: 03-CP - Building Blocks of Block Ciphers.

Success probability. The chance that an adversary succeeds at a specified attack or guessing task. Aliases: adversary success probability.

Source: 01-CP - Probability Refresher: Events and Outcomes.

Successor. The first peer at or after a key's identifier on the Chord ring, responsible for storing or answering for that key. Aliases: Successor (Chord), successor node.

Source: 02-DS - DHT Example: Chord.

Super Representative. One of 27 block producers elected by stakers' votes under Tron's delegated proof of stake. Aliases: SR, SRs, Super Representatives.

Source: 01-PC - 27 elected producers; final after 19 confirm.

Supernode. A high-capacity peer that acts as a hub for weaker peers by handling more indexing, routing, or relay work. Aliases: Supernode (P2P network), supernodes, ultrapeer, ultrapeers.

Source: 01-DS - Gnutella's Fix: Supernodes & "Smart Flooding".

Supernode hierarchy. A P2P design where more capable peers form a higher-capacity backbone that helps weaker peers route or search. Aliases: supernode model, supernodes.

Source: 02-DS - Scalability Challenges{.smaller}.

Superposition. The quantum behavior where a system can represent overlapping possible states until measurement. Aliases: Superposition (quantum computing).

Source: 05-CP - Quantum Primer: Why It Matters for Cryptography.

Supply cap. Bitcoin's protocol-designed limit on total issuance, commonly described as 21 million BTC. Aliases: 21 million BTC limit, 21 million bitcoin limit, Supply cap (Bitcoin monetary policy).

Source: 04-AS - Bitcoin Monetary Policy and Difficulty.

Supply chain traceability. The use of shared records to follow product origin, handling, location, or certification across organizations. Aliases: Supply chain traceability (blockchain use case), food traceability, traceability.

Source: 01-AS - Supply Chain Case.

Surround vote. A slashable voting pattern where one checkpoint vote improperly surrounds or wraps around a prior vote. Aliases: Surround vote (Ethereum slashing), Surround vote (PoS offense), protocol-violating vote pattern, surround votes, surround voting.

Source: 03-CM - Slashing — Making Misbehavior Expensive.

sUSD. A dollar-pegged token, the collateral the bZx attacker bought on thin pools to raise its price in ETH.

Source: 10-FO - bZx, Feb 2020: borrow, move a price, borrow.

Swap event. The log a Uniswap v2 pool writes on each trade: the amounts in and out and the address that got the output. Aliases: Swap log, Swap logs.

Source: 07-EF - A swap leaves Swap and Sync logs.

Swap fee. The 0.30% each Uniswap v2 trade pays, which stays in the pool for the liquidity providers. Aliases: 0.30% fee, pool fee, trading fee.

Source: 07-EF - The pool's reserves set its price.

Swapping-as-a-service. A service some laundering OTC desks offer that converts a client's crypto into several different assets. Aliases: swap service.

Source: 04-FO - Some OTC desks launder: small in, large out.

Sweeper contract. A copied contract that thieves delegate a wallet to after its key has leaked, so the wallet is emptied automatically; CrimeEnjoyor is a short, widely reused one. Aliases: CrimeEnjoyor, sweeper.

Source: 09-FO - After 7702, an outflow may not be the owner's.

Sweeping. An exchange moving the coins sent to customers' deposit addresses into other wallets it controls. Aliases: deposit sweep, exchange sweep, sweep, sweep transaction, sweeps.

Source: 05-BF - Whoever holds the key controls the coins.

SWIFT. A global bank-to-bank financial messaging network that carries payment instructions but does not settle funds itself. Aliases: SWIFT (payment messaging), Society for Worldwide Interbank Financial Telecommunication.

Source: 02-AS - Global Payment Infrastructure: Overview.

Sybil attack. An attack where one adversary creates many fake peer identities to distort peer selection, routing, or consensus influence. Aliases: Sybil attack (Bitcoin network), Sybil attack (P2P threat), Sybil attacks, Sybil flood, Sybil pressure, Sybils, fake identities, fake peer attack.

Source: 02-DS - The Great P2P Weakness: Sybil & Eclipse Attacks.

Sybil resistance. Mechanisms that make it expensive for one actor to create many identities and overwhelm an open network. Aliases: Sybil resistance through economics, anti-Sybil mechanism, sybil resistance.

Source: 04-DS - Permissionless Ledgers.

Sybil risk. The risk that one actor can join or impersonate many participants to gain outsized influence. Aliases: Sybil attack, Sybil-style risk.

Source: 05-IN - Why Decentralization Is Harder.

Sybil-resistant leader election. A method for choosing who may propose a block that limits fake identities by requiring an expensive scarce resource. Aliases: Sybil resistant leader election.

Source: 02-CM - Proof of Work: Purpose and Agenda.

Symmetric cryptography. A cryptographic approach where the same shared secret key is used to encrypt and decrypt data. Aliases: symmetric crypto, symmetric encryption.

Source: 03-CP - The Shared Key.

Sync event. The log a Uniswap v2 pool writes after every swap, mint or burn, giving both reserves; divide one by the other, after scaling by decimals, for the new price. Aliases: Sync log, Sync logs.

Source: 07-EF - A swap leaves Swap and Sync logs.

Systemic trust shock. A broad loss of confidence caused by failures that cascade through an entire financial or institutional system.

Source: 03-IN - A Crisis of Trust: 2008.

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Tactic. In ATT&CK and AADAPT, the goal of a step: the why behind a technique, such as Defense Evasion. Aliases: ATT&CK tactic, tactics.

Source: 12-FO - ATT&CK: tactics are why, techniques how.

Tamper evidence. The property that changing any committed item changes hashes along its path and therefore changes the Merkle root. Aliases: Tamper evidence (Merkle tree), tamper detection, tamper-evident.

Source: 02-CP - Merkle Trees: Motivation.

Tamper-evident. A property where unauthorized changes become detectable rather than silently hidden. Aliases: tamper evidence.

Source: 02-IN - A blockchain is blocks, each linked to the last.

Tamper-evident ledger. A record whose structure makes unauthorized changes detectable by participants who verify the cryptographic links and rules. Aliases: decentralized ledger, tamper evidence, tamper-evident history.

Source: 01-CP - How Blockchain Uses Cryptography.

Taproot. The output type (bc1p… addresses) that can be spent by a single key (key path) or by revealing a script (script path); a key-path spend hides its rule. Aliases: Taproot output, bc1p.

Source: 06-BF - Taproot multi-sig can look like one key.

Tapscript. The script language of Taproot script-path spends, in which OP_CHECKMULTISIG is disabled and m-of-n is written with OP_CHECKSIGADD. Aliases: BIP 342, BIP342.

Source: 06-BF - Taproot multi-sig can look like one key.

Target checkpoint. The later checkpoint named in a validator vote that a source checkpoint is being linked to. Aliases: Target checkpoint (PoS finality vote), target epoch.

Source: 03-CM - Justification: Supermajority Vote Links.

Target rule. The validity rule that a candidate block's hash must be numerically below the current target threshold. Aliases: Target rule (proof of work), hash < target, hash-below-target rule.

Source: 02-CM - Visualizing the Target Rule.

Technique. In ATT&CK and AADAPT, the means an adversary uses to reach a tactic's goal: the how. A technique ID is a shared name other analysts read the same way. Aliases: ATT&CK technique, technique ID, techniques.

Source: 12-FO - ATT&CK: tactics are why, techniques how.

Temporary fork. A short-lived split where different valid blocks at the same height are seen by different parts of the network. Aliases: Temporary fork (proof of work), competing branch, short-lived fork.

Source: 02-CM - Block Propagation and Stale/Orphan Blocks.

Test sale. A sale a team makes to check that a listing works. One or two sales at trivial prices point to a test, not a wash trade. Aliases: test sales.

Source: 11-FO - Look-alikes that are not wash trades.

The Bitcoin Standard. Saifedean Ammous's framing of Bitcoin as a modern digital successor to gold and hard-money economics. Aliases: Bitcoin Standard, The Bitcoin Standard (book and thesis).

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

The Merge. The September 15, 2022 Ethereum upgrade that moved block production from Proof of Work to Proof of Stake while preserving existing state. Aliases: Beacon Chain merge, Merge, The Merge (Ethereum upgrade), The Merge (Ethereum).

Source: 01-EF - Pending, then included, then final.

Thin history. A small amount of interaction evidence that is not yet enough to support high confidence in a peer's reputation. Aliases: low feedback scope.

Source: 03-DS - PeerTrust: Handling Manipulation.

Thin market. A pool with small reserves, where one trade moves the price a lot, so its price is one trade's word rather than the market's. Aliases: thin exchange pool, thin exchange pools, thin markets, thin pool, thin pools.

Source: 07-EF - Arbitrage holds a deep pool near the market.

THORChain. A cross-chain swap protocol where anyone can swap a coin on one chain for a coin on another.

Source: 08-EF - Liquidity bridges: someone pays you there.

Threat model. The set of failures, attackers, and assumptions a system is designed to withstand. Aliases: threat models.

Source: 05-IN - Fault Tolerance Spectrum.

Threat modeling. A structured way to identify what can go wrong in a system, who might attack it, and which weaknesses matter most.

Source: 03-IN - Why This History Matters.

Throughput. The rate at which a ledger or database can process accepted writes or transactions. Aliases: Throughput (ledger performance), transaction throughput, write throughput.

Source: 04-DS - Permissioned Ledgers.

Time lock. A spending condition that prevents an output from being spent until a specified block height, time, or sequence condition is met. Aliases: Time lock (Bitcoin Script), time-locked output, timelock.

Source: 03-BF - Why Bitcoin Script Exists.

Time-seeded PRNG. A pseudorandom generator seeded from the current time, making outputs guessable within a narrow time window. Aliases: clock-seeded PRNG, srand(time()).

Source: 05-CP - Case Study: Linux.Encoder RNG Failure.

Time-to-Live. A counter or limit that prevents a forwarded message from spreading forever through a network. Aliases: TTL, Time-to-Live (network messages).

Source: 01-DS - Case Study: Gnutella & "Flooding".

Time-warp attack. A coordinated timestamp manipulation attack that tricks the difficulty control loop into lowering difficulty or mismeasuring time. Aliases: time warp.

Source: 02-CM - Timestamp Manipulation and Time-Warp Attacks.

Timelock. A delay mechanism that forces an upgrade or administrative action to wait before it can take effect. Aliases: Timelock (upgrade governance), time lock.

Source: 04-EF - Upgradeability and Proxy Patterns.

Timestamp manipulation. Miner behavior that sets block timestamps within allowed limits to influence how the protocol measures elapsed time. Aliases: timestamp games.

Source: 02-CM - Timestamp Manipulation and Time-Warp Attacks.

Timing attack. An attack that exploits when transactions or blocks are first observed to infer origin, gain advantage, or affect confidence. Aliases: Timing attack (Bitcoin network), latency attack, timing and latency attack.

Source: 04-BF - Timing & Latency Attacks.

Timing correlation. An analytic method that compares transaction timing across flows, services, or chains to infer likely relationships. Aliases: Timing correlation (forensics), timing analysis.

Source: 04-AS - Obfuscation and Countermeasures.

Timing fault. A failure where a node returns correct information too late or outside the required ordering window. Aliases: timing faults.

Source: 05-IN - Timing Faults.

Timing window. A bounded time interval used to match likely related deposits and withdrawals across chains or services. Aliases: Timing window (cross-chain tracing), timing windows.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Tiny transfer. A poisoning lure: a small amount of the same token sent from the look-alike to the victim.

Source: 09-FO - Address poisoning plants a look-alike in history.

Tip. The part of an Ethereum transaction fee that goes to the block's fee recipient instead of being burned. Aliases: Tip (EIP-1559), priority fee, proposer tip.

Source: 01-EF - A transaction is a signed, numbered message.

Token. A blockchain-based representation of value, rights, or other state managed by platform rules or smart contracts. Aliases: Token (blockchain), tokens.

Source: 01-IN - Week 1 teaches how it works, week 2 traces.

Token account. A separate Solana account in which a wallet holds one token; a wallet needs one for each token it holds. Aliases: SPL token account, token accounts.

Source: 08-EF - Solana keeps each token in its own account.

Token contract. Smart-contract or program logic that records token balances and enforces transfer, minting, approval, freezing, upgrade, or governance rules on an existing chain. Aliases: Token contract (tokenization), contract-issued token, token contracts, token program, token programs.

Source: 05-EF - Coin or Token? Start With the Balance Record.

Token gate. An access rule that grants or denies entry based on whether a wallet holds a specified token or balance. Aliases: token gating, token-gated access.

Source: 06-EF - Token-Based Communities.

Token ID. The number that identifies one NFT within its contract; ownerOf returns the one address that owns it. Aliases: Token ID (NFT), identifier, token identifier, unique identifier, unique identifiers.

Source: 05-EF - An NFT is a token ID with one owner.

Token incentive. A reward or ownership structure created through tokens that shapes how participants behave in a blockchain system. Aliases: economic incentive, token incentives.

Source: 05-AS - Human & Economic Constraints.

Token standard. A shared contract interface that wallets, markets, indexers, and applications rely on to interpret and interact with tokens. Aliases: ERC standard, ERC standards, Token standard (Ethereum), standards, token standards.

Source: 06-EF - Fungible vs. Non-Fungible.

Token-based community. A community that uses token ownership to control membership, access, participation, or governance. Aliases: token community, token-based communities.

Source: 06-EF - Token-Based Communities.

Token-weighted governance. A governance model where voting power is based on the amount or number of governance tokens a participant controls. Aliases: token-weighted voting.

Source: 06-EF - Two-Layer Community Architecture.

token0 / token1. A pool's two tokens, in a fixed order; check which is which before dividing one reserve by the other. Aliases: token0, token1.

Source: 07-EF - A swap leaves Swap and Sync logs.

Tokenization. The process of linking blockchain token records to claims about assets, access, identity, provenance, or rights. Aliases: Tokenization (application layer).

Source: 04-AS - NFTs and Tokenization.

Tokenized asset. A real-world or digital claim represented as an on-chain token. Aliases: tokenized assets, tokenized claim.

Source: 01-AS - Finance (High-Level Overview).

Tokenized Financial Infrastructure. Financial infrastructure that represents claims, assets, or settlement processes as digital tokens. Aliases: tokenized finance, tokenized-finance pathway.

Source: 03-AS - Latin America: Impact Signals.

tokenURI. An optional ERC-721 function that returns where a token's metadata is, often off chain, for example on IPFS. Aliases: metadata pointer, token URI, tokenURI (NFT metadata), tokenURI function.

Source: 05-EF - An NFT is a token ID with one owner.

Topic. A 32-byte indexed field in an Ethereum log used by clients to filter events efficiently. Aliases: Topic (Ethereum log), indexed topic, log topic.

Source: 04-EF - Events and Logs.

Tornado Cash. An Ethereum privacy protocol whose 2022 sanctions became a major case study in relay and builder censorship pressure. Aliases: Tornado Cash (Ethereum policy case).

Source: 02-EF - Censorship.

Traceable execution. The ability to follow and verify actions recorded by blockchain transactions and smart contracts. Aliases: Traceable execution (blockchain systems), verifiable execution.

Source: 05-AS - Two Powerful Technologies Collide.

TraderTraitor. The North Korean actors the FBI attributed the Bybit theft to, in its public service announcement of 26 February 2025.

Source: 07-EF - Bybit: stolen tokens became ETH first.

Transaction. A proposed change to blockchain state, such as transferring value or invoking a contract, that must be validated by the network. Aliases: Transaction (blockchain), transactions.

Source: 01-IN - Start by saying what cases you work.

Transaction (Bitcoin). A signed message proposing that bitcoin value move from current control to a new owner or destination. Aliases: Bitcoin transaction, payment, transaction, transactions, tx.

Source: 01-BF - Primitive 1: Ownership via Digital Signatures.

transaction (Ethereum). A signed instruction from an account, which anyone watching the network can read before a block includes it. Aliases: Ethereum transaction (Ethereum), transaction.

Source: 03-EF - A DEX trade is visible before it lands.

Transaction batching. Running several operations in one transaction, which EIP-7702 allows an ordinary account to do. Aliases: batching.

Source: 09-FO - EIP-7702 lets a plain account run code.

Transaction bundle. A group of ordered transactions submitted together to a builder with an associated bid or value estimate. Aliases: Transaction bundle (MEV), bundle, bundles.

Source: 03-EF - Searchers, builders and relays set the order.

Transaction censorship. The deliberate delay or exclusion of selected transactions from relay or confirmation. Aliases: Transaction censorship (Bitcoin), censorship, selective censorship.

Source: 04-BF - Why Attack Bitcoin?.

Transaction fee. A cost paid to have a blockchain transaction included or processed by the network. Aliases: Transaction fee (Bitcoin), Transaction fee (mining incentives), fees, miner fees, transaction fees.

Source: 03-IN - Promises vs. Reality.

Transaction graph. The network of relationships among transactions, inputs, outputs, and addresses visible from blockchain data. Aliases: Transaction graph (Bitcoin), Transaction graph (forensics), graph structure, on-chain graph, payment graph, transaction graph structure.

Source: 04-BF - Mitigating Network Deanonymization.

Transaction input. A reference to a previous UTXO plus the unlocking data needed to spend it in a new transaction. Aliases: Transaction input (Bitcoin), input.

Source: 02-BF - A trace is read one transaction at a time.

Transaction lifecycle. The five steps every payment takes: signed in the wallet, broadcast, held in the mempool, mined into a block, then confirmed by blocks on top. Aliases: Transaction lifecycle (blockchain), end-to-end transaction, lifecycle of a transaction, transaction flow.

Source: 02-IN - Every payment takes the same five steps.

Transaction ordering. The proposer's ability to decide which pending transactions appear in a block and in what order. Aliases: Transaction ordering (Bitcoin), Transaction ordering (block proposal), global transaction ordering, order of transactions, ordering power, transaction inclusion.

Source: 03-CM - The Block Proposer: Selection and Building.

Transaction output. A newly created spendable value record that locks bitcoin to a recipient, change address, or other script condition. Aliases: Transaction output (Bitcoin), output.

Source: 02-BF - A trace is read one transaction at a time.

Transaction pipeline. The end-to-end path from a user action through wallet signing, RPC broadcast, node validation, propagation, and on-chain recording. Aliases: Transaction pipeline (blockchain), transaction flow, transaction path.

Source: 01-AS - Walking a Transaction Step-by-Step.

Transaction policy. Rules that determine which organizations or roles must approve a transaction before it can be committed. Aliases: approval policy, endorsement policy.

Source: 01-PC - Fabric demo: watch the gatekeeper work.

Transaction propagation. The spreading of a valid transaction across network peers after it is submitted to a node. Aliases: Transaction propagation (blockchain network), propagation, transaction relay.

Source: 01-AS - Walking a Transaction Step-by-Step.

Transaction receipt. What a node returns for a transaction once it is in a block: its status (1 success, 0 failure), gas used, effective gas price, logs and any new contract address. Aliases: Transaction receipt (Ethereum), receipt.

Source: 01-EF - The receipt says what actually happened.

Transaction relay. The peer-to-peer process where signed transactions move from wallets to nodes and then across the network. Aliases: Transaction relay (Bitcoin network), Transaction relay (Bitcoin), broadcast, gossip, transaction propagation.

Source: 02-BF - Replace-by-fee: unconfirmed can still change.

Transaction selection. The miner's choice of which valid transactions to include when assembling a block. Aliases: Transaction selection (mining), block assembly policy, miner discretion.

Source: 02-CM - Censorship and Transaction Selection.

Transaction signing. Using a private key to create cryptographic authorization for a blockchain action. Aliases: signed transaction, signing, wallet signature.

Source: 01-AS - Wallets: The Control Layer.

Transaction sponsorship. Under EIP-7702, another account sending and paying for a transaction on a delegated account's behalf. Aliases: sponsored fees.

Source: 09-FO - EIP-7702 lets a plain account run code.

Transaction validation. The software process nodes use to check whether a transaction follows Bitcoin's rules before accepting it. Aliases: Transaction validation (Bitcoin), Transaction validation (blockchain), transaction checks, validate transactions, validation, validation logic.

Source: 04-BF - Bitcoin Overflow Bug.

transactions_root. The header commitment to the exact ordered list of transactions included in an execution payload. Aliases: transaction root, transactions root, transactions_root (Ethereum execution payload).

Source: 02-EF - Execution Payload: Header Structure.

Transfer history. The immutable record of prior token transfers and owners visible from on-chain state and events. Aliases: Transfer history (token), ownership history.

Source: 06-EF - What Is a Token?.

transferFrom. The ERC-20 call a spender uses to move an owner's tokens under an approval; it emits a Transfer event in a transaction the spender sent.

Source: 09-FO - An approval lets someone else spend your tokens.

Transitive trust. Trust extended through another trusted relationship rather than verified directly.

Source: 05-IN - Trust.

Transparency. The ability for participants or permitted members to verify state changes and rule compliance. Aliases: Transparency (distributed ledgers), verifiability.

Source: 04-DS - Transparency vs. Privacy.

Transparency log. A public append-only record that uses Merkle roots and proofs to support auditing of inclusion and history. Aliases: append-only log, transparency logs.

Source: 02-CP - Where You'll See Them (Non-Exhaustive).

Transparent coin. A coin, such as Ether or Litecoin, whose ledger shows payouts publicly, so they can be traced with the usual methods; Monero is the contrast.

Source: 08-FO - Most cryptojacking mines Monero, on CPUs.

Transport cost. The monetary and resource cost of moving, relaying, storing, or processing data across a network. Aliases: Transport cost (distributed systems), egress cost, relay cost.

Source: 01-DS - Fallacy #7: Transport Cost Is Zero.

Travel Rule. The rule that for a transfer of $3,000 or more, or its equivalent in crypto, the sending institution and any intermediary must pass on required information about the sender and recipient. Aliases: FATF Travel Rule.

Source: 05-FO - The Travel Rule sends the sender's data along.

TRC-20. Tron's fungible token standard, compatible with Ethereum's ERC-20 at the interface level, so a USDT transfer on Tron is a contract call that emits a Transfer log. Aliases: TRC20.

Source: 01-PC - Tron works like Ethereum, with T-addresses.

Treasury disbursement. The release of community or protocol funds according to contract rules, votes, or other governance processes. Aliases: Treasury disbursement (DAO), treasury release, treasury spending.

Source: 06-EF - Two-Layer Community Architecture.

Trezor. A hardware wallet that keeps the key offline and signs inside the device.

Source: 05-BF - Most people hold bitcoin one of two ways.

Tron. A public account chain that works like Ethereum (balances per address, tokens as contracts, transfers as logs); its addresses start with T, and it charges fees in Energy and Bandwidth. Aliases: T-address, TRON, Tron (stablecoin network).

Source: 01-PC - Tron works like Ethereum, with T-addresses.

Trust. A system dependency on an entity behaving correctly under stated conditions. Aliases: Trust (P2P reputation), Trust (security engineering).

Source: 05-IN - Trust.

Trust anchor. A trusted starting point used to validate a chain of trust decisions. Aliases: trust-anchor model.

Source: 05-IN - Trust.

Trust assumption. A condition about which actors, code, infrastructure, or cryptography must behave correctly for a system to remain safe. Aliases: Trust assumption (blockchain design), trust assumptions.

Source: 05-AS - Technical Constraints.

Trust boundary. The line around the people, institutions, or mechanisms that the system assumes are trustworthy enough. Aliases: Trust boundary (ledger design), boundary of trust.

Source: 04-DS - Trade-offs.

Trust but verify. A workflow that uses reputation to choose peers, cryptographic checks to verify results, and verified outcomes to update reputation. Aliases: complete trust loop, trust but verify pattern.

Source: 03-DS - Properties of Reputation Systems.

Trust matrix. A structured table of normalized trust values showing how much each peer trusts other peers. Aliases: C matrix.

Source: 03-DS - EigenTrust: From Local Opinions to Global Scores.

Trust mechanism. A system feature that turns uncertain peer behavior into signals, scores, or rules that help participants choose safer partners. Aliases: trust mechanisms.

Source: 03-DS - Intro: "The Bad Download".

Trust minimization. A design goal that reduces how much users must rely on a trusted institution or leader for the system to work. Aliases: Trust minimization (blockchain), minimized trust, trust-minimized, trust-minimized alternative.

Source: 03-IN - A Crisis of Trust: 2008.

Trust model. The explicit answer to who or what participants must trust for the system's record to remain correct. Aliases: Trust model (system design), trust assumption, trust assumptions.

Source: 04-DS - Distributed Ledger vs. Centralized Database (Compare/Contrast).

Trust relocation. The idea that blockchain changes where trust is placed, moving it into code, operators, infrastructure, incentives, and institutions. Aliases: Trust relocation (blockchain systems), relocated trust, trust shifts.

Source: 05-AS - Key Lessons.

Trust-Minimized Protocol. A protocol designed to reduce the need to rely on a central institution or intermediary behaving correctly. Aliases: trust-minimized protocols.

Source: 04-IN - Design Trade-Offs: Efficiency vs. Autonomy.

Trusted bootstrap paradox. The tension that a decentralized node still needs some initial peer-discovery source that may be centralized or compromisable. Aliases: bootstrap dependency.

Source: 02-CM - Eclipse and Network-Level Attacks.

Trusted checkpoint. A recent finalized block reference obtained from trusted or diverse sources before syncing a PoS node. Aliases: Trusted checkpoint (PoS bootstrapping), checkpoint hash, recent trusted checkpoint.

Source: 03-CM - Weak Subjectivity.

Trusted Intermediary. An institution or actor that users rely on to process, validate, or settle transactions correctly. Aliases: financial intermediary, trusted arbiter, trusted intermediaries, trusted third party.

Source: 04-IN - The Bitcoin Standard: A Digital Return to "Hard Money".

Trusted third party. An external actor that others rely on to vouch for identity, attributes, or correct behavior. Aliases: TTP, certificate authority, certification authority, third party.

Source: 05-IN - Trust.

Trustless. A misleading shorthand for systems that reduce reliance on specific institutions while still depending on explicit assumptions and mechanisms. Aliases: Trustless (blockchain claim), trustless system.

Source: 05-IN - What is "trust" in digital systems?.

Trustless coordination. Coordination among parties who do not need personal trust because cryptographic checks and protocol rules make agreement verifiable.

Source: 01-CP - Why Cryptography Matters Today.

Trustless model. A system model where participants rely on verifiable rules and mathematical proof instead of a central authority or personal trust. Aliases: trustless system, trustless systems.

Source: 01-CP - Cryptography and Trustless Systems.

Trustlessness. The claim that a system reduces the need to trust specific people or institutions by relying on code, cryptography, and consensus. Aliases: Trustlessness (blockchain claims), trustless system, trustlessness claim.

Source: 05-AS - Key Lessons.

Turing complete. A language property that supports general computation with control flow such as loops, branching, and function calls. Aliases: Turing complete (smart contract language), Turing completeness.

Source: 04-EF - From Bitcoin Script to Solidity.

TWAP. A time-weighted average price: an average over many blocks, which a recent large trade cannot move much. Aliases: time-weighted average price.

Source: 10-FO - A price oracle can be moved with borrowed money.

Two-party trust. A direct trust relationship where one party expects another party to behave as promised. Aliases: bilateral trust.

Source: 05-IN - Trust.

Two-Phase Commit. A coordination protocol where a central coordinator first asks participants if they are ready and then issues a final commit only if all agree. Aliases: 2PC, two-phase commit.

Source: 05-IN - Classical Consensus Approaches.

Type 0x04 transaction. The Ethereum transaction type that carries EIP-7702 authorizations and sets an account's code. Aliases: 0x04 transaction.

Source: 09-FO - EIP-7702 lets a plain account run code.

Typology. The named type of crime a case fits, such as cryptojacking or ransomware.

Source: 08-FO - Cryptojacking: mining on someone else's machine.

U

U.S. person. Anyone who must comply with OFAC sanctions: U.S. citizens and lawful permanent residents wherever they are, everyone in the United States, and U.S. companies with their foreign branches. Aliases: U.S. persons.

Source: 05-FO - OFAC designates; every U.S. person blocks.

Unauthenticated messages. Messages that do not cryptographically prove who sent them or that they were not forged. Aliases: oral messages.

Source: 05-IN - The Byzantine Generals Problem (formal).

Unbonding period. A delay after validator exit during which stake remains locked before withdrawal is allowed. Aliases: Unbonding period (PoS validator lifecycle), cooldown, unbonding.

Source: 03-CM - Proof of Stake: The Full Lifecycle.

Unconfirmed transaction. A transaction the network has received but that has not yet become part of accepted blockchain history. Aliases: pending transaction, unconfirmed.

Source: 02-IN - Every payment takes the same five steps.

Unhosted wallet. A self-custody wallet where the user holds the keys on their own computer, phone or device and needs no third party to transact. Aliases: non-custodial wallet, self-hosted wallet, unhosted wallets.

Source: 05-BF - Whoever holds the key controls the coins.

Unified address. A newer Zcash address format that bundles several receivers in one address. Aliases: Zcash unified address, unified addresses.

Source: 07-FO - Zcash has two pools: one public, one shielded.

Unified provenance graph. A graph that links related asset movements across chains into one traceable history. Aliases: Unified provenance graph (blockchain forensics), provenance graph.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Uniform distribution. The expectation that hash outputs are spread evenly across the output space without useful patterns or biases. Aliases: Uniform distribution (hashing), uniform distribution of output, uniform output.

Source: 02-CP - Hash Function Properties.

Uniswap. A decentralized exchange that is an automated market maker; its v2 pools hold two tokens and issue ERC-20 LP tokens, while v3 and v4 positions are price ranges instead. Aliases: Uniswap v2, Uniswap v3, Uniswap v4.

Source: 07-EF - A DEX pool is the other side of the trade.

Unit of Account. A function of money that provides a common yardstick for pricing and comparing value.

Source: 04-IN - What is Money? A Social Technology.

Unlicensed money transmitter. A money services operator that transfers value without the authorization required by law.

Source: 03-IN - Early Attempts at Digital Money.

Unlocking data. Data such as a signature and public key that satisfies an output's locking script so the referenced UTXO can be spent. Aliases: Unlocking data (Bitcoin), scriptSig, witness.

Source: 02-BF - Every output is locked by a small script.

Unlocking script. The script or data supplied by a spender to satisfy a previous output's locking script. Aliases: Unlocking script (Bitcoin Script), scriptSig.

Source: 03-BF - Bitcoin Script Basics.

Unpredictability. A property where seeing part of a sequence does not make the next value feasibly guessable. Aliases: Unpredictability (PRG property), predictability.

Source: 01-CP - Probability Refresher: Independence & Random Variables.

Unsigned integer. A nonnegative integer type commonly used for balances, counters, timestamps, and range checks. Aliases: Unsigned integer (Solidity), uint, uint256, unsigned int.

Source: 04-EF - Types: Value Types.

Unspent transaction output. A transaction output that has not yet been spent and can be consumed only by satisfying its attached spending conditions. Aliases: UTXO, UTXOs, Unspent transaction output (Bitcoin), unspent output.

Source: 03-BF - Introduction to Bitcoin Script.

Unstructured overlay network. A P2P overlay where peers form mostly random neighbor connections rather than maintaining a precise searchable structure. Aliases: random mesh, unstructured overlay.

Source: 02-DS - Unstructured Overlay Networks.

Usability. How safely and easily ordinary users can understand, operate, and recover from interactions with blockchain systems. Aliases: UX, Usability (blockchain systems), user experience.

Source: 05-AS - The Real Question.

USDC. A dollar-referenced stablecoin issued by Circle with a US-regulated, sanctions-compliant posture. Aliases: Circle USDC, USD Coin, USDC (stablecoin).

Source: 02-AS - Types of Digital Money.

USDH. Huione's own stablecoin, advertised as one that cannot be frozen.

Source: 05-FO - Huione laundered at least $4 billion.

USDT. Tether's fiat-backed, dollar-referenced stablecoin, issued on several chains; its TetherToken contract gives the owner key owner-only functions to blacklist an address and destroy its balance, on Tron as on Ethereum. Aliases: Tether, TetherToken, USDT (stablecoin), USDT on Tron.

Source: 01-PC - Half of all USDT sits on Tron.

Use Anonymizing Services. The AADAPT technique that covers CoinJoin, tumblers and peer-to-peer exchanges, all under Defense Evasion. Aliases: ADT3030.

Source: 12-FO - AADAPT names most of the typologies.

User experience. How people actually encounter, understand, and navigate blockchain systems through products and interfaces. Aliases: UX, User experience (blockchain applications).

Source: 01-AS - From Theory to Reality.

Utility layer. Operational token uses such as product passports, supply-chain records, loyalty systems, credentials, and asset records. Aliases: Utility layer (token infrastructure), operational utility layer.

Source: 06-EF - Hidden Infrastructure Growth.

Utility-Driven Crypto Adoption. Crypto adoption motivated by practical needs such as remittances, dollar access, inflation pressure, or unreliable payment infrastructure. Aliases: high-utility adoption, necessity-driven use.

Source: 03-AS - Sub-Saharan Africa: Overview.

UTXO. A discrete unspent output from an earlier transaction that can be used as an input to a later transaction. Aliases: UTXO (Bitcoin), Unspent Transaction Output, spendable output, unspent output.

Source: 02-BF - P2SH and Taproot reveal the rule only on spend.

UTXO model (Bitcoin). An accounting model where value is tracked as discrete unspent outputs that transactions consume and create. Aliases: Bitcoin UTXO Model, UTXO, UTXO model (Bitcoin accounting), UTXO model (Bitcoin transaction model), unspent transaction output, unspent transaction output model.

Source: 01-EF - Bitcoin spends outputs; Ethereum edits balances.

V

Validation. The peer-side process that rechecks endorsement policy, MVCC constraints, and transaction well-formedness after ordering. Aliases: Validation (Hyperledger Fabric), commit-time validation, transaction validation.

Source: 02-PC - One Invoke: Proposal to Commit.

Validation against policy. The check that an ordered request still satisfies the relevant policy and state rules before commit. Aliases: Validation against policy (transaction path), policy validation.

Source: 01-PC - Fabric demo: watch the gatekeeper work.

Validator. A participant that orders transactions and helps produce new blocks under the network's consensus rules. Aliases: Validator (block production), block producer, miner, miners, validators.

Source: 01-AS - Nodes and Validators.

Validator (Ethereum). A Proof-of-Stake participant that helps propose, check, attest to, and finalize blocks using staked ETH. Aliases: Validator (Ethereum PoS role), Validator (Ethereum PoS), validators.

Source: 01-EF - Pending, then included, then final.

Validator (Proof of Stake). A participant that has staked capital and performs PoS duties such as proposing blocks, attesting, and voting on checkpoints. Aliases: Validator (PoS role), Validator (Proof of Stake role), validators.

Source: 03-CM - Orientation: Proof of Work vs Proof of Stake.

Validator / endorser. A participant role that approves transactions according to the network's policy. Aliases: Validator / endorser (permissioned ledger role), approver, endorser, validator.

Source: 01-PC - Fabric demo: watch the gatekeeper work.

Validator process. Software that signs the validator duties assigned by the consensus client. Aliases: Validator process (Ethereum client), validator client.

Source: 02-EF - PoS Overview: Validators, Committees, and Clients.

Validator selection. The consensus-layer process that assigns validators to proposer and committee duties for slots and epochs. Aliases: Validator selection (Ethereum PoS), duty assignment, validator assignment.

Source: 02-EF - Validator Selection & Randomness.

Validity. The consensus property that decided values must satisfy the protocol's rules. Aliases: Validity (consensus property).

Source: 04-AS - What Consensus Is Actually Trying to Guarantee.

Validity proof. A cryptographic proof that a Layer 2 result was computed correctly without requiring Layer 1 to replay every transaction. Aliases: Validity proof (Layer 2 verification), ZK proof, validity proofs, zero-knowledge proof.

Source: 01-AS - Scaling and Interoperability.

Value Stability. The ability of a monetary asset to maintain relatively stable purchasing power or price over time.

Source: 04-IN - The Institutional Counterpoint: The BIS Critique.

Value type. A Solidity type category whose values are copied when assigned or passed to functions. Aliases: Value type (Solidity), value types.

Source: 04-EF - Types: Value Types.

Value-form. AADAPT's word for any form of digital value, coin or token.

Source: 12-FO - AADAPT: ATT&CK's shape for digital assets.

VARA. Dubai's dedicated authority for licensing and supervising virtual-asset activity under a state-defined rulebook. Aliases: Dubai Virtual Assets Regulatory Authority, Virtual Assets Regulatory Authority.

Source: 03-AS - Gulf States: Impact Signals.

Variance. The unpredictability of mining income caused by the lottery-like timing of finding valid blocks. Aliases: Variance (mining payouts), payout variance, reward variance.

Source: 02-CM - Mining Pools, Variance, and Payouts.

Vendor assessment. An analytics firm's judgment of who likely did something, built from behavior and timing it observed. Quote it as the firm's view with its date, and give it only the confidence its lines support; it is not a government attribution. Aliases: analytics firm assessment, firm's assessment, vendor's assessment.

Source: 13-FO - You characterize; the government attributes.

Vendor label. A label or report from an analytics firm naming who holds an address. It adds speed and coverage, but its basis is often unstated and it may repeat another source. Aliases: vendor labels, vendor report, vendor tag.

Source: 14-FO - The chain says what moved; others say who.

Verifiability. The ability for anyone running appropriate software to check Bitcoin's rules and history for themselves. Aliases: Verifiability (Bitcoin), independent verification.

Source: 01-BF - Benefits of Bitcoin's Design.

Verifiable credential. A digitally checkable claim, such as a diploma or identity attribute, whose authenticity can be verified across organizations. Aliases: VC, VCs, Verifiable credential (digital identity), Verifiable credential (public-sector ledgers), verifiable credentials.

Source: 04-DS - Case Study B — Public Sector: EBSI & Estonia (KSI){.smaller}.

Verifiable Delay Function. A proposed time-locked computation that is slow to produce but easy to verify, reducing the ability to compute randomness outcomes early. Aliases: VDF, VDF beacon, Verifiable Delay Function (Ethereum randomness proposal).

Source: 02-EF - Global Randomness: RANDAO + VDF Beacons.

Verifiable randomness oracle. A service that supplies randomness with a proof that the output was generated correctly. Aliases: Chainlink VRF, VRF oracle.

Source: 05-CP - Case Study: Linux.Encoder RNG Failure.

Verification-weighted feedback. Feedback that is weighted more heavily when it is backed by checked outcomes rather than unsupported claims.

Source: 03-DS - Threats: Slander.

Verified outcome. An interaction result that has been checked with evidence such as hashes, signatures, or successful task completion. Aliases: verified outcomes, verified result.

Source: 03-DS - Trust but Verify: The Complete Trust Loop.

Verifier. The party that checks a presented verifiable credential and decides whether to accept the claim. Aliases: Verifier (verifiable credentials), credential verifier.

Source: 01-AS - Identity Case: Verifiable Credentials.

Version negotiation. A compatibility process where participants determine which protocol versions or features they can safely use together. Aliases: protocol negotiation.

Source: 01-DS - Fallacy #6: There Is One Administrator.

Versioned address format. A blockchain address design that marks which cryptographic scheme or account format the address uses. Aliases: Versioned address format (blockchain), address version, new address versions, versioned address formats.

Source: 05-CP - PQC Migration in Blockchain Systems.

Victim. In the Diamond Model, the target of an event, such as the platform or person robbed. Aliases: victim vertex.

Source: 12-FO - The Diamond: four features of one event.

Virtual Asset Service Provider. A business that provides services such as exchange, custody, transfer, or other virtual-asset activity for users. Aliases: CASP, VASP, crypto asset service provider, virtual asset service providers.

Source: 03-AS - Sub-Saharan Africa: Impact Signals.

Virtual coordinate space. CAN's logical map where keys hash to coordinates and peers own regions of the space. Aliases: Virtual coordinate space (CAN), coordinate space, grid, logical map.

Source: 02-DS - DHT Example: CAN.

Virtual Currency Exchange. A service where users trade or cash out crypto assets and where account records can become attribution evidence. Aliases: VCE, VCEs, Virtual Currency Exchange (blockchain investigations), crypto exchange.

Source: 02-FO - One false link merges two actors into one.

Volunteer computing. A distributed-computing model where many volunteers contribute idle machine resources to process large workloads. Aliases: volunteer distributed computing.

Source: 01-DS - Scientific Applications.

Vote phase. The step where validators or members signal support for a candidate, so that enough matching votes can make it final. Aliases: Vote phase (consensus), vote, voting.

Source: 01-CM - Open chains buy openness with slower finality.

W

Wallet. Software or hardware that manages blockchain keys and helps users create or authorize transactions. Aliases: Bitcoin wallet, Wallet (Bitcoin), Wallet (DApp architecture), Wallet (blockchain software), crypto wallet, hardware wallet, software wallet, wallets.

Source: 01-IN - Week 1 teaches how it works, week 2 traces.

Wallet application. A user-facing tool that manages keys, displays balances, and prepares or signs blockchain transactions. Aliases: crypto wallet, wallet, wallet app.

Source: 01-AS - What Does a User Actually Interact With?.

Wallet authentication. A login or access pattern where a user proves control of a blockchain address by signing a challenge. Aliases: wallet auth.

Source: 06-EF - How Token Gating Works.

Wallet Fingerprinting. Inferring wallet software or behavior from patterns such as output order, script type, or change address habits. Aliases: Wallet Fingerprinting (Bitcoin forensics), wallet behavior fingerprinting.

Source: 02-FO - Change is a guess from several weak clues.

Wallet graph. A network view of wallets and the transfers among them used to trace flows, clusters, and relationships. Aliases: Wallet graph (forensics), address graph, operator wallet graph.

Source: 06-EF - Rug Pull Operator Wallet Graph.

Wallet reuse. The repeated use of the same wallet or related wallet cluster across activities, creating a link analysts can trace. Aliases: Wallet reuse (forensics), address reuse.

Source: 06-EF - The Frosties Rug Pull.

Wallet seed. Secret seed material from which a cryptocurrency wallet derives private keys. Aliases: seed phrase, wallet seeds.

Source: 05-CP - When Good Crypto Goes Bad: Milk Sad (CVE-2023-39910).

Wallet software. Software that helps a user create transactions and apply the private key needed to authorize them. Aliases: wallet, wallets.

Source: 02-IN - The chain shows every payment, but no names.

Wasabi. A CoinJoin design. Wasabi 1.x makes many outputs of about 0.1 BTC each; Wasabi 2.x (WabiSabi) makes many inputs and outputs with arbitrary output denominations. Aliases: WabiSabi, Wasabi 1.x, Wasabi 2.x, Wasabi Wallet.

Source: 06-FO - Equal outputs mark a CoinJoin on sight.

Wash trading. Trading in which the same actor or related wallets pass an asset among themselves, so its trail goes in a circle. Aliases: Wash trading (NFT market), wash trade, wash trades.

Source: 05-EF - An NFT is a token ID with one owner.

Wash-trading loop. One token ID passing around a few related wallets, A to B to C and back to A, while the payment flows back the other way to the seller's group. Each hop records a sale, but no real ownership changes. Aliases: token cycle, token loop.

Source: 11-FO - The loop: one token ID among related wallets.

Weak subjectivity. The PoS property that new or long-offline nodes need a recent trusted checkpoint to identify the canonical chain safely. Aliases: Weak subjectivity (PoS bootstrapping), Weak subjectivity (Proof of Stake).

Source: 03-CM - Weak Subjectivity.

Welcoming bootstrap. An onboarding approach that lets honest newcomers build reputation gradually without giving attackers immediate influence. Aliases: fair onboarding, safe onboarding.

Source: 03-DS - Key Considerations in System Design.

Wholesale CBDC. A central-bank digital instrument intended for settlement among financial institutions rather than general public use. Aliases: wholesale CBDCs.

Source: 02-AS - Central Bank Digital Currencies (CBDCs).

Wholesale Digital Currency. A digital currency designed for use by banks or financial institutions rather than ordinary retail users. Aliases: shared wholesale digital currency, wholesale CBDC.

Source: 03-AS - Gulf States: Impact Signals.

Withdrawal key. The offline key or credential used to control validator fund withdrawal and recovery. Aliases: Withdrawal key (PoS validator operations), withdrawal credential.

Source: 03-CM - Becoming a Validator.

Witness. The private information that makes a statement true and lets the prover generate a convincing proof. Aliases: Witness (zero-knowledge proofs), secret witness.

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

Witness program. The data a SegWit output locks to; for P2WSH it is 0 followed by the 32-byte hash of the witness script. Aliases: SegWit program.

Source: 06-BF - One rule, three wrappers on chain.

Witness script. In P2WSH, the script that holds the spending rule; it is the last witness item when the output is spent, and its SHA-256 hash must match the output's witness program. Aliases: witness scripts, witnessScript.

Source: 06-BF - One rule, three wrappers on chain.

Work factor. The amount of computation an attacker is expected to need to succeed against a cryptographic property. Aliases: attack cost.

Source: 02-CP - Math Spotlight: Birthday Paradox and Work Factors.

Work unit. A packaged piece of computation assigned to a participant in a distributed volunteer-computing pipeline. Aliases: Work unit (volunteer computing), work units.

Source: 01-DS - Scientific Applications.

Workaround Rail. An alternative path used when mainstream financial infrastructure becomes unavailable, risky, or politically constrained. Aliases: alternative payment rail, alternative trade rail.

Source: 03-AS - Russia: Impact Signals.

World state. A query-friendly snapshot of the latest valid ledger values derived from the block log. Aliases: World state (Hyperledger Fabric), current state, current-value snapshot.

Source: 02-PC - From Permissioned Concepts to Fabric Terms.

Wormhole. A lock-and-mint bridge that lost 120,000 wETH (about $325 million) to an attacker. Aliases: Wormhole bridge.

Source: 08-EF - Lock-and-mint: the coin waits in a contract.

Wrapped asset. A token representation created on one chain to stand in for an asset locked, mirrored, or referenced from another system. Aliases: Wrapped asset (bridging), bridge-issued asset, linked token, wrapped token.

Source: 01-AS - Bridges: Linking Separate Chains.

X

XSema. A research system described in the lesson that automates cross-chain provenance extraction from bridge logs and related evidence.

Source: 01-FO - Future of Forensics: Automated Cross-Chain Tracing.

Z

Zcash. A cryptocurrency that supports optional shielded transactions backed by zero-knowledge proofs. Aliases: ZEC, Zcash (privacy coin).

Source: 04-AS - Privacy Coins.

Zero-decoy input. An early Monero input whose ring had no decoys, so it named the output it spent, as a Bitcoin input does. Aliases: 0-mixin input, no-decoy input, zero-mixin input.

Source: 07-FO - Old Monero leaked: zero decoys, newest is real.

Zero-Knowledge Proof. A proof that convinces a verifier a statement is true without revealing the secret witness or other private information behind it. Aliases: ZK proof, ZKP, zero-knowledge proof, zero-knowledge proofs, zero-knowledge proof.

Source: 05-CP - Zero-Knowledge Proofs: Prove Without Revealing.

Zero-value transfer. A poisoning lure: a transfer of zero of the same token that appears to go from the victim to the look-alike, though the attacker created it. Aliases: zero-value lure.

Source: 09-FO - Address poisoning plants a look-alike in history.

ZK circuit. A constraint representation of a computation that a zero-knowledge proof system can prove was satisfied. Aliases: ZK circuits, zero-knowledge circuit.

Source: 05-CP - Zero-Knowledge in Blockchain Practice.

ZK rollup. A scaling design that batches many transactions off chain and posts a succinct validity proof for on-chain verification. Aliases: ZK rollup (Ethereum scaling), ZK rollups, validity rollup, zero-knowledge rollup.

Source: 05-CP - Zero-Knowledge in Blockchain Practice.

zk-SNARK. A compact zero-knowledge proof that can verify a claim without revealing the underlying private data. Aliases: zero-knowledge succinct non-interactive argument of knowledge, zk-SNARK (privacy proof), zk-SNARKs.

Source: 04-AS - Privacy Coins.

zkSNACKs. The company that ran the Wasabi 2.x coordinator until it stopped on 1 June 2024, citing legal uncertainty.

Source: 06-FO - 2024: Samourai shut down, zkSNACKs stopped.